

Mathematical Aspects of Quantum Theories

A thesis submitted in partial fulfillment of the requirements for the
award of the degree of

MASTER OF SCIENCE

by

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June 2020

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Preface

The first complete description of Quantum mechanics as a mathematically rigorous theory is attributed to John von Neumann[5], via the Dirac - von Neumann axioms. Neumann, used the Spectral theory in functional analysis, to cast quantum mechanics in a rigid mathematical framework. Other formalisms like the C^* -algebra formalism came later following von Neumann's footsteps. In the following chapters, we study the mathematical structure that underlies quantum mechanics. We study Banach and Hilbert spaces in generality in the first chapter[1, 2], then specialize to the cases which are important to quantum mechanics. Measure theory is another necessary ingredient to understand quantum mechanics in its rigorous form[3, 2, 1]. It is introduced in chapter 2. Then we study the properties of the self-adjoint operators in chapter 3[1, 2]. The next chapter develops the two most important theorems in quantum mechanics, the spectral theorem and the stone's theorem[1, 2]. The applications of the formalism developed is explored in chapter 5[1, 2]. In the final chapter, we study the algebraic formulation of quantum field theory[4].

Chapter 1

Banach and Hilbert Spaces

Hilbert spaces and Banach spaces play a central role in the study of quantum mechanics. Like the phase space in classical physics, Hilbert space is the arena in which quantum mechanics plays out. Banach spaces also appear in the study of quantum mechanics quite often. We study them in detail in this chapter.

1.1 Banach Spaces[1]

Banach spaces are one very important class of spaces in functional analysis, and is heavily used in quantum mechanics.

Definition 1.1.1. A Banach space $(V, +, \cdot, \|\cdot\|)$ is a vector space with a norm,

$$\|\cdot\| : V \rightarrow \mathbb{R}$$

with, $\forall f, g \in V$ and $\forall \lambda \in \mathbb{C}$

1. $\|\lambda f\| = |\lambda| \cdot \|f\|$
2. $\|f + g\| \leq \|f\| + \|g\|$
3. $\|f\| \geq 0$ with $\|f\| = 0 \iff f = 0$

And, V is complete with respect to the norm defined.

Remark. ‘Complete’ means that any Cauchy sequence taken from V converges in V itself with respect to the defined norm.

Now, we study maps between spaces. Consider V a normed space (not necessarily Banach) and W a Banach space. Consider a map,

$$A : V \rightarrow W$$

Definition 1.1.2. A is called Bounded if,

$$\sup_{f \in V} \frac{\|Af\|}{\|f\|} < \infty$$

In that case, we define the operator norm as,

$$\|A\| := \sup_{f \in V} \frac{\|Af\|}{\|f\|} = \sup_{\|f\|=1} \|Af\|$$

Example 1.1.2.1. Let $V = W$, both Banach. Define,

$$id_W : W \rightarrow W \quad \text{such that} \quad f \mapsto f$$

Then,

$$\|id_W\| = \sup_{\|f\|=1} \|id_W f\| = \sup_{\|f\|=1} \|f\| = 1$$

This is an example for a bounded map.

Example 1.1.2.2. Let $W = C^0([0, 1]) := \left\{ f : [0, 1] \rightarrow \mathbb{C} \mid f \text{ is continuous} \right\}$. and define the norm on W as,

$$\forall f \in W, \quad \|f\| := \sup_{x \in [0, 1]} |f(x)|$$

Proposition 1.1.1. W is Banach with respect to the defined norm

Proof. Take any $\{f_n\}_{n \in \mathbb{N}} \subset W$ which is Cauchy. i.e,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : m, n \geq N \Rightarrow \|f_n - f_m\| < \epsilon$$

Now,

$$\|f_n - f_m\| = \sup_{x \in [0, 1]} |f_n(x) - f_m(x)|$$

Hence for any $x \in [0, 1]$ we have,

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\| < \epsilon$$

Which says $\{f_n(x)\}_{n \in \mathbb{N}} \subset C$ is Cauchy for every $x \in [0, 1]$. Since $\mathbb{C}$ is Banach (in fact Hilbert), this sequence converge in $\mathbb{C}$. Let, $f : W \rightarrow \mathbb{C}$ be

$$f(x) := \lim_{n \rightarrow \infty} f_n(x)$$

Then,

$$\|f_n - f\| = \sup_{x \in [0, 1]} \left| f_n(x) - \lim_{m \rightarrow \infty} f_m(x) \right| = \lim_{m \rightarrow \infty} \|f_n - f_m\| < \epsilon$$

Hence,

$$\lim_{m \rightarrow \infty} f_m = f$$

Which shows W is complete w.r.t. the norm and hence Banach. ■

To continue with the example, Define $V = C^1([0, 1])$, set of once continuously differentiable functions from $[0, 1]$ to $\mathbb{C}$. Define, $P : V \rightarrow W$ as,

$$\forall f \in V, \quad Pf := f'$$

Proposition 1.1.2. *P is unbounded.*

Proof. Choose $f_n(x) = \sin(2\pi nx)$, a sequence from V .

$$\|f_n\| = \sup_{x \in [0, 1]} |\sin(2\pi nx)| = 1$$

. Now,

$$\|P\| = \sup_{f \in V} \frac{\|Pf\|}{\|f\|} \geq \sup_{f_n \in V} \frac{\|Pf_n\|}{\|f_n\|}$$

But, $\|Pf_n\| = \sup_{x \in [0, 1]} |2\pi n \cos(2\pi nx)| = 2\pi n$ Hence,

$$\|P\| \geq \sup_{n \in \mathbb{N}} (2\pi n) \rightarrow \infty$$

Hence P is not Bounded. ■

Remark. Apart from some “minor” details (i.e, in QM we look at Hilbert spaces and do not really use the norm defined above) and the momentum operator is defined as $P\psi = -i\hbar\psi'$, the previous example tell us that the momentum operator in QM is unbounded!

Theorem 1.1.1. *Let V be a normed space and W a Banach space. Then the set,*

$$\mathcal{L}(V, W) := \left\{ A : V \rightarrow W \mid A \text{ is linear and bounded} \right\}$$

is a Banach space if equipped with the operator norm $\|\cdot\|_{\mathcal{L}}$.

Proof. We omit the minor steps, i.e, defining ‘+’ and ‘·’ and showing $\mathcal{L}(V, W)$ is a vector space. Also, one must show that $\|\cdot\|_{\mathcal{L}}$ does satisfy the properties of the norm. We need to show the completeness. For that, take $\{A_n\}_{n \in \mathbb{N}} \subset \mathcal{L}(V, W)$ a Cauchy sequence. i.e,

$$\forall \epsilon > 0 : \exists N \in \mathbb{N} : n, m \geq N \Rightarrow \|A_n - A_m\| < \epsilon$$

Now,

$$\|A_n - A_m\| = \sup_{f \in V} \frac{\|(A_n - A_m)f\|}{\|f\|} \geq \frac{\|A_n f - A_m f\|}{\|f\|}$$

For some $f \in V$. This would give,

$$\forall \epsilon > 0 : \exists N \in \mathbb{N} : n, m \geq N \Rightarrow \|A_n f - A_m f\| < \epsilon \cdot \|f\|$$

Hence, for any $f \in V$, $\{A_n f\}_{n \in \mathbb{N}} \subset W$ is Cauchy. But since W is Banach, the sequence converge in W . Then define, a candidate map, $A : V \rightarrow W$ as,

$$\forall f \in V : A f = \lim_{n \rightarrow \infty} A_n f$$

First check A is linear:

$$A(f + g) = \lim_{n \rightarrow \infty} A_n(f + g) = \lim_{n \rightarrow \infty} (A_n f + A_n g) = \lim_{n \rightarrow \infty} A_n f + \lim_{n \rightarrow \infty} A_n g = A f + A g$$

A similar analysis will show $(\lambda A)f = \lambda(Af)$. Now check if A is bounded:

$$\|A\| = \sup_{f \in V} \frac{\|A f\|}{\|f\|} = \sup_{f \in V} \frac{\|\lim_{n \rightarrow \infty} A_n f\|}{\|f\|} = \lim_{n \rightarrow \infty} \sup_{f \in V} \frac{\|A_n f\|}{\|f\|} = \lim_{n \rightarrow \infty} \|A_n\|$$

the third equal sign follows since the norm is continuous. We need to show $\|A_n\|$ is bounded. For that use the fact that $\{A_n\}$ is Cauchy.

$$\forall \epsilon > 0 : \exists N \in \mathbb{N} : n \geq N \Rightarrow \|A_n - A_N\| < \epsilon$$

Using triangle inequality of the norm, $\|A_n - A_N\| \geq \|A_n\| - \|A_N\|$, we get,

$$\|A_n\| < \epsilon + \|A_N\|$$

Hence, A_n is bounded, imply $\|A\| < \infty$, which gives us A is bounded. These two results gives us $A \in \mathcal{L}(V, W)$.

Now we need to show $\lim_{n \rightarrow \infty} A_n = A$:

Since A_n is Cauchy,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : \forall n, m \geq N : \|A_n - A_m\| < \epsilon$$

Which gives, $\forall f \in V$;

$$\|A_n - A_m\| \geq \frac{\|(A_n - A_m)f\|}{\|f\|}$$

$$\Rightarrow \forall \epsilon > 0, \exists N \in \mathbb{N} : \forall n, m \geq N : \forall f \in V : \|A_n f - A_m f\| < \epsilon \cdot \|f\|$$

Take $m \rightarrow \infty$ limit. As the norm is continuous, we can take the limit inside the norm. Also use $\lim_{m \rightarrow \infty} A_m f = A f$ to get,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : \forall n \geq N : \forall f \in V : \frac{\|A_n f - A f\|}{\|f\|} < \epsilon$$

Since the above is true for every $f \in V$,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : \forall n \geq N : \|A_n - A\| < \epsilon$$

■

Another important theorem we have is the *Bounded Linear Transformation*(BLT) theorem.

Theorem 1.1.2. Consider $A \in \mathcal{L}(D(A), W)$ which is densely defined. ($\overline{D(A)} = V$, where V is normed; and W is Banach). Then there is a unique extension of A to V . i.e, $\exists! \hat{A} \in \mathcal{L}(V, W)$ such that

$$\forall f \in D(A) : \hat{A}f = Af$$

Proof. Take any $f \in V$. Since $D(A)$ is densely defined, we can find $\{f_n\}_{n \in \mathbb{N}} \subset D(A)$ with

$$\lim_{n \rightarrow \infty} f_n = f$$

Then $\{f_n\}_{n \in \mathbb{N}}$ is Cauchy on $D(A)$. Since A is a bounded operator, $\{Af_n\}_{n \in \mathbb{N}}$ is Cauchy on W . Now, as W is Banach,

$$\lim_{n \rightarrow \infty} Af_n \in W \quad \text{exists}$$

Define the extension $\hat{A}$ as,

$$\begin{aligned}\hat{A}: V &\rightarrow W \\ f &\mapsto \hat{A}f := \lim_{n \rightarrow \infty} (Af_n)\end{aligned}$$

First, check this is at all an extension: Let $f \in D(A)$. Then choose $f_n = f$. Then,

$$\hat{A}f = \lim_{n \rightarrow \infty} Af_n = \lim_{n \rightarrow \infty} Af = Af \quad \text{Done!}$$

Now, check $\hat{A}$ is linear:

$$\begin{aligned}\hat{A}(f + g) &= \lim_{n \rightarrow \infty} A(f_n + g_n) \\ &= \lim_{n \rightarrow \infty} (Af_n + Ag_n) \\ &= \lim_{n \rightarrow \infty} Af_n + \lim_{n \rightarrow \infty} Ag_n \\ &= \hat{A}f + \hat{A}g\end{aligned}$$

(Here, f_n and g_n are some sequences in $D(A)$ which converge to f and g respectively). Similarly we get, $\hat{A}(\lambda f) = \lambda(\hat{A}f)$.

Now we need to show the well definition of $\hat{A}$: Take two sequences from $D(A)$ which converge to f . Then,

$$\begin{aligned}\hat{A}f &= \lim_{n \rightarrow \infty} Af_n \\ \hat{A}'f &= \lim_{n \rightarrow \infty} Af'_n\end{aligned}$$

Now,

$$\begin{aligned}\hat{A}f - \hat{A}'f &= \lim_{n \rightarrow \infty} Af_n - \lim_{n \rightarrow \infty} Af'_n \\ &= \lim_{n \rightarrow \infty} A(f_n - f'_n) \\ &= \hat{A}(f - f) \\ &= 0 \quad \text{Done!}\end{aligned}$$

Lastly, we need to show $\hat{A}$ is bounded:

$$\begin{aligned}
\|\hat{A}\| &= \sup_{f \in V} \frac{\|\hat{A}f\|}{\|f\|} \\
&= \sup_{f \in V} \frac{\left\| \lim_{n \rightarrow \infty} Af_n \right\|}{\left\| \lim_{n \rightarrow \infty} f_n \right\|} \\
&= \sup_{f \in V} \lim_{n \rightarrow \infty} \frac{\|Af_n\|}{\|f_n\|} \\
&= \lim_{n \rightarrow \infty} \sup_{f_n \in D(A)} \frac{\|Af_n\|}{\|f_n\|} \\
&= \lim_{n \rightarrow \infty} \|A\| \\
&= \|A\| < \infty \quad \text{Done!}
\end{aligned}$$

■

Definition 1.1.3. Let V be a normed space. Then the *Dual space* of V , denoted by V^* is defined as,

$$(V^*, \|\cdot\|_{V^*}) := (\mathcal{L}(V, \mathbb{C}), \|\cdot\|_{\mathcal{L}(V, \mathbb{C})})$$

Remark. Since $\mathbb{C}$ is a Banach space, the dual V^* is Banach, even if V is only normed.

1.2 Hilbert Spaces[1]

Hilbert spaces are yet another important class of spaces central to the study of quantum mechanics. In particular, the space called *separable Hilbert spaces* are all we need in quantum mechanics.

Definition 1.2.1. A Hilbert space $(\mathcal{H}, +, \cdot, (\cdot, \cdot))$ is a vector space, with a sesquilinear inner product defined on it, and is complete with respect to the norm induced by the inner product.

Remark. Hilbert spaces is a special case of Banach spaces. The norm we use here is,

$$\|\psi\| := \sqrt{(\psi, \psi)}$$

Remark. $(\mathcal{L}(\mathcal{H}, \mathcal{H}), \|\cdot\|_{\mathcal{L}(\mathcal{H}, \mathcal{H})})$ is a Banach space.

Theorem 1.2.1. *Let V be a normed space. The norm $\|\cdot\| : V \rightarrow \mathbb{R}$ is induced by a sesqui-linear inner product if and only if the parallelogram identity, $\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2$ is satisfied. In this case, the inner product is given by,*

$$(f, g) := \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2 + i\|f - ig\|^2 - i\|f + ig\|^2)$$

Proof. The only if part follows trivially from direct computation. For the if part, assume the norm satisfy the parallelogram identity and define the inner product as given in the theorem. Now,

$$\begin{aligned} ((f, g))^* &= \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2 + i\|f - ig\|^2 - i\|f + ig\|^2)^* \\ &= \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2 - i\|f - ig\|^2 + i\|f + ig\|^2) \\ &= \frac{1}{4}(\|g + f\|^2 - \|g - f\|^2 + i\|g - if\|^2 - i\|g + if\|^2) \\ &= (g, f) \end{aligned}$$

Now,

$$(f, g + h) = \frac{1}{4}(\|f + g + h\|^2 - \|f - g - h\|^2 + i\|f - ig - ih\|^2 - i\|f + ig + ih\|^2)$$

By repeated application of the parallelogram identity,

$$\begin{aligned} \operatorname{Re}((f, g + h)) &= \frac{1}{4}(\|f + g + h\|^2 - \|f - g - h\|^2) \\ &= \frac{1}{4}(\|f + g + h\|^2 + \|f + g - h\|^2 - \|f + g - h\|^2 - \|f - g - h\|^2) \\ &= \frac{1}{4}(2\|f + g\|^2 + 2\|h\|^2 - 2\|f - h\|^2 - 2\|g\|^2) \\ &= \frac{1}{4}(2\|f + g\|^2 + 2\|f\|^2 + 2\|h\|^2 - 2\|f - h\|^2 - 2\|f\|^2 - 2\|g\|^2) \\ &= \frac{1}{4}(2\|f + g\|^2 + \|f + h\|^2 + \|f - h\|^2 - 2\|f - h\|^2 - \|f + g\|^2 - \|f - g\|^2) \\ &= \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2 + \|f + h\|^2 - \|f - h\|^2) \\ &= \operatorname{Re}((f, g)) + \operatorname{Re}((f, h)) \end{aligned}$$

Replacing g, h with $-ig, -ih$ will give the result,

$$\operatorname{Im}((f, g + h)) = \operatorname{Im}((f, g)) + \operatorname{Im}((f, h))$$

Hence,

$$(f, g + h) = (f, g) + (f, h)$$

Now, since

$$(f, 0) = \frac{1}{4}(\|f\|^2 - \|f\|^2 + i\|f\|^2 - i\|f\|^2) = 0$$

We have $(f, 0g) = 0(f, g)$. Now by induction,

$$(f, (n+1)g) = (f, ng) + (f, g) = n(f, g) + (f, g) = (n+1)(f, g)$$

Now, since $(f, g) + (f, -g) = (f, g - g) = 0$, we have $(f, -g) = -(f, g)$. Hence by induction, $\forall m \in \mathbb{Z} : (f, mg) = m(f, g)$. Now,

$$\text{for } m \in \mathbb{Z} \setminus \{0\} : m \left(f, \frac{1}{m}g \right) = \left(f, m \frac{1}{m}g \right) = (f, g)$$

Which gives, $(f, \frac{1}{m}g) = \frac{1}{m}(f, g)$. Hence by induction, $\forall r \in \mathbb{Q} : (f, rg) = r(f, g)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$ and norms are continuous, we have $\forall r \in \mathbb{R} : (f, rg) = r(f, g)$. Now using the definition itself we get,

$$(f, ig) = i(f, g)$$

Hence,

$$\forall c \in \mathbb{C} : (f, cg) = c(f, g)$$

Hence the defined inner-product is sesqui-linear. Now, from the definition,

$$(f, f) = \|f\|^2$$

Hence the inner product is positive definite. Hence we have the proof! ■

Example 1.2.1.1. In this example, we will look at a norm which cannot be generated from an inner-product. Take $V = \mathbb{C}^n$ and define $\|x\| := \max_{1 \leq i \leq n} |x_i|, \forall x \in \mathbb{C}^n$. Choose,

$$f = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} -1/2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$\|f + g\| = 1/2$, $\|f - g\| = 3/2$. $\|f\| = 1$ and $\|g\| = 0$. Clearly, the parallelogram identity fails in this case. Hence the max norm cannot come from an inner-product.

Corollary 1.2.1.1. $\mathcal{H}^*$ is a Hilbert space.

Proof. We already know that $\mathcal{H}^*$ is Banach. The norm defined on it is the operator norm. i.e, for some $\ell \in \mathcal{H}^*$,

$$\|\ell\| := \sup_{\psi \in \mathcal{H}} \frac{|\ell(\psi)|}{\|\psi\|}$$

We need to show that the norm obeys the parallelogram identity. Take $m \in \mathcal{H}^*$.

$$\begin{aligned} \|\ell + m\|^2 + \|\ell - m\|^2 &= \sup_{\psi \in \mathcal{H}} \left(\frac{|\ell(\psi) + m(\psi)|}{\|\psi\|} \right)^2 + \sup_{\psi \in \mathcal{H}} \left(\frac{|\ell(\psi) - m(\psi)|}{\|\psi\|} \right)^2 \\ &= \sup_{\psi \in \mathcal{H}} \frac{|\ell(\psi) + m(\psi)|^2 + |\ell(\psi) - m(\psi)|^2}{\|\psi\|^2} \end{aligned}$$

Since $\mathbb{C}$ is a Hilbert space, we can use the parallelogram identity there. Hence we get,

$$\begin{aligned} \|\ell + m\|^2 + \|\ell - m\|^2 &= \sup_{\psi \in \mathcal{H}} \frac{2|\ell(\psi)|^2 + 2|m(\psi)|^2}{\|\psi\|^2} \\ &= 2\|\ell\|^2 + 2\|m\|^2 \end{aligned}$$

Which gives us that the norm is induced from a sesqui-linear inner product. Hence the proof. ■

Theorem 1.2.2. Cauchy - Schwarz inequality

$$\forall \psi, \phi \in \mathcal{H} : |(\psi, \phi)| \leq \|\psi\| \cdot \|\phi\|$$

Proof. Let $\lambda \in \mathbb{C}$. Then,

$$\begin{aligned} 0 &\leq \|\psi - \lambda \cdot \phi\|^2 \\ &= \|\psi\|^2 - \lambda \cdot (\psi, \phi) - \lambda^* \cdot (\phi, \psi) + |\lambda|^2 \cdot \|\phi\|^2 \end{aligned}$$

Choose $\lambda = \frac{(\phi, \psi)}{\|\phi\|^2}$. Then,

$$0 \leq \|\psi\|^2 - \frac{|(\psi, \phi)|^2}{\|\phi\|^2} - \frac{|(\psi, \phi)|^2}{\|\phi\|^2} + \frac{|(\psi, \phi)|^2}{\|\phi\|^4} \cdot \|\phi\|^2$$

Rearranging the above result, we get,

$$|(\psi, \phi)| \leq \|\psi\| \cdot \|\phi\|$$

■

Remark. The result holds even if $\mathcal{H}$ is just an inner-product space.

The Notion of a Basis

In linear algebra, we have a notion of basis for a vector space, with no additional structure on it. This is known as the *Hamel Basis*.

Definition 1.2.2. Let $(V, +, \cdot)$ be a vector space. The a subset, $\mathcal{B} \subseteq V$ is called a Hamel Basis if;

1. Any finite subset, $\{e_1, e_2, \dots, e_n\} \subseteq \mathcal{B}$ is linearly independent. i.e,

$$\sum_{i=1}^n \lambda^i e_i = 0 \Rightarrow \lambda^i = 0 \quad \forall i = 0, 1, \dots, n$$

2. For any $v \in V$, there exists a finite subset $\{e_1, e_2, \dots, e_n\} \subseteq \mathcal{B}$ and complex numbers $v^1, v^2, \dots, v^n$ such that ;

$$v = \sum_{i=1}^n v^i e_i$$

If there is a finite basis, then we define the dimension of the vector space V as

$$\dim(V) = |\mathcal{B}|$$

Otherwise the dimensionality is infinite.

But when we have a Hilbert space at hand, using the additional structure it contains, one may define a refined notion of basis, called the *Schauder Basis*

Definition 1.2.3. Let $(\mathcal{H}, +, \cdot, (\cdot, \cdot))$ be a Hilbert space. The a subset, $\mathcal{S} \subseteq \mathcal{H}$ is called a Schauder Basis if;

1. Any finite subset, $\{e_1, e_2, \dots, e_n\} \subseteq \mathcal{S}$ is linearly independent. i.e,

$$\sum_{i=1}^n \lambda^i e_i = 0 \Rightarrow \lambda^i = 0 \quad \forall i = 0, 1, \dots, n$$

2. For any $\psi \in \mathcal{H}$, there exists a countable subset $\{e_1, e_2, \dots\} \subseteq \mathcal{S}$ and a sequence of complex numbers $\psi^1, \psi^2, \dots$ such that ;

$$\psi = \begin{cases} \sum_{i=1}^d \psi^i e_i & \text{if } |\mathcal{S}| < \infty \\ \sum_{i=1}^{\infty} \psi^i e_i & \text{otherwise} \end{cases}$$

3. $\mathcal{S}$ is the smallest such set

Remark. We did not define a notion of Schauder basis in linear algebra because we cannot talk about convergence of sequences in that case.

Remark. In general, $\mathcal{S} \subseteq \mathcal{B}$. This is because if ψ is written as a limit in the Schauder basis, then to write ψ using the Hamel basis, we need to have this limit point in $\mathcal{B}$, since ψ can only be represented as a finite sum of elements from the Hamel basis.

Remark. We can rephrase the condition (2) for Schauder basis as follows. First define the *span*, i.e, for any $M \subseteq \mathcal{H}$,

$$\text{span}(M) := \{\text{Finite linear combinations of elements from } M\}$$

Using this, we rephrase the condition (2) as,

$$\text{span}(\mathcal{S}) \text{ is dense in } \mathcal{H} \iff \overline{\text{span}(\mathcal{S})} = \mathcal{H}$$

Also,

$$\text{span}(\mathcal{B}) = \mathcal{H}$$

Remark. If we take some $\sum_{i=1}^{\infty} \psi^i e_i$ and add it to $\mathcal{S}$, then we have two ways of representing the same thing, namely, 1. (the new element) and the infinite sum. Then the uniqueness is lost. One might object that due to condition (1) the uniqueness stay intact. But no! (1) talks about only finite subsets. Hence we add the third property.

Definition and Classification of Separable Hilbert Spaces

Definition 1.2.4. A Hilbert space is *Separable* if it has a Schauder Basis $\mathcal{S}$ which is,

1. Countable
2. Orthonormal. i.e, $\forall e_i, e_j \in \mathcal{S} : (e_i, e_j) = \delta_{ij}$

Now to classify the Hilbert spaces, we define the following.

Definition 1.2.5. Let $\mathcal{H}$ and $\mathcal{G}$ be two Hilbert spaces. Then a bijection $U \in \mathcal{L}(\mathcal{H}, \mathcal{G})$, preserving the inner product structure (i.e, $\forall \psi, \phi \in \mathcal{H} : (U\psi, U\phi) = (\psi, \phi)$) is called a *Unitary map*.

Remark. If such a U exists between two Hilbert spaces, then these two spaces cannot be distinguished using only $+, \cdot, (\cdot, \cdot)$.

Remark. Since we have the polarisation identity, it suffices to check whether U preserves the norm.

Definition 1.2.6. If a Unitary map exists between $\mathcal{H}$ and $\mathcal{G}$, then these two Hilbert spaces are called *Unitarily isomorphic*. We may write it as,

$$\mathcal{H} \cong_{\text{Hil}} \mathcal{G}$$

There is one particular Hilbert space we would be interested. It is the space of all square summable sequences.

Definition 1.2.7. The space of all square summable sequences in $\mathbb{C}$,

$$\ell^2(\mathbb{N}) := \left\{ a : \mathbb{N} \rightarrow \mathbb{C} \mid \sum_{i=1}^{\infty} |a_i|^2 < \infty \right\}$$

We equip $\ell^2(\mathbb{N})$ with point wise addition and s multiplication. And define a sesquilinear inner product as,

$$(a, b) := \sum_{i=1}^{\infty} a_i^* b_i$$

Lemma 1.2.3. $(\ell^2(\mathbb{N}), +, \cdot, (\cdot, \cdot))$ is a separable Hilbert space.

Proof. Take any Cauchy sequence, $\{a^n\}_{n \in \mathbb{N}} \subseteq \ell^2(\mathbb{N})$. i.e,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : n, m \geq N \Rightarrow \|a^n - a^m\| < \epsilon$$

Which means,

$$\sum_{i=1}^{\infty} |a_i^n - a_i^m|^2 < \epsilon^2 \Rightarrow |a_i^n - a_i^m| < \epsilon \quad \forall i \in \mathbb{N}$$

Hence $\{a_i^n\}_{n \in \mathbb{N}}$ is Cauchy in $\mathbb{C}$ for all $i \in \mathbb{N}$. Since $\mathbb{C}$ is a Hilbert space, this sequence converge in $\mathbb{C}$. Let,

$$\lim_{n \rightarrow \infty} a_i^n = a_i$$

And define $a \in \ell^2(\mathbb{N})$ as $a(i) := a_i$. Then trivially,

$$\lim_{n \rightarrow \infty} a^n = a$$

Hence, $\ell^2(\mathbb{N})$ is a Hilbert space. Now we have to find a countable orthonormal Schauder basis. Consider,

$$\mathcal{S} = \{e^1, e^2, \dots\}$$

with

$$e_i^n = \begin{cases} 0 & \text{if } i \neq n \\ 1 & \text{otherwise} \end{cases}$$

Then $\mathcal{S}$ is countable. And it is orthonormal. Also, each element is square summable. Take any $a \in \ell^2(\mathbb{N})$, then

$$a = \sum_{i=1}^{\infty} a_i e^i$$

Hence, $\mathcal{S}$ is a Schauder basis. Hence, the proof. ■

Now we are ready to classify all possible different Hilbert spaces. Remarkably, it turns out that there is only one kind of separable Hilbert space!

Theorem 1.2.4. *Any separable Hilbert space $(\mathcal{H}, +, \cdot, (\cdot, \cdot))$ is unitarily isomorphic to $(\ell^2(\mathbb{N}), +, \cdot, (\cdot, \cdot))$. i.e.,*

$$(\mathcal{H}, +, \cdot, (\cdot, \cdot)) \cong_{Hil} (\ell^2(\mathbb{N}), +, \cdot, (\cdot, \cdot))$$

Proof. We construct the bijective map $U \in \mathcal{L}(\mathcal{H}, \ell^2(\mathbb{N}))$ directly.

$$U : \psi \mapsto U\psi := \{(e_n, \psi)\}_{n \in \mathbb{N}} \subseteq \mathbb{C}$$

Where $\{e_n\}$ is an orthonormal basis of $\mathcal{H}$. Since,

$$\psi = \sum_{i=1}^{\infty} (e_i, \psi) e_i$$

and

$$\|\psi\|^2 = \sum_{i=1}^{\infty} |(e_i, \psi)|^2 < \infty$$

$U\psi \in \ell^2(\mathbb{N})$.

Define,

$$U^{-1} : \ell^2(\mathbb{N}) \rightarrow \mathcal{H}$$

$$\{a_n\}_{n \in \mathbb{N}} \mapsto \sum_{i=1}^{\infty} a_i e_i$$

$$U^{-1}(U\psi) = U^{-1}(\{(e_n, \psi)\}_{n \in \mathbb{N}}) = \sum_{i=1}^{\infty} (e_i, \psi) e_i = \psi$$

Which gives,

$$U^{-1} \circ U = id_{\mathcal{H}}$$

Similarly,

$$U \circ U^{-1} = id_{\ell^2}$$

Hence U is a bijection. Now, if U is unitary, then $\|U\| = 1$. To show this, take some $\psi \in \mathcal{H}$

$$\begin{aligned} \|\psi\|^2 &= \left\| \sum_{i=1}^{\infty} (e_i, \psi) e_i \right\|^2 \\ &= \left\| \lim_{N \rightarrow \infty} \sum_{i=1}^N (e_i, \psi) e_i \right\|^2 \\ &= \lim_{N \rightarrow \infty} \left\| \sum_{i=1}^N (e_i, \psi) e_i \right\|^2 \\ &= \lim_{N \rightarrow \infty} \sum_{i=1}^N \|(e_i, \psi) e_i\|^2 \\ &= \lim_{N \rightarrow \infty} \sum_{i=1}^N |(e_i, \psi)|^2 \\ &= \|U\psi\|^2 \end{aligned}$$

Here, the third equal sign follows because the norm is continuous and the fourth equal sign follows from applying the Pythagoras theorem. Hence, we have

$$\|U\| = 1$$

Which completes the proof. ■

1.3 Closed, Linear Subspaces and the Riesz Representation Theorem[1, 2]

Let $\mathcal{H}$ be a separable Hilbert space.

Definition 1.3.1. Let $\psi \in \mathcal{H}$, and $e \in \mathcal{H}$ a unit vector. Then,

$$\psi_{||} := (e, \psi) e$$

is called the *projection* of ψ to e and

$$\psi_{\perp} := \psi - \psi_{||}$$

the *orthogonal compliment*

We would like to extend this definition to atmost countable set $\{e_i\}_{i \in I}$ where I is some countable index set.

Proposition 1.3.1. Let $\psi \in \mathcal{H}$ and $\{e_i\}_{i \in I}$ be an orthonormal set on $\mathcal{H}$, for atmost countable I . Then,

1.

$$\psi = \psi_{||} + \psi_{\perp}$$

where

$$\psi_{||} := \sum_{i \in I} (e_i, \psi) e_i$$

$$\psi_{\perp} := \psi - \psi_{||}$$

also,

$$(\psi_{\perp}, e_j) = 0 \quad \forall j \in I$$

2. *Pythagoras Theorem:*

$$\|\psi\|^2 = \|\psi_{\perp}\|^2 + \|\psi_{||}\|^2$$

3. For any $\gamma \in \text{span}(\{e_i\}_{i \in I})$,

$$\|\psi - \gamma\| \geq \|\psi_{\perp}\|$$

Remark. The above is a proposition not a definition. This is because we need to check for the existence of limits before even defining the projection and orthogonal complement.

Proof. First assume I is finite.

1. The convergence in this definition is not an issue. Now,

$$\begin{aligned} (\psi_\perp, e_j) &= \left(\psi - \sum_{i=1}^n (e_i, \psi) e_i, e_j \right) \\ &= (\psi, e_j) - \sum_{i=1}^n (e_i, \psi)^* (e_i, e_j) \\ &= 0 \end{aligned}$$

2. Using (1), we can say, $(\psi_\perp, \psi_\parallel) = 0$. Hence,

$$\begin{aligned} \|\psi\|^2 &= (\psi_\perp + \psi_\parallel, \psi_\perp + \psi_\parallel) \\ &= \|\psi_\perp\|^2 + \|\psi_\parallel\|^2 \end{aligned}$$

3. For some $\gamma \in \text{span}(\{e_i\}_{i \in I})$,

$$\begin{aligned} \|\psi - \gamma\|^2 &= \left\| \psi_\perp + \psi_\parallel - \sum_{i=1}^n \gamma_i e_i \right\|^2 \\ &= \left\| \psi_\perp + \sum_{i=1}^n ((e_i, \psi) - \gamma_i) e_i \right\|^2 \\ &= \|\psi_\perp\|^2 + \sum_{i=1}^n |(e_i, \psi) - \gamma_i|^2 \\ &\geq \|\psi_\perp\|^2 \end{aligned}$$

To complete the proof, we just need to show that all limits exists. i.e,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (e_i, \psi) e_i \quad \text{exists}$$

Since the above is a sequence of partial sums in $\mathcal{H}$, we just need to show that it is

Cauchy. For $n \geq m$,

$$\begin{aligned} \left\| \sum_{i=1}^n (e_i, \psi) e_i - \sum_{i=1}^m (e_i, \psi) e_i \right\|^2 &= \left\| \sum_{i=m}^n (e_i, \psi) e_i \right\|^2 \\ &= \sum_{i=m}^n |(e_i, \psi)|^2 \\ &\leq \|\psi\|^2 < \infty \end{aligned}$$

For a fixed m , the above sequence (of n) is monotonically increasing and bounded from above. Hence, it converges. Hence the above sequence for fixed m is Cauchy. Which means,

$$\forall \epsilon > 0, \exists N \in \mathbb{N} : n, n' \geq N \Rightarrow \left| \sum_{i=m}^n |(e_i, \psi)|^2 - \sum_{i=m}^{n'} |(e_i, \psi)|^2 \right| = \left| \sum_{i=n}^{n'} |(e_i, \psi)|^2 \right| < \epsilon$$

Which gives,

$$\left\| \sum_{i=1}^{n'} (e_i, \psi) e_i - \sum_{i=1}^n (e_i, \psi) e_i \right\|^2 < \epsilon$$

Hence, our original sequence we wanted to prove is convergent is Cauchy, hence convergent! ■

Definition 1.3.2. A set $M \subseteq \mathcal{H}$ is called a *Linear Subspace* of $\mathcal{H}$ if

$$\forall \psi, \phi \in M, \forall a \in \mathbb{C} : a\psi + \phi \in M$$

Remark. Not all linear subspaces are sub Hilbert spaces, since it may not be complete.

Lemma 1.3.1. A closed subset $M \subseteq \mathcal{H}$ is Complete.

Proof. Consider a Cauchy sequence, $\{\psi_n\}_{n \in \mathbb{N}} \subseteq M$. We need to show that this sequence converge in M . Since this sequence is in $\mathcal{H}$, it converges in $\mathcal{H}$. Now assume that

$$\lim_{n \rightarrow \infty} \psi_n \notin M$$

Then,

$$\lim_{n \rightarrow \infty} \psi_n \in \mathcal{H} \setminus M$$

Since M is closed, $\mathcal{H} \setminus M$ is open. Hence,

$$\exists u \in \mathcal{H} \setminus M : \lim_{n \rightarrow \infty} \psi_n \in u$$

Then,

$$\exists N \in \mathbb{N} : \forall n \geq N : \psi_n \in u$$

But $\forall n \in \mathbb{N} : \psi_n \in M$, which is a contradiction. Hence,

$$\lim_{n \rightarrow \infty} \psi_n \in M$$

■

Corollary 1.3.1.1. *If $M \subseteq \mathcal{H}$ is closed and linear, then it is a sub Hilbert space of $\mathcal{H}$*

Remark. A singleton set is a closed subset of $\mathcal{H}$. But it is not a sub Hilbert space, since it is not linear. The only possible singleton set which can be a sub Hilbert space is the set with the zero only.

Orthogonal Projections

Definition 1.3.3. Let $M \subseteq \mathcal{H}$ be a subset of $\mathcal{H}$ (we are not demanding closure). Then,

$$M^\perp := \left\{ \psi \in \mathcal{H} \mid \forall \mu \in M : (\mu, \psi) = 0 \right\}$$

Proposition 1.3.2. $M^\perp$ is a sub Hilbert space

Proof. Since the inner product is linear in its second entry, $M^\perp$ is a linear subspace already. We have to show closure. For that, define a map,

$$\omega_\mu : \mathcal{H} \rightarrow \mathbb{C}$$

as,

$$\omega_\mu(\psi) := (\mu, \psi)$$

Then we write,

$$M^\perp = \bigcap_{\mu \in M} \text{preim}_{\omega_\mu}(\{0\})$$

We need to show that $\text{preim}_{\omega_\mu}(\{0\})$ is closed for some arbitrary $\mu \in M$, then as arbitrary intersection of closed sets is closed, $M^\perp$ will automatically be closed. For that,

$$\begin{aligned}\mathcal{H} \setminus \text{preim}_{\omega_\mu}(\{0\}) &= \left\{ \psi \in \mathcal{H} \mid (\mu, \psi) \neq 0 \right\} \\ &= \left\{ \psi \in \mathcal{H} \mid (\mu, \psi) \in \mathbb{C} \setminus \{0\} \right\} \\ &= \text{preim}_{\omega_\mu}(\mathbb{C} \setminus \{0\})\end{aligned}$$

Since $\mathbb{C} \setminus \{0\}$ is open in the standard topology of $\mathbb{C}$, we just need to show ω_μ is continuous. For that,

$$\begin{aligned}|\omega_\mu(\psi) - \omega_\mu(\psi_0)| &= |(\mu, \psi) - (\mu, \psi_0)| \\ &= |(\mu, \psi - \psi_0)| \\ &\leq \|\mu\| \cdot \|\psi - \psi_0\|\end{aligned}$$

Choose,

$$\delta = \frac{\epsilon}{\|\mu\|}$$

Then,

$$\forall \epsilon > 0, \exists \delta > 0 : \|\psi - \psi_0\| < \delta \Rightarrow |\omega_\mu(\psi) - \omega_\mu(\psi_0)| < \epsilon$$

Hence, ω_μ is continuous. Hence $\text{preim}_{\omega_\mu}(\{0\})$ is closed. This gives $M^\perp$ is closed and hence complete. ■

Remark. $M^{\perp\perp} \neq M$ in general, for if M is just a subset, even then $M^{\perp\perp}$ is a sub Hilbert space.

Proposition 1.3.3. *If $M \subseteq \mathcal{H}$ is a closed, linear subspace of $\mathcal{H}$, then*

$$\mathcal{H} = M \oplus M^\perp$$

Proof. Take any $x \in M$ and $y \in M^\perp$, then trivially, $x + y \in \mathcal{H}$. Hence,

$$M \oplus M^\perp \subseteq \mathcal{H}$$

Now take any $\psi \in \mathcal{H}$. As M is closed and linear, it is a sub Hilbert space. Choose a Schauder basis for M , say $\{m_1, m_2, \dots\}$. Then define,

$$x := \sum_{i=1}^{\infty} (m_i, \psi) m_i$$

The limits are guaranteed to exist in M as we are on a sub Hilbert space. Also define,

$$y := \psi - x$$

take any $z \in M$,

$$\begin{aligned} (z, y) &= (z, \psi - x) \\ &= \left(\sum_{i=1}^{\infty} z_i m_i, \psi \right) - \left(\sum_{i=1}^{\infty} z_i m_i, \sum_{j=1}^{\infty} (m_j, \psi) m_j \right) \\ &= \sum_{i=1}^{\infty} z_i^* (m_i, \psi) - \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} z_i^* (m_j, \psi) \delta_{ij} \\ &= 0 \end{aligned}$$

Hence, $y \in M^{\perp}$ and trivially, $\psi = x + y$. Hence,

$$\mathcal{H} \subseteq M \oplus M^{\perp}$$

Therefore,

$$\mathcal{H} = M \oplus M^{\perp}$$

■

Proposition 1.3.4. *If $M \subseteq \mathcal{H}$ is a closed, linear subspace, then,*

$$M^{\perp\perp} = M$$

Proof. Take any $\psi \in M$. Then

$$\forall \alpha \in M^{\perp} : (\psi, \alpha) = 0$$

Which gives, $\psi \in M^{\perp\perp}$. Hence,

$$M \subseteq M^{\perp\perp}$$

Now take any $\phi \in M^{\perp\perp}$. Since M is closed and linear,

$$\mathcal{H} = M \oplus M^{\perp}$$

Hence,

$$\phi = x + y$$

For some $x \in M$ and $y \in M^\perp$. Therefore, $(\phi, y) = 0$.

$$0 = (\phi, y) = (x + y, y) = \|y\|^2$$

This means $y = 0$, or $\phi = x \in M$. Hence,

$$M^{\perp\perp} \subseteq M$$

Combining the results, we have the proof. ■

Proposition 1.3.5. *Let $M \subseteq \mathcal{H}$ be a linear subspace. Then,*

$$M^{\perp\perp} = \overline{M}$$

Proof. Using the same argument as in the last proof, we get

$$M \subseteq M^{\perp\perp}$$

(This part of the proof didn't use the closure property). Notice,

$$M_1 \subseteq M_2 \Rightarrow M_2^\perp \subseteq M_1^\perp$$

This can be proven quickly. Take $\phi \in M_2^\perp$. Then,

$$\begin{aligned} & \forall m_2 \in M_2 : (m_2, \phi) = 0 \\ \Rightarrow & \forall m_1 \in M_1 \subseteq M_2 : (m_1, \phi) = 0 \\ \Rightarrow & \phi \in M_1^\perp \end{aligned}$$

Now,

$$M \subseteq \overline{M} \Rightarrow \overline{M}^\perp \subseteq M^\perp \Rightarrow M^{\perp\perp} \subseteq \overline{M}^{\perp\perp} = \overline{M}$$

The last step follows from the last proposition. Also,

$$M \subseteq M^{\perp\perp} \Rightarrow \overline{M} \subseteq \overline{M^{\perp\perp}} = M^{\perp\perp}$$

The last line follows since $M^{\perp\perp}$ is closed. Combining the results, we have the proof. ■

Definition 1.3.4. Let $M \subseteq \mathcal{H}$ be a sub Hilbert space. Define,

$$P_M : \mathcal{H} \rightarrow M$$

as,

$$P_M \psi \equiv \psi_{||} := \sum_{i \in I} (e_i, \psi) e_i$$

(Where $\{e_i\}$ is a basis of M) is called the *orthogonal projection* to M .

Theorem 1.3.2. *The orthogonal projection satisfy the following properties,*

1. $P_M \circ P_M = P_M$
2. $\forall \psi, \phi \in \mathcal{H} : (\psi, P_M \psi) = (P_M \psi, \psi)$
3. $P_{M^\perp} \psi = \psi_\perp$
4. $P_M \in \mathcal{L}(\mathcal{H}, M)$

Proof. .

1. $P_M(P_M \psi) = \sum_{i \in I} (e_i, \psi) P_M e_i = \sum_{i \in I} (e_i, \psi) e_i = P_M \psi$
2. $(\psi, P_M \psi) = \sum_{i \in I} (e_i, \psi) (\psi, e_i) = \left(\sum_{i \in I} (\psi, e_i) e_i, \psi \right) = (P_M \psi, \psi)$
3. Since, $\mathcal{H} = M \oplus M^\perp$, $P_M \psi + P_{M^\perp} \psi = \psi$. Which gives, $P_{M^\perp} \psi = \psi - P_M \psi =: \psi_\perp$
4. $\|P_M\| = \sup_{\psi \in \mathcal{H}} \frac{\|P_M \psi\|}{\|\psi\|} < \infty$

■

The converse of the above theorem is heavily used.

Theorem 1.3.3. *Let $P \in \mathcal{L}(\mathcal{H}, \mathcal{H})$ such that,*

1. $P \circ P = P$
2. $\forall \psi, \phi \in \mathcal{H} : (P \psi, \phi) = (\psi, P \phi)$

Then, $P(\mathcal{H})$ is closed and $P = P_{P(\mathcal{H})}$

Proof.

$$P(\mathcal{H}) = \left\{ P \psi \in \mathcal{H} \mid \forall \psi \in \mathcal{H} \right\}$$

For any $\psi \in \mathcal{H}$, we can write,

$$\psi = P \psi + (\psi - P \psi)$$

Now, $\psi - P\psi \in P(\mathcal{H})^\perp$. To show this, take any $\phi \in P(\mathcal{H})$

$$\begin{aligned} (\phi, \psi - P\psi) &= (\phi, \psi) - (\phi, P\psi) \\ &= (\phi, \psi) - (P\phi, \psi) \\ &= (\phi, \psi) - (\phi, \psi) \\ &= 0 \end{aligned}$$

This then gives,

$$\mathcal{H} = P(\mathcal{H}) \oplus P(\mathcal{H})^\perp$$

Which gives us that $P(\mathcal{H})$ is closed. ■

Bra, Ket and the Riesz Representation Theorem

For any $\psi \in \mathcal{H}$, define,

$$\ell_\psi : \mathcal{H} \rightarrow \mathbb{C}$$

as,

$$\ell_\psi(\phi) := (\psi, \phi)$$

Now,

$$\|\ell_\psi\| = \sup_{\phi \in \mathcal{H}} \frac{|(\psi, \phi)|}{\|\phi\|} \leq \sup_{\phi \in \mathcal{H}} \frac{\|\phi\| \|\psi\|}{\|\phi\|} = \|\psi\| < \infty$$

Hence, $\ell_\psi \in \mathcal{L}(\mathcal{H}, \mathbb{C}) \equiv \mathcal{H}^*$. Hence, we have,

$$\forall \psi \in \mathcal{H}, \exists! \ell_\psi \in \mathcal{H}^*$$

In the quantum mechanics jargon, we say, for every ket there is a unique bra. It turns out that the converse is also true. That is where Riesz Representation Theorem comes in.

Theorem 1.3.4. *Riesz Representation Theorem:*

$$\forall \ell \in \mathcal{H}^*, \exists! \lambda \in \mathcal{H} : \ell = \ell_\lambda$$

Proof. If $\ell = 0$, then $\lambda = 0$ is the unique choice, which sends every vector in $\mathcal{H}$ to zero.

If $\ell \neq 0$, then consider,

$$\ker(\ell) = \left\{ \phi \in \mathcal{H} \mid \ell(\phi) = 0 \right\} \equiv \text{preim}_\ell(\{0\})$$

Now, $\ker(\ell)$ is closed. To show this, consider,

$$\begin{aligned}\mathcal{H} \setminus \ker(\ell) &= \left\{ \phi \in \mathcal{H} \mid \ell(\phi) \neq 0 \right\} \\ &= \left\{ \phi \in \mathcal{H} \mid \ell(\phi) \in \mathbb{C} \setminus \{0\} \right\} \\ &= \text{preim}_\ell(\mathbb{C} \setminus \{0\})\end{aligned}$$

Now since $\mathbb{C} \setminus \{0\}$ is open in the standard topology on $\mathbb{C}$, it suffices to show that ℓ is continuous. For that,

$$|\ell(\psi) - \ell(\psi_0)| = |\ell(\psi - \psi_0)| \leq \|\ell\| \cdot \|\psi - \psi_0\|$$

The last step follows since ℓ is bounded. Now by choosing $\|\psi - \psi_0\| < \delta = \frac{\epsilon}{\|\ell\|}$, we get the continuity. The upshot is that $\ker(\ell)$ is closed, and hence,

$$\mathcal{H} = \ker(\ell) \oplus \ker(\ell)^\perp$$

Now since $\ell \neq 0$, $\ker(\ell) \neq \mathcal{H}$, hence $\ker(\ell)^\perp \neq \{0\}$. So, without loss of any generality, choose,

$$\xi \in \ker(\ell)^\perp \setminus \{0\} : \|\xi\| = 1$$

Take,

$$\lambda := (\ell(\xi))^* \xi \in \ker(\ell)^\perp$$

Claim : $\ell = \ell_\lambda$

To show this, take some $\psi \in \mathcal{H}$,

$$\begin{aligned}\ell_\lambda(\psi) - \ell(\psi) &= ((\ell(\xi))^* \xi, \psi) - \ell(\psi) (\xi, \xi) \\ &= (\xi, \ell(\xi)\psi) - (\xi, \ell(\psi)\xi) \\ &= (\xi, \ell(\xi)\psi - \ell(\psi)\xi)\end{aligned}$$

For the left hand side to be zero, we just need $\ell(\xi)\psi - \ell(\psi)\xi \in \ker(\ell)$ since $\xi \in \ker(\ell)^\perp$. This is trivially so because,

$$\ell(\ell(\xi)\psi - \ell(\psi)\xi) = 0$$

Hence, $\ell_\lambda = \ell$. Now to show the uniqueness, assume,

$$\ell_{\lambda_1} = \ell = \ell_{\lambda_2}$$

Then,

$$0 = \ell_{\lambda_1}(\psi) - \ell_{\lambda_2}(\psi) = (\lambda_1 - \lambda_2, \psi)$$

Which shows,

$$\lambda_1 = \lambda_2$$

Hence the proof. ■

Remark. Consider $\mathcal{H} = L^2(\mathbb{R})$, the space of square integrable functions. Define, ℓ as,

$$\ell(\psi(x)) := \psi(0)$$

ℓ just maps the wave function to its value at the origin. The sesquilinear inner product we have is,

$$(\psi, \phi) := \int_{\mathbb{R}} dx \psi^*(x) \phi(x)$$

Is there a corresponding ket for the “bra” we defined? i.e, a function that maps any wave function to its value at the origin when taken an inner product with it? The delta function certainly does the job as,

$$\int_{\mathbb{R}} dx (\delta(x))^* \phi(x) = \phi(0)$$

but hold on! The delta function is not square integrable and hence not a member of $L^2(\mathbb{R})$. Where did Riesz Representation Theorem failed? The answer is it didn't! We simply haven't managed to define an ℓ which meets the conditions of Riesz Representation Theorem. Here, the ℓ we defined is linear. But,

$$\|\ell\| = \sup_{\psi \in \mathcal{H}} \frac{|\ell(\psi)|}{\|\psi\|} = \sup_{\psi \in \mathcal{H}} \frac{|\psi(0)|}{\|\psi\|}$$

As $\psi(0)$ can be arbitrarily large, ℓ is not bounded, and hence $\ell \notin \mathcal{L}(L^2(\mathbb{R}), \mathbb{C})$. Hence we cannot use Riesz Representation Theorem here!

Remark. In the proof of Riesz lemma, the way we define λ tells us that λ cannot be unique unless $\ker(\ell)^\perp$ is one dimensional no matter what the dimension of $\mathcal{H}$ is. In finite dimensions,

$$\psi = \sum_{i=1}^d \psi^i e_i$$

Then

$$\ell(\psi) = 0 \quad \Rightarrow \quad \sum_{i=1}^d \psi^i \ell(e_i) = 0$$

Which tells us that $\ker(\ell)$ is $d - 1$ dimensional, and hence $\ker(\ell)^\perp$ is 1 dimensional.

We need to extend this proof to ∞ dimensional case also.

Chapter 2

Basics of Measure Theory

The whole idea of measure theory is to assign a notion of “size” to sets. The “size” or measure, is expected to have certain properties, like the measure of the union of two sets is the sum of individual measures and so on. But it turns out that the notion of measure cannot be consistently given to all subsets of any given sets. Only a certain class of sets, called “measurable sets” can have a notion of “volume”. We discuss this in detail in this chapter.

Definition 2.0.1. Let M be any set. Then a collection of subsets of M , $\mathcal{A} \subseteq 2^M$ is called an *Algebra* if

1. $M \in \mathcal{A}$
2. If $U, V \in \mathcal{A}$, then $U \cup V \in \mathcal{A}$
3. If $U \in \mathcal{A}$, then $M \setminus U \in \mathcal{A}$

Remark. It is immediately clear that $\emptyset \in \mathcal{A}$. Also, by de Morgan’s rule, $\mathcal{A}$ is closed under finite intersections.

Definition 2.0.2. If an algebra is closed under countable unions, then it is called a σ -algebra.

Remark. A set with a σ -algebra is called a measurable space.

Example 2.0.2.1. Let $M = \{1, 2, 3\}$. Then $\mathcal{A} = \{\emptyset, M, \{1\}, \{2, 3\}\}$ is an algebra.

Theorem 2.0.1. *The intersection of a family of σ -algebras, $\{\mathcal{A}_\alpha\}$ on a set M is also a σ -algebra*

Proof. Let $\mathcal{A} = \bigcap_{\alpha} \mathcal{A}_{\alpha}$. As $M \in \mathcal{A}_{\alpha}$ for all α , $M \in \mathcal{A}$. Now consider $\Sigma \subseteq \mathcal{A}$. Let $\beta = \bigcup_{u \in \Sigma} u$. By definition of $\mathcal{A}$, we have $\Sigma \subseteq \mathcal{A}_{\alpha}$. Which gives $\beta \in \mathcal{A}_{\alpha} \Rightarrow \beta \in \mathcal{A}$. Take any $u \in \mathcal{A}$. Hence $u \in \mathcal{A}_{\alpha} \Rightarrow M \setminus u \in \mathcal{A}_{\alpha} \Rightarrow M \setminus u \in \mathcal{A}$ ■

Definition 2.0.3. The smallest σ -algebra $\sigma(S)$ for a set M that contains a collection of subsets $S \subseteq 2^M$ is defined as,

$$\sigma(S) := \text{intersection of all } \sigma\text{-algebras, } \sigma \text{ of } M \text{ with } S \in \sigma$$

Remark. By the last theorem, $\sigma(S)$ is indeed a σ -algebra. It is the smallest such by definition.

This gives us a natural way to associate a topology and σ -algebra.

Definition 2.0.4. Let $(M, \mathcal{T})$ be a topological space. Then $\sigma(\mathcal{T})$ is called the *Borel σ -algebra* on M

Example 2.0.4.1. Take $M = \mathbb{R}$, and $\mathcal{T}$ as the standard topology. Since $(a, b) \in \mathcal{T}$, we have $(a, b) \in \sigma(\mathcal{T})$. Also,

$$\bigcap_{n \geq 1} \left(a - \frac{1}{n}, b \right) = [a, b) \in \sigma(\mathcal{T})$$

Similarly, $[a, b] \in \sigma(\mathcal{T})$

Definition 2.0.5. A measure, $\mu : \sigma \rightarrow \overline{\mathbb{R}}_0^+$ on a measurable space (M, σ) is a map satisfying,

1. $\mu(\emptyset) = 0$
2. If $A_1, A_2, \dots \in \sigma$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$\mu\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} \mu(A_n)$$

Remark. (M, σ, μ) is called a measure space.

Theorem 2.0.2. Measure satisfies the following properties,

1. *Monotony*

$$A_1 \subseteq A_2 \in \sigma \Rightarrow \mu(A_1) \leq \mu(A_2)$$

2. *Sub-additivity*

$$\{A_1, A_2, \dots\} \subseteq \sigma \Rightarrow \mu\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n \geq 1} \mu(A_n)$$

3. *Continuity from below*

$$A_1 \subseteq A_2 \subseteq \dots \quad \text{with} \quad \bigcup_{n \geq 1} A_n = A \Rightarrow \lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$$

4. *Continuity from above*

$$A_1 \supseteq A_2 \supseteq \dots \quad \text{with} \quad \bigcap_{n \geq 1} A_n = A \quad \text{and} \quad \mu(A_1) < \infty \Rightarrow \lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$$

Proof. 1. $A_2 = A_1 \cup A_2 \setminus A_1 \Rightarrow \mu(A_2) = \mu(A_1) + \mu(A_2 \setminus A_1) \Rightarrow \mu(A_2) \geq \mu(A_1)$

2. Proof is immediate if we use property 1 and axiom 2 of the measure.

3. Define $A_n := A_n \setminus A_{n-1}$. Then

$$\bigcup_{k=1}^n \tilde{A}_k = A_n \quad \text{and} \quad \tilde{A}_i \cap \tilde{A}_j = \emptyset \quad \forall i \neq j$$

Hence,

$$\mu\left(\bigcup_{k=1}^n \tilde{A}_k\right) = \mu(A_n) \Rightarrow \lim_{n \rightarrow \infty} \mu\left(\bigcup_{k=1}^n \tilde{A}_k\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

But,

$$\begin{aligned} \lim_{n \rightarrow \infty} \mu\left(\bigcup_{k=1}^n \tilde{A}_k\right) &= \lim_{n \rightarrow \infty} \mu(\tilde{A}_1) + \dots + \mu(\tilde{A}_n) \\ &= \mu(\tilde{A}_1) + \dots \\ &= \mu\left(\bigcup_{k=1}^{\infty} \tilde{A}_k\right) \\ &= \mu(A) \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$$

4. Define $\tilde{A}_n := A_1 \setminus A_n$. Then we have,

$$\tilde{A}_1 \subseteq \tilde{A}_2 \subseteq \dots$$

Also,

$$\bigcup_{n \geq 1} \tilde{A}_n = A_1 \setminus A$$

Hence,

$$\lim_{n \rightarrow \infty} \mu(\tilde{A}_n) = \mu(A_1 \setminus A)$$

Which gives,

$$\lim_{n \rightarrow \infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \mu(A)$$

i.e.,

$$\lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$$

■

Definition 2.0.6. Let (M, σ, μ) be a measure space and P some property. Then we say P holds “almost every where” on M if

$$\exists A \in \sigma : \mu(A) = 0 \quad \text{and} \quad \forall a \in M \setminus A : P(a)$$

Definition 2.0.7. Let (M, σ, μ) be a measure space, where $(M, +, \cdot)$ is a vector space. Then the measure μ is called translational invariant if,

$$\forall m \in M, \forall A \in \sigma, A + m \in \sigma \quad \text{and} \quad \mu(A + m) = \mu(A)$$

Definition 2.0.8. Lebesgue Measure: Let $M = \mathbb{R}^d$ with $\sigma = \sigma(\mathcal{T}_{\mathbb{R}^d})$. Then the measure defined by,

$$\lambda^d([a_1, b_1) \times [a_2, b_2) \times \dots \times [a_d, b_d)) := \prod_{k=1}^d (b_k - a_k)$$

is called the Lebesgue measure.

Remark. We chose the semi open intervals to define the Lebesgue measure because they generate the σ -algebra and they can be used to cover the entire real line. Also, it is trivial to see that the measure is translational invariant.

Definition 2.0.9. Measurable Maps: Let (M, σ_M) and (N, σ_N) be two measurable spaces. Then $f : M \rightarrow N$ is called measurable if,

$$\forall A \in \sigma_N : \text{preim}_f(A) \in \sigma_M$$

Lemma 2.0.3. f is measurable iff $\forall A \in \varepsilon : \text{preim}_f(A) \in \sigma_M$ where $\sigma(\varepsilon) = \sigma_N$

Corollary 2.0.3.1. Any continuous map is measurable with respect to the Borel σ -algebra.

Corollary 2.0.3.2. Any monotonous map is measurable with respect to the Borel σ -algebra.

2.1 The Lebesgue Integral[1, 2, 3]

Definition 2.1.1. Characteristic Function: Let M be any set. Then for any $A \subseteq M$, define $\chi_A : M \rightarrow \mathbb{R}$ as,

$$\chi_A(x) := \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases}$$

Definition 2.1.2. Simple Function: Let M be any set. Then $s : M \rightarrow \mathbb{R}$ is called simple if,

$$s(M) = \{s_1, s_2, \dots, s_N\} \quad \text{for some } N \in \mathbb{N}$$

or if $\exists s_1, \dots, s_N \in \mathbb{R}$ and $\exists A_1, \dots, A_N \subseteq M$ such that

$$s = \sum_{i=1}^N s_i \chi_{A_i}$$

We define the integral for a non negative simple function first.

Definition 2.1.3. Let (M, σ, μ) be a measure space and $s : M \rightarrow \mathbb{R}_0^+$ be a non negative measurable simple function. Then For any $A \in \sigma$,

$$\int_A s d\mu := \sum_{i=1}^N s_i \mu(A_i \cap A)$$

Where

$$s = \sum_{i=1}^N s_i \chi_{A_i}$$

Remark. The convention we use above is $0 \cdot \infty = 0$

Corollary 2.1.0.1. $\int_A s \, d\mu = \int_M s \cdot \chi_A \, d\mu$

Proposition 2.1.1.

$$\int_{\bigcup_{j=1}^{\infty} B_j} s \, d\mu = \sum_{j=1}^{\infty} \int_{B_j} s \, d\mu$$

Proof.

$$\begin{aligned} \int_{\bigcup_{j=1}^{\infty} B_j} s \, d\mu &= \sum_{i=1}^n s_i \mu \left(A_i \bigcup \left(\bigcup_{j=1}^{\infty} B_j \right) \right) \\ &= \sum_{i=1}^n s_i \mu \left(\bigcup_{j=1}^{\infty} (A_i \bigcup B_j) \right) \\ &= \sum_{i=1}^n \sum_{j=1}^{\infty} s_i \mu (A_i \bigcup B_j) \\ &= \sum_{j=1}^{\infty} \int_{B_j} s \, d\mu \end{aligned}$$

■

Proposition 2.1.2.

$$\int_A \alpha s \, d\mu = \alpha \int_A s \, d\mu$$

Proof. It is trivial from the definition. ■

Proposition 2.1.3. For two non negative measurable simple functions s and t ,

$$\int_A (s + t) \, d\mu = \int_A s \, d\mu + \int_A t \, d\mu$$

Proof. Let

$$\begin{aligned} s &= \sum_{i=1}^n s_i \chi_{A_i} \\ t &= \sum_{i=1}^m t_i \chi_{B_i} \end{aligned}$$

Then

$$s + t = \sum_{i=1}^n \sum_{j=1}^m (s_i + t_j) \chi_{A_i \cap B_j}$$

Then,

$$\begin{aligned}
\int_A (s + t) d\mu &= \sum_{i,j} (s_i + t_j) \mu((A_i \cap B_j) \cap A) \\
&= \sum_{i,j} s_i \mu((A_i \cap B_j) \cap A) + \sum_{i,j} t_j \mu((A_i \cap B_j) \cap A) \\
&= \sum_i s_i \sum_j \mu((A_i \cap B_j) \cap A) + \sum_j t_j \sum_i \mu((A_i \cap B_j) \cap A) \\
&= \sum_i s_i \mu(\cup_j (A_i \cap B_j) \cap A) + \sum_i t_j \mu(\cup_i (A_i \cap B_j) \cap A) \\
&= \sum_i s_i \mu(A_i \cap A) + \sum_j t_j \mu(B_j \cap A) \\
&= \int_A s d\mu + \int_A t d\mu
\end{aligned}$$

■

Proposition 2.1.4. *For two non negative measurable simple functions s and t ,*

$$s \leq t \Rightarrow \int_A s d\mu \leq \int_A t d\mu$$

Proof. Write $t = s + (t - s)$. As both s and $t - s$ are non negative, use the previous proposition to get the result. ■

Proposition 2.1.5.

$$A \subseteq B \Rightarrow \int_A s d\mu \leq \int_B s d\mu$$

Proof. The monotonicity of the measure gives the proof immediately. ■

Definition 2.1.4. Let $f : M \rightarrow \mathbb{R}_0^+$ be a non negative measurable function. Then for any $A \in \sigma$,

$$\int_A f d\mu := \sup_{s \leq f} \int_A s d\mu$$

Theorem 2.1.1. Theorem on Monotone Convergence: *Let f_n be a monotone non decreasing sequence of non negative measurable functions, i.e., $0 \leq f_1 \leq f_2 \leq \dots$, with*

$$f := \sup_{n \geq 1} f_n$$

Then,

$$\lim_{n \rightarrow \infty} \int_A f_n d\mu = \int_A f d\mu$$

Proof. let

$$\int_A f_n d\mu = \alpha_n$$

Since $f_n \leq f$ for all $n \in \mathbb{N}$, and $\int_A f d\mu$ exists, we have $\lim \alpha_n := \alpha$ exists. Also, $\alpha \leq \int_A f d\mu$. Now take a simple function such that $0 \leq s \leq f$, and $\theta \in (0, 1)$. Define,

$$A_n := \left\{ x \in A \mid f_n(x) \geq \theta s(x) \right\}$$

Then

$$\lim_{n \rightarrow \infty} A_n = \left\{ x \in A \mid f(x) \geq \theta s(x) \right\} =: A$$

Hence,

$$\int_A f_n d\mu \geq \int_{A_n} f_n d\mu \geq \theta \int_{A_n} s d\mu$$

Taking the limit will give,

$$\alpha \geq \theta \int_A s d\mu$$

This will work for $\theta = 1$. As s is arbitrary, we get,

$$\alpha \geq \int_A f d\mu$$

Hence we have,

$$\alpha := \lim_{n \rightarrow \infty} \int_A f_n d\mu = \int_A f d\mu$$

■

Theorem 2.1.2. *Fatou's Lemma:* Let f_n be a sequence of non negative measurable functions. Then,

$$\int_A \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_A f_n d\mu$$

Proof. Let $g_n := \inf_{k \geq n} f_k$. Then $g_n \leq f_n$. This gives,

$$\int_A g_n d\mu \leq \int_A f_n d\mu$$

but,

$$\liminf_{n \rightarrow \infty} \int_A g_n d\mu = \lim_{n \rightarrow \infty} \int_A g_n d\mu = \int_A \lim_{n \rightarrow \infty} g_n d\mu = \int_A \liminf_{n \rightarrow \infty} f_n d\mu$$

Hence the proof. ■

Remark. It can be shown that every non negative function can be written as a limit of a sequence of non negative simple functions. Hence the propositions we had for the non negative simple functions will continue to hold for non negative functions.

Theorem 2.1.3. *Let $f : M \rightarrow \mathbb{R}_0^+$ be non negative measurable. Then,*

$$\int_M f d\mu = 0 \iff f =_{a.e.} 0$$

Proof. For the only if part, Assume $\int_M f d\mu = 0$. Define,

$$A_n := \left\{ m \in M \mid f(m) > \frac{1}{n+1} \right\}$$

and

$$s_n := \frac{1}{n+1} \chi_{A_n}$$

By definition, $s_n \leq f$ for all n . This gives,

$$0 \leq \int_M s_n d\mu \leq \int_M f d\mu = 0$$

This gives $\int_M s_n d\mu = 0$ for all n . But since $\int_M s_n d\mu = \frac{1}{n+1} \mu(A_n)$, we get $\mu(A_n) = 0$ for all n . Now,

$$\bigcup_{n \geq 1} A_n = A := \left\{ m \in M \mid f(m) \geq 0 \right\} \quad \text{with} \quad A_n \subseteq A_{n+1}$$

And using the properties of the measure,

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n) = 0$$

Since on A , f takes non zero values, we have $f =_{a.e.} 0$

Now for the if part, assume $f =_{a.e.} 0$. Then for any simple function $s \leq f$, $s =_{a.e.} 0$.

As $s = \sum_i s_i \chi_{A_i}$, we have either $s_i = 0$ or $\mu(A_i) = 0$. Hence the integral of s is zero.

Hence,

$$\int_M f d\mu := \sup_{s \leq f} \int_M s d\mu = 0$$

■

Corollary 2.1.3.1. *For non negative f and g , measurable,*

$$f =_{a.e.} g \iff \int_M f d\mu = \int_M g d\mu$$

Theorem 2.1.4. *For non negative f, g measurable,*

$$f \leq_{a.e.} g \iff \int_M f d\mu \leq \int_M g d\mu$$

Proof.

$$B := \{m \in M : f(m) > g(m)\}$$

Hence, $\mu(B) = 0$ and $f \leq g$ on $M \setminus B$. Then,

$$\begin{aligned} \int_M f d\mu &= \int_B f d\mu + \int_{M \setminus B} f d\mu \\ &\leq \int_{M \setminus B} f d\mu \\ &\leq \int_{M \setminus B} g d\mu \\ &= \int_B g d\mu + \int_{M \setminus B} g d\mu \\ &= \int_M g d\mu \end{aligned}$$

■

Definition 2.1.5. Lebesgue Integral for any Measurable function: Let $f : M \rightarrow \mathbb{R}$ be any measurable function. Then,

$$\int_A f d\mu := \int_A f^+ d\mu - \int_A f^- d\mu$$

Where $f^+ := \max(0, f)$ and $f^- := \max(-f, 0)$

Remark. We say that a function f is integrable if the above integral exists. i.e, $\int_A |f| d\mu < \infty$

Definition 2.1.6. Let $f : M \rightarrow \mathbb{C}$ be measurable. Then

$$\int_A f d\mu := \int_A \operatorname{Re}(f) d\mu + i \int_A \operatorname{Im}(f) d\mu$$

Remark. Since we have defined the integrals using non negative functions, all the properties remain intact.

Corollary 2.1.4.1. *f, g integrable, then,*

$$f =_{a.e.} g \Rightarrow \int_M f d\mu = \int_M g d\mu$$

Corollary 2.1.4.2. *f, g integrable, then,*

$$f \leq_{a.e.} g \Rightarrow \int_M f d\mu \leq \int_M g d\mu$$

Theorem 2.1.5. Theorem on Dominated Convergence: *Let $f_n : M \rightarrow \mathbb{R}$ be a sequence of integrable functions, and $f : M \rightarrow \mathbb{R}$ be some function such that f_n converge to f point wise. Let g be an integrable function such that $|f_n| \leq g$ for all $n \in \mathbb{N}$. Then, f is integrable and*

$$\int_A f d\mu = \lim_{n \rightarrow \infty} \int_A f_n d\mu$$

Proof. Since $|f_n| \leq g$, taking limits will give $|f| \leq g$. As g is integrable, f is also integrable.

Now $|f_n| \leq g$ gives $g + f_n \geq 0$. Using Fatou's Lemma, we get,

$$\int_A \liminf (f_n + g) d\mu \leq \liminf \int_A (f_n + g) d\mu$$

This gives,

$$\int_A (f + g) d\mu \leq \liminf \int_A f_n d\mu + \int_A g d\mu$$

Hence,

$$\int_A f d\mu \leq \liminf \int_A f_n d\mu$$

change f to $-f$. Then we will have,

$$\int_A (-f) d\mu \leq \liminf \int_A (-f_n) d\mu$$

Which is equivalent to,

$$\int_A f d\mu \geq \limsup \int_A f_n d\mu$$

Since

$$\limsup f_n = f = \liminf f_n$$

We have,

$$\int_A f d\mu = \lim_{n \rightarrow \infty} \int_A f_n d\mu$$

■

2.2 The Function Space, $L^p(M, \sigma, \mu)[1]$

The space of “square integrable functions” play a very important role in quantum mechanics. We study that here in detail. Let (M, σ, μ) be any general measure space.

Definition 2.2.1. For any $1 \leq p < \infty$,

$$\mathcal{L}^p(M, \sigma, \mu) := \left\{ f : M \rightarrow \mathbb{C} \mid f \text{ is measurable and } \int_M |f|^p d\mu < \infty \right\}$$

We now equip $\mathcal{L}^p(M, \sigma, \mu)$ with pointwise addition and s-multiplication.

Proposition 2.2.1. $\mathcal{L}^p(M, \sigma, \mu)$ is then a complex vector space.

Proof. For any $z \in \mathbb{C}$ and $f \in \mathcal{L}^p(M, \sigma, \mu)$, $zf \in \mathcal{L}^p(M, \sigma, \mu)$ trivially. Now take $g \in \mathcal{L}^p(M, \sigma, \mu)$.

$$|f + g|^p \leq 2^{p-1}(|f|^p + |g|^p)$$

This gives $f + g \in \mathcal{L}^p(M, \sigma, \mu)$ ■

For the above proposition, we used an inequality.

Lemma 2.2.1. For $a, b \in \mathbb{C}$

$$|a + b|^p \leq 2^{p-1}(|a|^p + |b|^p)$$

Proof. The map $x \rightarrow x^p$ for $x \geq 0$ is convex, since the second derivative $p(p-1)x^{p-2} \geq 0$. Hence,

$$\left| \frac{a+b}{2} \right|^p \leq \left(\frac{|a|+|b|}{2} \right)^p \leq \frac{|a|^p + |b|^p}{2}$$
■

Lemma 2.2.2. Young’s Inequality: Let $a, b \in \mathbb{R}_0^+$ and $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Proof. Define,

$$f(x) := \frac{x^p}{p} + \frac{1}{q} - x \quad x \in [0, \infty)$$

Then,

$$f'(x) = x^{p-1} - 1 \quad \text{and} \quad f''(x) = (p-1)x^{p-2}$$

From this we get f is at a minimum for $x = 1$. And notice $f(1) = 0$. Hence, $\forall x \in [0, \infty)$,

$$f(x) \geq 0 \Rightarrow \frac{x^p}{p} + \frac{1}{q} - x \geq 0$$

Since x is non negative, $\exists a, b \in \mathbb{R}_0^+$ such that, $x = ab^{\frac{1}{1-p}}$. Then we get,

$$\frac{a^p b^{\frac{p}{1-p}}}{p} + \frac{1}{q} - ab^{\frac{1}{1-p}} \geq 0$$

Rearranging and by using $q = \frac{p}{p-1}$, we get

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

■

Remark. Young's Inequality becomes trivial if we take $p = 1$ and $q = \infty$. Hence, the inequality can be extended to these values also.

Theorem 2.2.3. Holder's Inequality: Let (M, σ, μ) be a measure space, $p, q \in [1, \infty]$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Let $f, g : M \rightarrow \mathbb{C}$ be measurable. Then,

$$\int_M |fg| d\mu \leq \left(\int_M |f|^p d\mu \right)^{1/p} \left(\int_M |g|^q d\mu \right)^{1/q}$$

or,

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

Proof. Whenever the integrals diverge, the inequality is trivial. Hence we take the integrals to converge. Now define,

$$A := \|f\|_p \quad \text{and} \quad B := \|g\|_q$$

Also,

$$a := \frac{|f|}{A} \quad \text{and} \quad b := \frac{|g|}{B}$$

Use Young's inequality for a and b ,

$$\frac{|fg|}{AB} \leq \frac{|f|^p}{pA} + \frac{|g|^q}{qB}$$

Integrate both sides,

$$\frac{1}{AB} \int_M |fg| d\mu \leq \frac{1}{pA} \int_M |f|^p d\mu + \frac{1}{qB} \int_M |g|^q d\mu$$

Using the definition of A and B , we have,

$$\frac{1}{AB} \int_M |fg| d\mu \leq \frac{1}{p} + \frac{1}{q} = 1$$

Hence,

$$\|fg\|_1 \leq AB = \|f\|_p \|g\|_q$$

■

Now, in the hopes of turning $\mathcal{L}^p(M, \sigma, \mu)$ into a Banach space, we define a norm on it.

Proposition 2.2.2. *The map $\|\cdot\|_p : \mathcal{L}^p(M, \sigma, \mu) \rightarrow \mathbb{R}$ defined as,*

$$\|f\|_p := \left(\int_M |f|^p d\mu \right)^{1/p}$$

Fails to be a norm.

Proof.

$$\|\lambda \cdot f\|_p = |\alpha| \|f\|_p \quad \text{holds}$$

Now for the triangle inequality (known as the Minkowski Inequality),

$$|f + g|^p = |f + g| \cdot |f + g|^{p-1} \leq |f + g|^{p-1} (|f| + |g|)$$

$$\begin{aligned} \|f + g\|_p^p &= \int_M |f + g|^p d\mu \\ &\leq \int_M |f + g|^{p-1} (|f| + |g|) d\mu \\ &= \int_M |f| \cdot |f + g|^{p-1} d\mu + \int_M |g| \cdot |f + g|^{p-1} d\mu \end{aligned}$$

Now use Holder's Inequality,

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \left(\int_M |f + g|^p d\mu \right)^{1-\frac{1}{p}} = (\|f\|_p + \|g\|_p) \frac{\|f + g\|_p^p}{\|f + g\|_p}$$

Rearrange to get the triangle inequality. Now, the map is positive definite by definition. However, $\|f\|_p = 0 \Rightarrow f =_{a.e.} 0$ only. Hence the map fails to be a norm. ■

To circumvent this issue, we define an equivalence class,

$$f \sim g \iff f =_{a.e.} g$$

Then define,

$$L^p(M, \sigma, \mu) := \mathcal{L}^p(M, \sigma, \mu) / \sim$$

Hence, the map $\|\cdot\|_p : L^p(M, \sigma, \mu) \rightarrow \mathbb{R}$

$$\|f\|_p := \left(\int_M |[f]|^p d\mu \right)^{1/p}$$

is well defined and qualifies as a norm since $f =_{a.e.} g \Rightarrow \|f\|_p = \|g\|_p$.

Now, we would like to turn the normed space $(L^p(M, \sigma, \mu), \|\cdot\|_p)$ into Banach. For that, we first prove the following Lemma.

Lemma 2.2.4. *Let $\{g_k \in L^p(M, \sigma, \mu)\}_{k \in \mathbb{N}}$ be a sequence such that, $\sum_{k=1}^{\infty} \|g_k\|_p < \infty$.*

Then, $\exists f \in L^p(M, \sigma, \mu) : \sum_{k=1}^{\infty} g_k = f$

Proof. Define, $h_n, h : M \rightarrow \overline{\mathbb{R}}_0^+$ as,

$$h_n := \sum_{k=1}^n |g_k| \quad \text{and} \quad h := \sum_{k=1}^{\infty} |g_k|$$

Then, $\{h_n\}_{n \in \mathbb{N}}$ is an increasing sequence which converges to h . Hence using monotone convergence theorem,

$$\int_M h^p d\mu = \lim_{n \rightarrow \infty} \int_M h_n^p d\mu$$

Using the Minkowski Inequality,

$$\|h_n\|_p \leq \sum_{k=1}^n \|g_k\|_p \leq \alpha := \sum_{k=1}^{\infty} \|g_k\|_p$$

Hence, $\lim_{n \rightarrow \infty} \int_M h_n^p d\mu$ exists, hence $\|h\|_p < \infty$, which gives us $h \in L^p(M, \sigma, \mu)$. Also,

$$\|h\|_p = \sum_{k=1}^{\infty} |g_k| \leq \alpha$$

As h_n is an increasing and bounded sequence, it converges. This means,

$$\sum_{k=1}^{\infty} g_k \text{ is absolutely convergent}$$

Hence, the sequence is convergent. ■

Theorem 2.2.5. Riesz-Fischer Theorem: $(L^p(M, \sigma, \mu), \|\cdot\|_p)$ is complete.

Proof. Let $\{f_k \in L^p(M, \sigma, \mu)\}_{k \in \mathbb{N}}$ be a Cauchy sequence. Choose a sub sequence such that,

$$\|f_{k_{j+1}} - f_{k_j}\|_p \leq \frac{1}{2^j}$$

Define,

$$g_j := f_{k_{j+1}} - f_{k_j}$$

Then

$$\sum_{j=1}^{\infty} \|g_j\|_p \leq \sum_{j=1}^{\infty} \frac{1}{2^j} < \infty$$

Hence, by last lemma we have $f_{k_1} + \sum_{j=1}^{\infty} g_j$ convergent in $L^p(M, \sigma, \mu)$. Let,

$$f_{k_1} + \sum_{j=1}^{\infty} g_j = f$$

$$\lim_{j \rightarrow \infty} f_{k_j} = \lim_{j \rightarrow \infty} \left(f_{k_1} + \sum_{j=1}^{j-1} g_j \right) = f_{k_1} + \sum_{j=1}^{\infty} g_j = f$$

Since f_k is Cauchy, and its sub sequence converge, the sequence converge to the same limit. Hence the result. ■

Now, in quantum mechanics, the $p = 2$ case is of atmost important. For $p = 2$, we can define an inner product.

Definition 2.2.2. Define the map $(\cdot, \cdot) : L^2(M, \sigma, \mu) \rightarrow \mathbb{C}$ as,

$$(f, g) := \int_M f^* g d\mu$$

Proposition 2.2.3. The above defined map is a sesquilinear inner product.

Corollary 2.2.5.1. $L^2(M, \sigma, \mu)$ equipped with the inner product is a Hilbert Space.

Chapter 3

Self - Adjoint Operators and Spectrum

3.1 Self-Adjoint Operators[1, 2]

Self Adjoint Operators play a central role in quantum mechanics. These are the operators we will associate with “physical observables” in the theory.

Definition 3.1.1. A linear map, $A : D(A) \rightarrow \mathcal{H}$ is said to be densely defined if $D(A)$ is dense in $\mathcal{H}$

Definition 3.1.2. Adjoint of a Map: Let $A : D(A) \rightarrow \mathcal{H}$ be densely defined. Then the *adjoint* of A , denoted by $A^* : D(A^*) \rightarrow \mathcal{H}$ is defined by,

$$D(A^*) := \left\{ \psi \in \mathcal{H} \mid \exists \eta \in \mathcal{H} : \forall \alpha \in D(A), (\psi, A\alpha) = (\eta, \alpha) \right\}$$

And,

$$A^*\psi := \eta$$

Proposition 3.1.1. A^* is well defined

Proof. Let $\psi \in \mathcal{H}$ and $\eta, \eta' \in \mathcal{H}$ such that $\forall \alpha \in D(A)$,

$$(\psi, A\alpha) = (\eta, \alpha) \quad \text{and} \quad (\psi, A\alpha) = (\eta', \alpha)$$

Hence, $\forall \alpha \in D(A)$, $(\eta - \eta', \alpha) = 0$. This means $\eta - \eta' \in D(A)^\perp$. But since $D(A)$ is dense in $\mathcal{H}$, we have $D(A)^\perp = \{0\}$. Hence, $\eta = \eta'$ ■

Proposition 3.1.2. *If A is densely defined, then $\ker(A^*) = (\text{ran } A)^\perp$*

Proof.

$$\psi \in \ker(A^*) \iff \forall \alpha \in D(A), (A^*\psi, \alpha) = 0 \iff \alpha \in D(A), (\psi, A\alpha) = 0 \iff \psi \in (\text{ran } A)^\perp$$

■

Definition 3.1.3. Let A, B be two operators from their respective domains to the Hilbert space. We say B is an extension of A and write $A \subseteq B$ if,

1. $D(A) \subseteq D(B)$
2. $\forall \alpha \in D(A) : A\alpha = B\alpha$

Lemma 3.1.1. *Let A, B be densely defined. Then,*

$$A \subseteq B \Rightarrow B^* \subseteq A^*$$

Proof. Let $\psi \in D(B^*)$. Then,

$$\exists \eta \in \mathcal{H} : \forall \beta \in D(B), (\psi, B\beta) = (\eta, \beta)$$

This means,

$$\forall \alpha \in D(A), (\psi, B\alpha) = (\eta, \alpha)$$

Hence $\psi \in D(A^*)$. Also. by defining $\eta =: A^*\psi$, we have our result. ■

Definition 3.1.4. Symmetric Map: A map A is called symmetric if,

$$\forall \alpha, \beta \in D(A) : (\alpha, A\beta) = (A\alpha, \beta)$$

Remark. In Quantum Mechanics, “symmetric” is synonymous with “Hermitian”

Lemma 3.1.2. *If A is densely defined and symmetric, then $A \subseteq A^*$*

Proof. Let $\psi \in D(A)$. Since A is symmetric,

$$\forall \alpha \in D(A) : (\psi, A\alpha) = (A\psi, \alpha)$$

Define $\eta := A\psi$. Hence, $\psi \in D(A^*)$ and $\eta =: A^*\psi$ ■

Definition 3.1.5. Self-Adjoint Map: A densely defined map A is called self adjoint if $A = A^*$

Corollary 3.1.2.1. *Self-adjointness imply symmetry, but not the other way round.*

Corollary 3.1.2.2. *A self-adjoint map cannot be extended further by another symmetric map.*

Proof. Let A be self-adjoint and B be symmetric such that $A \subseteq B$.

$$A \subseteq B \subseteq B^* \subseteq A^* = A \Rightarrow A = B$$

■

Remark. Since all self-adjoint operators are symmetric, no self-adjoint extension is possible for an already self-adjoint operator.

Definition 3.1.6. Let $A : D(A) \rightarrow \mathcal{H}$ be a densely defined operator. Then,

1. A is said to be *Closable* if $A^* : D(A^*) \rightarrow \mathcal{H}$ is also densely defined
2. *Closure* of a closable map $\bar{A} := A^{**}$
3. A is called *Closed* if $A = \bar{A}$

Lemma 3.1.3. *If A is symmetric, then it is closable*

Proof. Since A is symmetric, we have $D(A) \subseteq D(A^*)$. Taking topological closure,

$$\mathcal{H} = \overline{D(A)} \subseteq \overline{D(A^*)} \subseteq \mathcal{H}$$

Hence, $\overline{D(A^*)} = \mathcal{H}$ which gives A is closable.

■

Lemma 3.1.4. *If A is symmetric, then $A^{**} \subseteq A^*$*

Proof.

$$A \subseteq A^* \Rightarrow A^{**} \subseteq A^*$$

■

Lemma 3.1.5. *If A is closable, then $A \subseteq A^{**}$*

Proof. From the definition of $D(A^*)$, we have

$$\forall \alpha \in D(A), \forall \psi \in D(A^*) : (\psi, A\alpha) = (A^*\psi, \alpha)$$

Switching places of α and ϕ ,

$$\forall \psi \in D(A), \forall \alpha \in D(A^*) : (\alpha, A\psi) = (A^*\alpha, \psi)$$

This is equivalent to,

$$\forall \psi \in D(A), \forall \alpha \in D(A^*) : (\psi, A^*\alpha) = (A\psi, \alpha)$$

Define $\eta := A\psi$. Then,

$$\forall \psi \in D(A), \exists \eta \in \mathcal{H} : \forall \alpha \in D(A^*), (\psi, A^*\alpha) = (\eta, \alpha)$$

Hence, $\psi \in D(A^{**})$, and $A^{**}\psi := \eta$. This gives the result. ■

Corollary 3.1.5.1. *If A is symmetric, then $A \subseteq A^{**} \subseteq A^*$*

Corollary 3.1.5.2. *If A is closable, then A^{**} is densely defined.*

Corollary 3.1.5.3. *If A is symmetric, then the closure of A is also symmetric.*

Proof. Since $A \subseteq \overline{A} \subseteq A^*$,

$$\forall \psi \in D(\overline{A}), \phi \in D(A^*) : (\psi, A^*\phi) = (\overline{A}\psi, \phi)$$

Which immediately gives,

$$\forall \psi \in D(\overline{A}), \phi \in D(\overline{A}) \subseteq D(A^*) : (\psi, \overline{A}\phi) = (\overline{A}\psi, \phi)$$
■

Definition 3.1.7. Essential Self-Adjointness: Let $A : D(A) \rightarrow \mathcal{H}$ be symmetric. A is called essentially self-adjoint if $\overline{A}$ is self-adjoint

Remark. Notice that self-adjointness imply essential self-adjointness, not the otherway round.

Theorem 3.1.6. *If A is essentially self-adjoint, then $\overline{A}$ is the unique self-adjoint extension of A*

Proof. We already know, $A \subseteq \overline{A}$. We need to show uniqueness. Let $A \subseteq B = B^*$. Then, $B^{**} \subseteq A^* \Rightarrow A^{**} \subseteq B^{***} = B$. Since A^{**} is self-adjoint, we have $A^{**} \equiv \overline{A} = B$ ■

Graphical Representation of Maps

Let $A : D(A) \rightarrow \mathcal{H}$ be any map. Then, we define

Definition 3.1.8. The Graph of A

$$\Gamma(A) := \left\{ [\psi, A\psi] \in \mathcal{H}^2 \mid \psi \in D(A) \right\}$$

Remark. The notation $[\cdot, \cdot]$ stands for an ordered pair. We use this to avoid confusion with the inner product.

Example 3.1.8.1. Consider a bounded map A . Then we know,

$$\|A\| = \sup_{\psi \in D(A)} \frac{\|A\psi\|}{\|\psi\|} \leq M \in \mathbb{R}$$

To rephrase this in terms of the graph, define,

Definition 3.1.9. Graph Norm: $\|\cdot\| : \Gamma(A) \rightarrow \mathbb{R}$ as,

$$\left\| [\psi, \tilde{\psi}] \right\| := \|\psi\| + \|\tilde{\psi}\|$$

If A is bounded,

$$\left\| [\psi, A\psi] \right\| = \|\psi\| + \|A\psi\| \leq \|\psi\| + \|A\|\|\psi\| = M\|\psi\| \quad \text{for some } M \in \mathbb{R}$$

Hence, A bounded imply, $\exists M \in \mathbb{R} : \forall [\psi, \tilde{\psi}] \in \Gamma(A), \left\| [\psi, \tilde{\psi}] \right\| \leq M$. Trivially, the converse is also true. Hence, we can write,

Proposition 3.1.3. A is bounded iff $\exists M \in \mathbb{R} : \forall [\psi, \tilde{\psi}] \in \Gamma(A), \left\| [\psi, \tilde{\psi}] \right\| \leq M$.

Example 3.1.9.1. Now say U is a unitary map. Then,

$$\forall \psi, \phi \in D(U) : (\psi, \phi) = (U\psi, U\phi)$$

Define $(\cdot, \cdot) : \Gamma(U) \times \Gamma(U) \rightarrow \mathbb{C}$ as

Definition 3.1.10. Graph Inner product: $(\cdot, \cdot) : \Gamma(U) \times \Gamma(U) \rightarrow \mathbb{C}$

$$\left([\psi, \tilde{\psi}], [\phi, \tilde{\phi}] \right) := (\psi, \phi) + (\tilde{\psi}, \tilde{\phi})$$

Proposition 3.1.4. U is unitary iff $\forall [\psi, \tilde{\psi}], [\phi, \tilde{\phi}] \in \Gamma(U) : \left([\psi, \tilde{\psi}], [\phi, -\tilde{\phi}] \right) = 0$

The proof is trivial!

Now, we go on to redefine closure in terms of graphs. For that, first consider the topological closure of a graph,

$$\overline{\Gamma(A)} = \left\{ [\psi, \tilde{\psi}] \in \mathcal{H}^2 \mid \exists \{[\psi_n, A\psi_n]\}_{n \in \mathbb{N}} \subseteq \Gamma(A) : [\psi_n, A\psi_n] \rightarrow [\psi, \tilde{\psi}] \right\}$$

We want to define an operator $\overline{A} : D(\overline{A}) \rightarrow \mathcal{H}$ in such a way that $\Gamma(\overline{A}) = \overline{\Gamma(A)}$. i.e., $\forall [\psi, \tilde{\psi}] \in \overline{\Gamma(A)}$ define $\overline{A}\psi := \tilde{\psi}$. Hence it is possible to define it in such a way that $D(\overline{A}) = \overline{D(A)}$, as a result $A \subseteq \overline{A}$. Now for the well definition of $\overline{A}$, we need

$$\psi, \phi \in \overline{D(A)} : \overline{A}\psi = \overline{A}\phi \Rightarrow \psi = \phi$$

or,

$$[\psi_n, A\psi_n] \rightarrow [0, \tilde{\psi}] \Rightarrow \tilde{\psi} = 0$$

Hence, we make the following definition,

Definition 3.1.11. Let $A : D(A) \rightarrow \mathcal{H}$ be any map.

1. A is called closable if $[\psi_n, A\psi_n] \rightarrow (0, \tilde{\psi}) \Rightarrow \tilde{\psi} = 0$
2. The closure of a closable map A denoted by $\overline{A}$ satisfy $\Gamma(\overline{A}) = \overline{\Gamma(A)}$
3. A is closed if $\Gamma(A) = \overline{\Gamma(A)}$

Remark. As of now, these are new definitions and do not have any connections to the previous definitions. We will make connections now via the following theorem.

Theorem 3.1.7. Let A be densely defined. Then,

1. A^* is closed.
2. A is closable iff A^* is densely defined and $\overline{A} = A^{**}$. Also, $\overline{A}^* = A^*$
3. If A is injective and $\text{ran } A$ is dense, then $(A^*)^{-1} = (A^{-1})^*$. If A is closable and $\overline{A}$ is injective, then $(\overline{A})^{-1} = \overline{A^{-1}}$

Proof. .

1.

$$\begin{aligned}\Gamma(A^*) &= \left\{ [\phi, \tilde{\phi}] \in \mathcal{H}^2 \mid \forall \psi \in D(A) : (\phi, A\psi) = (\tilde{\phi}, \psi) \right\} \\ &= \left\{ [\phi, \tilde{\phi}] \in \mathcal{H}^2 \mid \forall [\psi, \tilde{\psi}] \in \Gamma(A) : ([\phi, \tilde{\phi}], [\tilde{\psi}, -\psi]) = 0 \right\}\end{aligned}$$

Define a unitary map $U : \mathcal{H}^2 \rightarrow \mathcal{H}^2$,

$$U[\phi, \psi] := [\psi, -\phi]$$

then,

$$\Gamma(A^*) = \left\{ [\phi, \tilde{\phi}] \in \mathcal{H}^2 \mid \forall [\psi, \tilde{\psi}] \in \Gamma(A) : ([\phi, \tilde{\phi}], U[\psi, \tilde{\psi}]) = 0 \right\} = (U\Gamma(A))^\perp$$

Hence $\Gamma(A^*)$ is closed, which means A^* is closed.

2. Notice $\overline{\Gamma(A)} = \Gamma(A)^{\perp\perp}$. For any unitary map U and $M \subseteq \mathcal{H}$, we know $UM^\perp = (UM)^\perp$. From part 1, we have $\Gamma(A)^\perp = U^{-1}\Gamma(A^*) = -U\Gamma(A^*)$. Since U and $-U$ acting on the same space produce the same output we can forget the negative sign. Hence,

$$\begin{aligned}\overline{\Gamma(A)} &= (U\Gamma(A^*))^\perp \\ &= \left\{ [\psi, \tilde{\psi}] \in \mathcal{H}^2 \mid \forall [\phi, \tilde{\phi}] \in \Gamma(A^*) : ([\psi, \tilde{\psi}], U[\phi, \tilde{\phi}]) = 0 \right\} \\ &= \left\{ [\psi, \tilde{\psi}] \in \mathcal{H}^2 \mid \forall \phi \in D(A^*) : (\psi, A^*\phi) - (\tilde{\psi}, \phi) = 0 \right\}\end{aligned}$$

Hence,

$$[0, \tilde{\psi}] \in \overline{\Gamma(A)} \iff \forall \phi \in D(A^*) : (\tilde{\psi}, \phi) = 0 \iff \tilde{\psi} \in D(A^*)^\perp$$

Now, assume A is closable. Then $[0, \tilde{\psi}] \in \overline{\Gamma(A)} \Rightarrow \tilde{\psi} = 0$. Then we get $\tilde{\psi} \in D(A^*)^\perp$. But if $\exists \tilde{\alpha} \neq 0 \in D(A^*)^\perp$ this would mean $[0, \tilde{\alpha}] \in \overline{\Gamma(A)}$ (from the iff condition) which violates our assumption. Hence $D(A^*)^\perp = \{0\}$ which means that $D(A^*)$ is dense. Now assume the converse. If $D(A^*)$ is dense then $D(A^*)^\perp = \{0\}$ which in turn gives (from the iff condition) $[0, \tilde{\psi}] \in \overline{\Gamma(A)} \Rightarrow \tilde{\psi} = 0$. Thus, A is closable iff $D(A^*)$ is dense.

Now replace A by A^* in the $\Gamma(A^*)$ equation.

$$\Gamma(A^{**}) = (U\Gamma(A^*))^\perp = \overline{\Gamma(A)}$$

This gives us,

$$A^{**} = \overline{A}$$

Now replace A by A^* one more time,

$$\Gamma(A^{***}) = \overline{\Gamma(A^*)}$$

Which gives,

$$\overline{A^*} = \overline{A^*} = A^* \quad (\text{since } A^* \text{ is closed already})$$

3. If A is injective, we can define the inverse via the graph,

$$\Gamma(A^{-1}) = V\Gamma(A)$$

where $V : \mathcal{H}^2 \rightarrow \mathcal{H}^2$ is defined as $V(\psi, \phi) := (\phi, \psi)$ is another unitary map. Given that $\text{ran } A$ is dense. Hence

$$\ker A^* = (\text{ran } A)^\perp = \{0\} \Rightarrow (A^*)^{-1} \text{ exists}$$

Now,

$$\Gamma((A^*)^{-1}) = V\Gamma(A^*) = VU\Gamma(A)^\perp = UV\Gamma(A)^\perp = U(V\Gamma(A))^\perp = U\Gamma(A^{-1})^\perp$$

Since we had from part 2 that $U\Gamma(A)^\perp = \Gamma(A^*)$,

$$\Gamma((A^*)^{-1}) = \Gamma((A^{-1})^*) \Rightarrow (A^*)^{-1} = (A^{-1})^*$$

Now let A closable and $\overline{A}$ injective. Then

$$\Gamma((\overline{A})^{-1}) = V\Gamma(\overline{A}) = V\overline{\Gamma(A)} = \overline{V\Gamma(A)} = \overline{\Gamma(A^{-1})}$$

Hence,

$$(\overline{A})^{-1} = \overline{A^{-1}}$$

■

Corollary 3.1.7.1. *If A is self-adjoint and injective, then A^{-1} is self-adjoint.*

Proof. Since $A = A^*$, $(\text{ran } A)^\perp = \ker A^* = \ker A = \{0\}$. Hence part 3 of the above proof applies. Then $(A^*)^{-1} = (A^{-1})^* \Rightarrow A^{-1} = (A^{-1})^*$ ■

3.2 Resolvents, Spectra and all That[1, 2]

Definition 3.2.1. The **Resolvent map** of an operator A is the map,

$$R_A : \rho(A) \rightarrow \mathcal{L}(\mathcal{H})$$

given by,

$$R_A(z) := (A - z)^{-1}$$

The set $\rho(A)$ is called the **Resolvent set** of A , and is defined as,

$$\rho(A) := \left\{ z \in \mathbb{C} \mid (A - z)^{-1} \in \mathcal{L}(\mathcal{H}) \right\}$$

Definition 3.2.2. The spectrum of A is defined as,

$$\sigma(A) := \mathbb{C} \setminus \rho(A)$$

Definition 3.2.3. A number $\lambda \in \mathbb{C}$ is called an eigenvalue of A if $\exists \psi \in D(A) \setminus \{0\} : A\psi = \lambda\psi$

Corollary 3.2.0.1. *If λ is an eigenvalue of A , then it is in the spectrum of A*

Proof. If λ is an eigenvalue of A , then,

$$(A - \lambda)\psi = 0$$

Since ψ by definition is not the zero vector, the kernel of $A - \lambda$ is non empty. This means that $A - \lambda$ is not invertible. Hence $\lambda \notin \rho(A)$, which means that λ is in the spectrum of A . ■

Spectrum of a Self-Adjoint Operator

Definition 3.2.4. Let A be self-adjoint. Then,

1. The pure point spectrum,

$$\sigma_{pp}(A) := \left\{ z \in \mathbb{C} \mid \text{ran}(A - z) = \text{ran}(\overline{A - z}) \neq \mathcal{H} \right\}$$

2. The point embedded in continuum spectrum,

$$\sigma_{pec}(A) := \left\{ z \in \mathbb{C} \mid \text{ran}(A - z) \neq \text{ran}(\overline{A - z}) \neq \mathcal{H} \right\}$$

3. The purely continuous spectrum,

$$\sigma_{pc}(A) := \left\{ z \in \mathbb{C} \mid \text{ran}(A - z) \neq \overline{\text{ran}(A - z)} = \mathcal{H} \right\}$$

We can further define a point spectrum $\sigma_p(A)$ which is the union of $\sigma_{pp}(A)$ and $\sigma_{pec}(A)$, and a continuous spectrum.

Lemma 3.2.1. *Self-adjoint operators have all eigen values real*

Proof. Let ψ be an eigenvector for λ . Then,

$$\lambda(\psi, \psi) = (\psi, \lambda\psi) = (\psi, A\psi) = (A\psi, \psi) = \lambda^*(\psi, \psi)$$

Since the norm is non zero, $\lambda = \lambda^*$ ■

Theorem 3.2.2. *For a self-adjoint operator, the pure point spectrum contains only the eigenvalues.*

Proof. Let λ be an eigenvalue of A , which is a self-adjoint operator. Now,

$$\{0\} \neq \ker(A - \lambda) = \ker((A - \lambda)^*) = \text{ran}(A - \lambda)^\perp$$

Hence,

$$\overline{\text{ran}(A - \lambda)} = \text{ran}(A - \lambda)^{\perp\perp} \neq \mathcal{H}$$

This gives $\lambda \in \sigma_p(A)$.

Now to show the converse, we show the contrapositive. Let λ be not an eigen value of A . Hence λ^* is also not an eigenvalue, because if it were, then λ^* have to be real which contradicts our assumption that λ is not an eigenvalue. This means that there is no nonzero vector such that $A\psi = \lambda^*\psi$. Hence,

$$\{0\} = \ker(A - \lambda^*) = \ker((A - \lambda)^*) = \text{ran}(A - \lambda)^\perp$$

Hence,

$$\overline{\text{ran}(A - \lambda)} = \text{ran}(A - \lambda)^{\perp\perp} = \mathcal{H}$$

Hence $\lambda \notin \sigma_p(A)$ ■

Chapter 4

The Spectral Theorem and The Stone's Theorem

4.1 The Spectral Theorem[1, 2]

In the first few subsections, we develop the necessary definitions and proofs that are required to state and prove the Spectral Theorem.

Projection Valued Measures (PVMs)

Definition 4.1.1. A PVM is a map $P : \sigma(\mathcal{O}_{\mathbb{R}}) \rightarrow \mathcal{L}(\mathcal{H})$ satisfying the following;
 $\forall \Omega \in \sigma(\mathcal{O}_{\mathbb{R}})$,

1. $P(\Omega)^* = P(\Omega)$
2. $P(\Omega)^2 = P(\Omega)$
3. $P(\mathbb{R}) = id_{\mathcal{H}}$
4. For pairwise disjoint $\Omega_1, \Omega_2, \dots \in \sigma(\mathcal{O}_{\mathbb{R}})$ and $\psi \in \mathcal{H}$: $\sum_{n \geq 1} P(\Omega_n)\psi = P(\cup_{n \geq 1} \Omega_n)\psi$

Remark. The convergence used above is in $\mathcal{H}$. This is called strong convergence, and this implies the weaker convergence on $\mathcal{L}(\mathcal{H})$

Proposition 4.1.1. *PVMs satisfy the following properties,*

1. $P(\emptyset) = 0$

2. $P(\mathbb{R} \setminus \Omega) = id_{\mathcal{H}} - P(\Omega)$
3. $P(\Omega_1 \cup \Omega_2) + P(\Omega_1 \cap \Omega_2) = P(\Omega_1) + P(\Omega_2)$
4. $P(\Omega_1 \cap \Omega_2) = P(\Omega_1) \circ P(\Omega_2)$
5. $\Omega_1 \subseteq \Omega_2 \Rightarrow ran(P(\Omega_1)) \subseteq ran(P(\Omega_2))$

Proof. We prove the results in sequential order.

1. $\forall \psi \in \mathcal{H} : P(\emptyset)\psi = P(\emptyset \cup \emptyset)\psi = 2P(\emptyset)\psi \Rightarrow P(\emptyset)\psi = 0 \Rightarrow P(\emptyset) = 0$
2. $P(\mathbb{R}) = P((\mathbb{R} \setminus \Omega) \cup \Omega) \Rightarrow id_{\mathcal{H}} = P(\mathbb{R} \setminus \Omega) + P(\Omega)$
3. Write $\Omega_1 = (\Omega_1 \setminus \Omega_2) \cup (\Omega_1 \cap \Omega_2)$. The decomposition is in terms of mutually disjoint sets. Then using similar expression for Ω_2 ,

$$\begin{aligned}
 P(\Omega_1) + P(\Omega_2) &= P(\Omega_1 \setminus \Omega_2) + P(\Omega_1 \cap \Omega_2) + P(\Omega_2 \setminus \Omega_1) + P(\Omega_1 \cap \Omega_2) \\
 &= P((\Omega_1 \setminus \Omega_2) \cup (\Omega_1 \cap \Omega_2) \cup (\Omega_2 \setminus \Omega_1)) + P(\Omega_1 \cap \Omega_2) \\
 &= P(\Omega_1 \cup \Omega_2) + P(\Omega_1 \cap \Omega_2)
 \end{aligned}$$

4. First let $\Omega_1 \cap \Omega_2 = \emptyset$. Then the above result give,

$$P(\Omega_1 \cup \Omega_2) = P(\Omega_1) + P(\Omega_2)$$

squaring this and using the idempotency of P , we get,

$$P(\Omega_1) \circ P(\Omega_2) = -P(\Omega_2) \circ P(\Omega_1)$$

Applying $P(\Omega_2)$ from the right on the above equation and using the above equation again on it will give,

$$P(\Omega_1) \circ P(\Omega_2) = P(\Omega_2) \circ P(\Omega_1)$$

Hence, $P(\Omega_1) \circ P(\Omega_2) = 0$. Now for any general Ω_1 and Ω_2 ,

$$\begin{aligned}
 P(\Omega_1) \circ P(\Omega_2) &= [P(\Omega_1 \setminus \Omega_2) + P(\Omega_1 \cap \Omega_2)] \circ [P(\Omega_2 \setminus \Omega_1) + P(\Omega_1 \cap \Omega_2)] \\
 &= P(\Omega_1 \cap \Omega_2)
 \end{aligned}$$

5. If $\Omega_1 \subseteq \Omega_2$, then $\Omega_2 = (\Omega_2 \setminus \Omega_1) \cup \Omega_1$. Which implies the result.



Remark. Since $\sigma(\mathcal{O}_{\mathbb{R}})$ is generated by $(-\infty, \lambda], \forall \lambda \in \mathbb{R}$, it suffices to specify,

$$P(\lambda) := P((-\infty, \lambda])$$

$P(\lambda)$ is called the *Resolution of the Identity*

Definition 4.1.2. For any $\psi, \phi \in \mathcal{H}$, we can define a complex valued measure, $\mu_{\psi, \phi} : \sigma(\mathcal{O}_{\mathbb{R}}) \rightarrow \mathbb{C}$ as,

$$\mu_{\psi, \phi}(\Omega) := (\psi, P(\Omega)\phi)$$

Definition 4.1.3. For any $\psi \in \mathcal{H}$, we define a real valued measure, $\mu_{\psi} : \sigma(\mathcal{O}_{\mathbb{R}}) \rightarrow \mathbb{R}$ as,

$$\mu_{\psi} := \mu_{\psi, \psi}$$

Remark. The real valued measure is guaranteed to be real because the PVM is self adjoint.

Remark. To store the information of a single PVM, we need many(may be infinite) number of real (complex) valued measures!

Integration w.r.t. a PVM

We make sense of the integration w.r.t. a PVM in three steps.

Simple Functions

A function $f : \mathbb{R} \rightarrow \mathbb{C}$ is called simple if it has a finite range. Then it can be written as,

$$f = \sum_{n=1}^N f_n \chi_{\Omega_n} \quad \text{for some } N \in \mathbb{N}$$

where $\Omega_n = \text{preim}_f(\{f_n\})$. It is obvious that we can choose these Ω_n 's in such a way that they are all mutually disjoint.

Definition 4.1.4. $\text{Sim}(\mathbb{R}) := \left\{ f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is simple and measurable} \right\}$

Definition 4.1.5. Let $f \in \text{Sim}(\mathbb{R})$. Then,

$$\int_{\mathbb{R}} f dP := \sum_{n=1}^N f_n P(\Omega_n)$$

Corollary 4.1.0.1. $\int_{\mathbb{R}} \chi_{\Omega} dP = P(\Omega)$

Corollary 4.1.0.2. $(\psi, (\int_{\mathbb{R}} f dP) \psi) = \int_{\mathbb{R}} f d\mu_{\psi, \psi}$

Proposition 4.1.2. Equip $\text{Sim}(\mathbb{R})$ with the supremum norm, and define the map $\int_{\mathbb{R}} dP : \text{Sim}(\mathbb{R}) \rightarrow \mathcal{L}(\mathcal{H})$ as $f \mapsto \int_{\mathbb{R}} f dP$. Then this map is linear with $\|\int_{\mathbb{R}} dP\| = 1$

Proof. The proof for the linearity is exactly the same as that of the linearity of the Lebesgue integral for simple functions. For the norm,

$$\left\| \int_{\mathbb{R}} dP \right\| = \sup_{f \in \text{Sim}(\mathbb{R})} \frac{\left\| \int_{\mathbb{R}} f dP \right\|}{\|f\|_{\infty}} = \sup_{f \in \text{Sim}(\mathbb{R})} \sup_{\psi \in \mathcal{H}} \frac{\left\| \left(\int_{\mathbb{R}} f dP \right) \psi \right\|}{\|f\|_{\infty} \cdot \|\psi\|}$$

Hence we calculate,

$$\begin{aligned} \left\| \left(\int_{\mathbb{R}} f dP \right) \psi \right\|^2 &= \left(\left(\int_{\mathbb{R}} f dP \right) \psi, \left(\int_{\mathbb{R}} f dP \right) \psi \right) \\ &= \left(\sum_{n=1}^N f_n P(\Omega_n) \psi, \sum_{m=1}^N f_m P(\Omega_m) \psi \right) \\ &= \left(\psi, \sum_{m=1}^N \sum_{n=1}^N f_n^* f_m P(\Omega_n) \circ P(\Omega_m) \psi \right) \\ &= \left(\psi, \sum_{n=1}^N |f_n|^2 P(\Omega_n) \psi \right) \\ &= \int_{\mathbb{R}} |f|^2 d\mu_{\psi} \end{aligned}$$

Hence we get the estimate,

$$\left\| \left(\int_{\mathbb{R}} f dP \right) \psi \right\| \leq \|f\|_{\infty} \cdot \|\psi\|$$

Notice that $\|f\|_{\infty} \cdot \chi_{\mathbb{R}} \in \text{Sim}(\mathbb{R})$ achieves the equality. Hence we have,

$$\left\| \int_{\mathbb{R}} dP \right\| = \sup_{f \in \text{Sim}(\mathbb{R})} \sup_{\psi \in \mathcal{H}} \frac{\|f\|_{\infty} \cdot \|\psi\|}{\|f\|_{\infty} \cdot \|\psi\|} = 1$$

■

Bounded Borel Functions

Definition 4.1.6. $B(\mathbb{R}) := \left\{ f : \mathbb{R} \rightarrow \mathbb{C} \mid f \text{ is measurable and } \|f\|_{\infty} < \infty \right\}$

Remark. It can be shown rather easily that $(B(\mathbb{R}), \|\cdot\|_\infty)$ is a Banach space. But it is not necessary for what follows.

We already had $\text{Sim}(\mathbb{R}) \subseteq B(\mathbb{R})$ and the inclusion is dense. Then we can use the BLT theorem to extend the operator $\int_{\mathbb{R}} dP$ to $B(\mathbb{R})$, and the operator norm stays the same.

Proposition 4.1.3. *The map $\int_{\mathbb{R}} dP : B(\mathbb{R}) \rightarrow \mathcal{L}(\mathcal{H})$ satisfies the following,*

1. $\int_{\mathbb{R}} id_{\mathbb{R}} dP = id_{\mathcal{H}}$
2. $\int_{\mathbb{R}} f \cdot g dP = \left(\int_{\mathbb{R}} f dP\right) \circ \left(\int_{\mathbb{R}} g dP\right)$
3. $\int_{\mathbb{R}} f^* dP = \left(\int_{\mathbb{R}} f dP\right)^*$

Proof. .

1. $\int_{\mathbb{R}} id_{\mathbb{R}} dP = \int_{\mathbb{R}} \chi_{\mathbb{R}} dP = P(\mathbb{R}) = id_{\mathcal{H}}$
2. We prove the next two properties for simple functions, and then it continues to hold in $B(\mathbb{R})$ because $\text{Sim}(\mathbb{R})$ is dense in $B(\mathbb{R})$ and the dominated convergence theorem. Let,

$$f = \sum_{n=1}^N f_n \chi_{\Omega_n} \quad g = \sum_{m=1}^M g_m \chi_{\Theta_m}$$

$$\begin{aligned} \left(\int_{\mathbb{R}} f dP\right) \circ \left(\int_{\mathbb{R}} g dP\right) &= \left(\sum_{n=1}^N f_n P(\Omega_n)\right) \circ \left(\sum_{m=1}^M g_m P(\Theta_m)\right) \\ &= \sum_{n=1}^N \sum_{m=1}^M f_n g_m P(\Omega_n \cap \Theta_m) \end{aligned}$$

but,

$$f \cdot g = \sum_{n=1}^N \sum_{m=1}^M f_n g_m \chi_{\Omega_n \cap \Theta_m}$$

Hence the result

3.

$$\begin{aligned}
\int_{\mathbb{R}} f^* dP &= \sum_{n=1}^N f_n^* P(\Omega_n) \\
&= \left(\sum_{n=1}^N f_n P(\Omega_n) \right)^* \\
&= \left(\int_{\mathbb{R}} f dP \right)^*
\end{aligned}$$

■

Remark. With the above three properties, $\int_{\mathbb{R}} dP : B(\mathbb{R}) \rightarrow \mathcal{L}(\mathcal{H})$ is a C^* -algebra homomorphism.

Any Borel Function

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be any measurable Borel function, not necessarily bounded. Then we expect the resulting operator $\int_{\mathbb{R}} f dP$ also to be unbounded in $\mathcal{H}$. Hence we look for a suitable domain in which such an operator can be defined without trouble. Previously we had,

$$\forall f \in B(\mathbb{R}), \psi \in \mathcal{H} : \left\| \left(\int_{\mathbb{R}} f dP \right) \psi \right\|^2 = \int_{\mathbb{R}} |f|^2 d\mu_{\psi}$$

Hence for any general $f : \mathbb{R} \rightarrow \mathbb{C}$, measurable, we choose a domain,

$$D(f) \equiv D\left(\int_{\mathbb{R}} f dP\right) := \left\{ \psi \in \mathcal{H} \mid \int_{\mathbb{R}} |f|^2 d\mu_{\psi} < \infty \right\}$$

Lemma 4.1.1. $D(f)$ is a linear subspace of $\mathcal{H}$

Proof. If $\psi \in D(f)$, then $\mu_{\alpha\psi}(\Omega) = |\alpha|^2 \mu_{\psi}(\Omega)$ implies that $\int_{\mathbb{R}} |f|^2 d\mu_{\alpha\psi} < \infty$ hence $\alpha\psi \in D(f)$. Likewise for $\psi, \phi \in D(f)$,

$$\mu_{\psi+\phi}(\Omega) = \|P(\Omega)(\psi + \phi)\|^2 \leq 2(\|P(\Omega)\psi\|^2 + \|P(\Omega)\phi\|^2) = 2(\mu_{\psi}(\Omega) + \mu_{\phi}(\Omega))$$

which says $\psi + \phi \in D(f)$

■

Now we make use of what we have learned for the bounded functions to define $\int_{\mathbb{R}} f dP$ for any general borel funcion f . First we define,

$$f_n := \chi_{\Omega_n} f$$

where

$$\Omega_n := \left\{ x \in \mathbb{R} \mid |f(x)| \leq n \right\}$$

Each f_n are bounded by definition and it is clear that,

$$\lim_{n \rightarrow \infty} f_n = f$$

Which tells us that $\{f_n\}_{n \in \mathbb{N}} \subseteq L^2(\mathbb{R}, \mu_\psi)$ is Cauchy. If we can manage to show that $\left\{ \left(\int_{\mathbb{R}} f_n dP \right) \psi \right\}_{n \in \mathbb{N}}$ is Cauchy in $\mathcal{H}$ for all $\psi \in D(f)$, we can use this for the definition.

Lemma 4.1.2. *If $\{f_n\}_{n \in \mathbb{N}} \subseteq L^2(\mathbb{R}, \mu_\psi)$ is Cauchy, then $\left\{ \left(\int_{\mathbb{R}} f_n dP \right) \psi \right\}_{n \in \mathbb{N}} \subseteq \mathcal{H}$ is Cauchy for all $\psi \in D(f)$.*

Proof. Take any $\psi \in D(f)$. Then,

$$\begin{aligned} \left\| \left(\int_{\mathbb{R}} f_n dP \right) \psi - \left(\int_{\mathbb{R}} f_m dP \right) \psi \right\|^2 &= \left(\left(\int_{\mathbb{R}} f_n dP - \int_{\mathbb{R}} f_m dP \right) \psi, \left(\int_{\mathbb{R}} f_n dP - \int_{\mathbb{R}} f_m dP \right) \psi \right) \\ &= \left(\psi, \left(\int_{\mathbb{R}} (|f_n|^2 + |f_m|^2 - f_n^* f_m - f_m^* f_n) dP \right) \psi \right) \\ &= \int_{\mathbb{R}} (|f_n|^2 + |f_m|^2 - f_n^* f_m - f_m^* f_n) d\mu_\psi \\ &= \int_{\mathbb{R}} |f_n - f_m|^2 d\mu_\psi \\ &\leq \|f_n - f_m\|_\infty^2 \cdot \|\psi\|^2 \end{aligned}$$

Since the $\|\psi\|$ is fixed, f_n being Cauchy imply that $\left(\int_{\mathbb{R}} f_n dP \right) \psi$ is also Cauchy. ■

Definition 4.1.7. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be any borel funcion.

$$\forall \psi \in D(f) : \left(\int_{\mathbb{R}} f dP \right) \psi := \lim_{n \rightarrow \infty} \left(\int_{\mathbb{R}} f_n dP \right) \psi$$

Remark. By the construction of the domain,

$$\forall \psi \in D(f) : \left\| \left(\int_{\mathbb{R}} f dP \right) \psi \right\|^2 = \int_{\mathbb{R}} |f|^2 d\mu_\psi < \infty$$

Now, to define the adjoint of the operator, we need the domain to be dense.

Lemma 4.1.3. *$D(f)$ is dense in $\mathcal{H}$*

Proof. For any $\psi \in \mathcal{H}$ construct a sequence,

$$\psi_n := P(\Omega_n)\psi$$

Clearly, the sequence converge to ψ as larger and larger n will make $\Omega_n \rightarrow \mathbb{R}$. We just have to show that $\psi_n \in D(f)$. For that,

$$\begin{aligned} d\mu_{\psi_n}(\Omega) &= (\psi_n, P(\Omega)\psi_n) \\ &= (P(\Omega_n)\psi, P(\Omega) \circ P(\Omega_n)\psi) \\ &= (\psi, P(\Omega \cap \Omega_n \cap \Omega_n)\psi) \\ &= (\psi, P(\Omega \cap \Omega_n)\psi) \\ &= d\mu_{\psi}(\Omega \cap \Omega_n) \\ &= \chi_{\Omega_n} d\mu_{\psi}(\Omega) \end{aligned}$$

Therefore,

$$\int_{\mathbb{R}} |f|^2 d\mu_{\psi_n} = \int_{\mathbb{R}} |f|^2 \chi_{\Omega_n} d\mu_{\psi} = \int_{\Omega_n} |f|^2 d\mu_{\psi}$$

Since f is bounded on all Ω_n , the above integral is finite, and hence $\psi_n \in D(f)$. ■

Proposition 4.1.4. *For any Borel function f ,*

$$\int_{\mathbb{R}} f^* dP = \left(\int_{\mathbb{R}} f dP \right)^*$$

Proof. Define the sequence of bounded functions f_n as before. For the bounded functions, we already have this result proved. Hence,

$$\forall \psi, \phi \in D(f_n) = D(f_n^*) : \left(\phi, \left(\int_{\mathbb{R}} f_n dP \right) \psi \right) = \left(\left(\int_{\mathbb{R}} f_n^* dP \right) \phi, \psi \right)$$

Since the inner-product is continuous, if we take the limits we get,

$$\forall \psi, \phi \in D(f) = D(f^*) : \left(\phi, \left(\int_{\mathbb{R}} f dP \right) \psi \right) = \left(\left(\int_{\mathbb{R}} f^* dP \right) \phi, \psi \right)$$

Now if we can show that $D((\int_{\mathbb{R}} f dP)^*) \subseteq D(f)$, this would mean

$$\forall \psi \in D(f), \phi \in D\left(\left(\int_{\mathbb{R}} f dP\right)^*\right) : \left(\phi, \left(\int_{\mathbb{R}} f dP \right) \psi \right) = \left(\left(\int_{\mathbb{R}} f^* dP \right) \phi, \psi \right)$$

which is the result we want to prove. To prove this, take $\psi \in D((\int_{\mathbb{R}} f dP)^*)$. Then by definition of the adjoint,

$$\forall \phi \in D(f), \exists \tilde{\psi} \in L^2(\mathbb{R}, \mu_{\psi}) : \left(\psi, \left(\int_{\mathbb{R}} f dP \right) \phi \right) = (\tilde{\psi}, \phi)$$

Now using,

$$\int_{\mathbb{R}} f_n dP = \int_{\mathbb{R}} f \cdot \chi_{\Omega_n} dP = \left(\int_{\mathbb{R}} f dP \right) \circ P(\Omega_n)$$

We have $\forall \phi \in \mathcal{H}$,

$$\begin{aligned} \left(\left(\int_{\mathbb{R}} f_n^* dP \right) \psi, \phi \right) &= \left(\psi, \left(\int_{\mathbb{R}} f_n dP \right) \phi \right) \\ &= \left(\psi, \left(\int_{\mathbb{R}} f dP \right) \circ P(\Omega_n) \phi \right) \\ &= \left(P(\Omega_n) \tilde{\psi}, \phi \right) \end{aligned}$$

Since the above statement is for all $\phi \in \mathcal{H}$,

$$\left(\int_{\mathbb{R}} f_n^* dP \right) \psi = P(\Omega_n) \tilde{\psi}$$

Therefore,

$$\lim_{n \rightarrow \infty} \left\| \left(\int_{\mathbb{R}} f_n^* dP \right) \psi \right\|^2 = \lim_{n \rightarrow \infty} \left\| P(\Omega_n) \tilde{\psi} \right\|^2 = \left\| P(\mathbb{R}) \tilde{\psi} \right\|^2 = \left\| \tilde{\psi} \right\|^2$$

Hence

$$\int_{\mathbb{R}} |f|^2 d\mu_{\psi} = \lim_{n \rightarrow \infty} \left[\int_{\mathbb{R}} |f_n|^2 d\mu_{\psi} \right] = \lim_{n \rightarrow \infty} \left\| \left(\int_{\mathbb{R}} f_n^* dP \right) \psi \right\|^2$$

exists. Which means that $\psi \in D(f)$. ■

Proposition 4.1.5. *For any Borel functions f and g ,*

1. $\alpha \int_{\mathbb{R}} f dP + \beta \int_{\mathbb{R}} g dP \subseteq \int_{\mathbb{R}} (\alpha f + \beta g) dP$ with $D(\alpha \int_{\mathbb{R}} f dP + \beta \int_{\mathbb{R}} g dP) = D(|f| + |g|)$
2. $(\int_{\mathbb{R}} f dP) \circ (\int_{\mathbb{R}} g dP) \subseteq \int_{\mathbb{R}} f \cdot g dP$ with $D((\int_{\mathbb{R}} f dP) \circ (\int_{\mathbb{R}} g dP)) = D(g) \cap D(fg)$

The “Inverse” Spectral Theorem

This is an intermediate result which is trivial from the definitions and proofs we had till now. The statement is almost the inverse of the Spectral Theorem, hence this name.

Theorem 4.1.4. Inverse Spectral Theorem: *Given a PVM, P , we can construct a self-adjoint operator A_P via,*

$$A_P = \int_{\mathbb{R}} id_{\mathbb{R}} dP$$

Where $id_{\mathbb{R}} : \mathbb{R} \rightarrow \mathbb{C}$ embeds the real line into the complex plane.

Proof. The existence is proven in the last section. Self adjointness is immediate,

$$A_P^* = \left(\int_{\mathbb{R}} id_{\mathbb{R}} dP \right)^* = \int_{\mathbb{R}} id_{\mathbb{R}}^* dP = \int_{\mathbb{R}} id_{\mathbb{R}} dP = A_P$$

■

The Stieltje's Inversion Formula

This is yet another ingredient we need to prove the spectral theorem. The formula allows us to reconstruct the PVM from a “spectrally decomposable” self-adjoint operator.

Definition 4.1.8. A self adjoint operator A is called spectrally decomposable if there is a PVM, P such that,

$$A = \int_{\mathbb{R}} id_{\mathbb{R}} dP$$

Definition 4.1.9. For any measurable $f : \mathbb{R} \rightarrow \mathbb{C}$, and a spectrally decomposable self adjoint A , define,

$$f(A) : D(f) \rightarrow \mathcal{L}(\mathcal{H})$$

as,

$$f(A) := \int_{\mathbb{R}} f(\lambda) P(d\lambda) \equiv \int_{\mathbb{R}} (f \cdot id_{\mathbb{R}}) dP$$

Definition 4.1.10. For any operator A , the Resolvent of A is a map,

$$R_A : \rho(A) \rightarrow \mathcal{L}(\mathcal{H})$$

where $\rho(A) \subseteq \mathbb{C}$ is called the resolvent set defined via,

$$\rho(A) := \left\{ z \in \mathbb{C} \mid (A - z)^{-1} \in \mathcal{L}(\mathcal{H}) \right\}$$

and

$$R_A(z) := (A - z)^{-1}$$

Now for any spectrally decomposable self adjoint A ,

$$\forall z \in \rho(A) : R_A(z) = (A - z)^{-1} =: r_z(A)$$

with

$$r_z(\lambda) := \frac{1}{\lambda - z}$$

Hence we can write,

$$R_A(z) = \int_{\mathbb{R}} \frac{1}{\lambda - z} P(d\lambda)$$

For any $\psi \in \mathcal{H}$,

$$(\psi, R_A(z)\psi) = \int_{\mathbb{R}} \frac{1}{\lambda - z} \mu_{\psi}(d\lambda)$$

Definition 4.1.11. A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is called a Herglotz function if it maps the upper half of the complex plane to itself.

Theorem 4.1.5. $(\psi, R_A(\cdot)\psi) : \mathbb{C} \rightarrow \mathbb{C}$ is Herglotz

Proof.

$$\begin{aligned} \operatorname{Im}(\psi, R_A(z)\psi) &= \operatorname{Im} \left(\int_{\mathbb{R}} \frac{1}{\lambda - z} \mu_{\psi}(d\lambda) \right) \\ &= \int_{\mathbb{R}} \operatorname{Im} \left(\frac{1}{\lambda - z} \right) \mu_{\psi}(d\lambda) \\ &= \int_{\mathbb{R}} \frac{\operatorname{Im}(z)}{|\lambda - z|^2} \mu_{\psi}(d\lambda) \\ &= \operatorname{Im}(z) \int_{\mathbb{R}} \frac{1}{|\lambda - z|^2} \mu_{\psi}(d\lambda) \end{aligned}$$

The integral is a positive real number. Hence $\operatorname{Im}(z) > 0 \Rightarrow \operatorname{Im}[(\psi, R_A(z)\psi)] > 0$ ■

Now before proving the Stieltje's Inversion formula, consider the integral,

$$I = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{t_1}^{t_2} dt \operatorname{Im}(\psi, R_A(t + i\epsilon)\psi)$$

Since the spectrum of a self adjoint operator is always real, $t + i\epsilon \in \rho(A)$ is guaranteed. Hence,

$$I = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{t_1}^{t_2} dt \left[\int_{\mathbb{R}} \frac{\epsilon}{|\lambda - t - i\epsilon|^2} \mu_{\psi}(d\lambda) \right]$$

The integrals can be interchanged using Fubini's theorem. Hence,

$$I = \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}} \left[\frac{1}{\pi} \int_{t_1}^{t_2} dt \frac{\epsilon}{|\lambda - t - i\epsilon|^2} \right] \mu_{\psi}(d\lambda)$$

Now,

$$\frac{1}{\pi} \int_{t_1}^{t_2} dt \frac{\epsilon}{|\lambda - t - i\epsilon|^2} = \frac{1}{\pi} \left[\arctan \left(\frac{t - \lambda}{\epsilon} \right) \right]_{t_1}^{t_2}$$

Taking the limit we get,

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{t_1}^{t_2} dt \frac{\epsilon}{|\lambda - t - i\epsilon|^2} = \frac{1}{2} \chi_{(t_1, t_2)} + \frac{1}{2} \chi_{[t_1, t_2]}$$

Now we are ready to prove the Stieltje's inversion.

Theorem 4.1.6. *Stieltje's Inversion Formula:* For a spectrally decomposable self adjoint A ,

$$\mu_{\psi}((-\infty, \lambda]) = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{-\infty}^{\lambda + \delta} dt \operatorname{Im}(\psi, R_A(t + i\epsilon)\psi)$$

Proof.

$$\begin{aligned} \text{RHS} &= \lim_{\delta \rightarrow 0^+} \int_{\mathbb{R}} \left[\frac{1}{2} \chi_{(-\infty, \lambda + \delta)} + \frac{1}{2} \chi_{(-\infty, \lambda]} \right] \mu_{\psi}(d\lambda) \\ &= \int_{\mathbb{R}} \chi_{(-\infty, \lambda]} \mu_{\psi}(d\lambda) \\ &= \mu_{\psi}((-\infty, \lambda]) \\ &= \text{LHS} \end{aligned}$$

■

Towards the Spectral Theorem

To free the Stieltje's Inversion Formula from the “spectrally decomposable” condition, we have to do some more work. We first state a theorem without proof.

Theorem 4.1.7. If F is a Herglotz function with $|F(z)| \leq \frac{M}{\operatorname{Im}(z)}$ for some $M \in \mathbb{R}$, then there exists a real value measure μ with $\mu(\mathbb{R}) \leq M$ and,

$$F(z) = \int_{\mathbb{R}} \frac{1}{\lambda - z} \mu(d\lambda)$$

This theorem opens up a way. We now just have to prove that $(\psi, R_A(\cdot)\psi) : \rho(A) \rightarrow \mathbb{C}$ is Herglotz. For that, we first prove the following lemma.

Lemma 4.1.8. First Resolvent Formula: For any $A : D(A) \rightarrow \mathcal{H}$ and $a, b \in \rho(A)$,

$$R_A(a) - R_A(b) = \begin{cases} (a - b)R_A(a)R_A(b) \\ \text{or} \\ (a - b)R_A(b)R_A(a) \end{cases}$$

Proof.

$$\begin{aligned} R_A(a) - R_A(b) &= (A - a)^{-1} - (A - b)^{-1} \\ &= (A - a)^{-1}[1 - (A - a)(A - b)^{-1}] \\ &= (A - a)^{-1}[(A - b)(A - b)^{-1} - (A - a)(A - b)^{-1}] \\ &= (A - a)^{-1}[(A - b) - (A - a)](A - b)^{-1} \\ &= (a - b)(A - a)^{-1}(A - b)^{-1} \\ &= (a - b)R_A(a)R_A(b) \end{aligned}$$

The ‘or’ part is exactly the same, except we take $R_A(a)$ outside to the right in the first step. ■

Lemma 4.1.9. For any self adjoint A and $\psi \in \mathcal{H}$, $(\psi, R_A(\cdot)\psi) : \rho(A) \rightarrow \mathbb{C}$ is Herglotz.

Proof. Take some $z \in \rho(A)$. As the spectrum of a self adjoint operator is always real, $\rho(A)$ covers the upper half of the complex plane for sure. Now,

$$\begin{aligned} \operatorname{Im}(\psi, R_A(z)\psi) &= \frac{1}{2i}[(\psi, R_A(z)\psi) - (R_A(z)\psi, \psi)] \\ &= \frac{1}{2i}[(\psi, R_A(z)\psi) - (\psi, R_A(z^*)\psi)] \\ &= \frac{1}{2i}[(\psi, (R_A(z) - R_A(z^*))\psi)] \\ &= \frac{z - z^*}{2i}[(\psi, R_A(z)R_A(z^*)\psi)] \\ &= \operatorname{Im}(z) \cdot \|R_A(z)\|^2 \end{aligned}$$

Hence the proof. ■

Now, as $(\psi, R_A(z)\psi)$ is bounded, there exists a real valued measure μ_ψ such that,

$$(\psi, R_A(z)\psi) = \int_{\mathbb{R}} \frac{1}{\lambda - z} \mu_\psi(d\lambda)$$

Now, we can use the Steiltje's inversion formula to reconstruct the measure μ_ψ . Then using the polarisation identity, we can construct a complex valued measure as follows,

$$\forall \psi, \phi \in \mathcal{H} : \mu_{\psi, \phi} := \frac{1}{4}(\mu_{\psi+\phi} - \mu_{\psi-\phi} + i\mu_{\psi-i\phi} - i\mu_{\psi+i\phi})$$

Using this, we define a sesquilinear form,

$$s_\Omega(\psi, \phi) := \int_{\mathbb{R}} \chi_\Omega(\lambda) \mu_{\psi, \phi}(d\lambda)$$

Lemma 4.1.10. *If s is a sesquilinear form with,*

$$|s(\psi, \phi)| \leq C \|\psi\| \|\phi\|$$

for some $C \geq 0$, then $\exists! A \in \mathcal{L}(\mathcal{H}) : s(\psi, \phi) = (A\psi, \phi)$

Proof. Define a linear functional,

$$\ell(\cdot) := s(\psi, \cdot)$$

For a fixed $\psi \in \mathcal{H}$. Since its bounded, Riesz theorem tells us that,

$$\exists! \chi \in \mathcal{H} : \ell(\phi) = s(\psi, \phi) = (\chi, \phi)$$

Define, $A\psi := \chi$. To show the linearity of A ,

$$\begin{aligned} (A(\alpha\psi_1 + \psi_2), \phi) &= s(\alpha\psi_1 + \psi_2, \phi) \\ &= \alpha^* s(\psi_1, \phi) + s(\psi_2, \phi) \\ &= \alpha^* (A\psi_1, \phi) + (A\psi_2, \phi) \\ &= (\alpha A\psi_1 + A\psi_2, \phi) \end{aligned}$$

Now,

$$\begin{aligned} \|A\| &= \sup_{\psi \in \mathcal{H}} \frac{\|A\psi\|}{\|\psi\|} \\ &= \sup_{\psi \in \mathcal{H}} \sup_{A\psi \in \mathcal{H}} \frac{|(A\psi, A\psi)|}{\|\psi\| \cdot \|A\psi\|} \\ &\leq \sup_{\psi \in \mathcal{H}} \sup_{\phi \in \mathcal{H}} \frac{|(A\psi, \phi)|}{\|\psi\| \cdot \|\phi\|} \\ &\leq C \end{aligned}$$

■

Now,

$$\begin{aligned}
|s_\Omega(\psi, \phi)| &\leq |s_\Omega(\psi, \psi)|^{\frac{1}{2}} |s_\Omega(\phi, \phi)|^{\frac{1}{2}} \\
&= (\mu_\psi(\Omega))^{\frac{1}{2}} (\mu_\phi(\Omega))^{\frac{1}{2}} \\
&\leq (\mu_\psi(\mathbb{R}))^{\frac{1}{2}} (\mu_\phi(\mathbb{R}))^{\frac{1}{2}} \\
&= \|\psi\| \|\phi\|
\end{aligned}$$

Hence for all $\Omega \in \sigma(\mathcal{O}_\mathbb{R})$ we have a unique $P_A(\Omega)$, a bounded operator such that,

$$s_\Omega(\psi, \phi) = \int_{\mathbb{R}} \chi_\Omega(\lambda) \mu_{\psi, \phi}(d\lambda) = (\psi, P_A(\Omega)\phi)$$

Also, notice that,

$$\forall \psi \in \mathcal{H} : 0 \leq (\psi, P_A(\Omega)\psi) \leq 1$$

Lemma 4.1.11. $P_A(\Omega)$ is a PVM

Proof. To prove self adjointness,

$$(\psi, P_A(\Omega)\phi) = s_\Omega(\psi, \phi) = (s_\Omega(\phi, \psi))^* = (\phi, P_A(\Omega)\psi)^* = (P_A(\Omega)\psi, \phi)$$

Now to prove the rest of the properties, we first do,

$$\begin{aligned}
\int_{\mathbb{R}} \frac{1}{\lambda - \tilde{z}} \mu_{R_A(z^*)\psi, \phi}(d\lambda) &= (R_A(z^*)\psi, R_A(\tilde{z})\phi) \\
&= (\psi, R_A(z)R_A(\tilde{z})\phi) \\
&= \frac{1}{z - \tilde{z}} [(\psi, R_A(z)\phi) - (\psi, R_A(\tilde{z})\phi)] \\
&= \frac{1}{z - \tilde{z}} \int_{\mathbb{R}} \left(\frac{1}{\lambda - z} - \frac{1}{\lambda - \tilde{z}} \right) \mu_{\psi, \phi}(d\lambda) \\
&= \int_{\mathbb{R}} \frac{1}{\lambda - \tilde{z}} \frac{\mu_{\psi, \phi}(d\lambda)}{\lambda - z}
\end{aligned}$$

Hence,

$$\mu_{R_A(z^*)\psi, \phi} = \frac{\mu_{\psi, \phi}(d\lambda)}{\lambda - z}$$

Now,

$$\begin{aligned}
\int_{\mathbb{R}} \frac{1}{\lambda - z} \mu_{\psi, P_A(\Omega)\phi}(d\lambda) &= (\psi, R_A(z)P_A(\Omega)\phi) \\
&= (R_A(z^*)\psi, P_A(\Omega)\phi) \\
&= \int_{\mathbb{R}} \chi_\Omega(\lambda) \mu_{R_A(z^*)\psi, \phi}(d\lambda) \\
&= \int_{\mathbb{R}} \chi_\Omega(\lambda) \frac{\mu_{\psi, \phi}(d\lambda)}{\lambda - z}
\end{aligned}$$

Hence,

$$\mu_{\psi, P_A(\Omega)\phi} = \chi_{\Omega} \mu_{\psi, \phi}$$

Or,

$$(\psi, P_A(\Omega)\phi) = (\psi, \chi_{\Omega} \phi)$$

Hence,

$$\begin{aligned} (\psi, P_A(\Omega_1 \cap \Omega_2)\phi) &= (\psi, \chi_{\Omega_1 \cap \Omega_2} \phi) \\ &= (\psi, \chi_{\Omega_1} \chi_{\Omega_2} \phi) \\ &= (\psi, P_A(\Omega_1)(\chi_{\Omega_2} \phi)) \\ &= (\psi, P_A(\Omega_1) \circ P_A(\Omega_2)\phi) \end{aligned}$$

Hence,

$$P_A(\Omega_1 \cap \Omega_2) = P_A(\Omega_1) \circ P_A(\Omega_2)$$

Idempotency is immediate once we choose $\Omega_1 = \Omega_2$. Since we already have idempotency, to show $P_A(\mathbb{R}) = id_{\mathcal{H}}$, we just need $\ker(P_A(\mathbb{R})) = \{0\}$. Take $\psi \in \ker(P_A(\mathbb{R}))$. Then,

$$0 = (\psi, P_A(\mathbb{R})\psi) = \mu_{\psi}(\mathbb{R}) = \|\psi\|^2 \Rightarrow \psi = 0$$

Now let $\{\Omega_n\}$ be disjoint.

$$\begin{aligned} \sum_{i=1}^{\infty} (\psi, P_A(\Omega_i)\psi) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (\psi, P_A(\Omega_i)\psi) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu_{\psi}(\Omega_i) \\ &= \lim_{n \rightarrow \infty} \mu_{\psi}(\cup_{i=1}^n \Omega_i) \\ &= \mu_{\psi}(\cup_{i=1}^{\infty} \Omega_i) \\ &= (\psi, P_A(\cup_{i=1}^{\infty} \Omega_i)\psi) \end{aligned}$$

Hence the proof! ■

The Spectral Theorem

Now we have everything in place to prove the spectral theorem.

Theorem 4.1.12. Spectral Theorem: For every self adjoint operator A , there is a unique projection valued measure P_A such that,

$$A = \int_{\mathbb{R}} id_{\mathbb{R}} dP_A$$

Proof. We already have proven everything. But lets outline the proof in the correct order here. Since A is self adjoint, we have $\forall \psi \in \mathcal{H}$, $(\psi, R_A(\cdot)\psi)$ a Herglotz function. Hence there exists a real valued measure μ_{ψ} such that

$$(\psi, R_A(z)\psi) = \int_{\mathbb{R}} \frac{1}{\lambda - z} \mu_{\psi}(d\lambda)$$

Using Steiltje's inversion, reconstruct the μ_{ψ} from the above equation. Use this to construct a complex valued measure via the polarisation identity. Then, there exists a unique PVM via the construction,

$$\int_{\mathbb{R}} \chi_{\Omega}(\lambda) \mu_{\psi, \phi}(d\lambda) = (\psi, P_A(\Omega)\phi)$$

$$A_{P_A} = \int_{\mathbb{R}} id_{\mathbb{R}} dP_A$$

is self adjoint. Since A and A_{P_A} has exactly the same PVM, they are essentially equal. ■

Example 4.1.11.1. $A = id_{\mathcal{H}}$. The resolvent set is,

$$\rho(id_{\mathcal{H}}) = \mathbb{C} \setminus \{1\}$$

Hence,

$$\text{Im}(\psi, R_A(z)\psi) = \text{Im}\left(\psi, \frac{id_{\mathcal{H}}}{1-z}\psi\right) = \frac{\text{Im}(z)}{|1-z|^2} \|\psi\|^2$$

Then using Steiltje's inversion,

$$\mu_{\psi}((-\infty, \lambda]) = \lim_{\delta \rightarrow 0^+} \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \int_{-\infty}^{\lambda+\delta} dt \frac{\epsilon}{(1-t)^2 + \epsilon^2} \|\psi\|^2 = \|\psi\|^2 \chi_{(-\infty, \lambda]}$$

This gives,

$$\mu_{\psi, \phi}(\Omega) = (\psi, \phi) \chi_{\Omega}$$

Hence,

$$(\psi, P_{id_{\mathcal{H}}}(\Omega)\phi) = \chi_{\Omega}(\psi, \phi)$$

Which tells us,

$$P_{id_{\mathcal{H}}}(\Omega) = id_{\mathcal{H}} \chi_{\Omega}$$

Another Version of the Spectral Theorem

The spectral theorem is usually stated in a different way. To understand that, we first have to do some background work.

Orthogonal Sums of Operators

Let $\mathcal{H}_1, \mathcal{H}_2$ be two Hilbert spaces and $A_{1,2} : D(A_{1,2}) \rightarrow \mathcal{H}_{1,2}$ be two operators. Then we define,

$$A := A_1 \oplus A_2 \quad \text{with} \quad D(A) := D(A_1) \oplus D(A_2)$$

such that,

$$A(\psi_1 + \psi_2) := A_1\psi_1 + A_2\psi_2$$

Clearly, only if both A_1 and A_2 are closed, essentially self adjoint, etc. is A so. We can extend this definition to countable sums. Now, let $\mathcal{H}_1 \subseteq \mathcal{H}$ be a closed subspace, and P_1 the projector to $\mathcal{H}_1$.

Definition 4.1.12. $\mathcal{H}_1$ is said to reduce an operator A if $P_1 D(A) \subseteq D(A)$ and $\forall \psi \in D(A) : P_1 A\psi = AP_1\psi$

Remark. Choose $\mathcal{H}_2 = \mathcal{H}_1^\perp$. Then $P_2 = id_{\mathcal{H}} - P_1$ also reduces A

Lemma 4.1.13. Suppose $\mathcal{H} = \bigoplus_j \mathcal{H}_j$ where all $\mathcal{H}_j$ reduced A . Then $A = \bigoplus_j A_j$ where $A_j\psi = A\psi$ for all $\psi \in P_j D(A)$

Proof. Since each $\mathcal{H}_j$ reduce A , we have $P_j D(A) \subseteq D(A)$. Also since $\mathcal{H} = \bigoplus_j \mathcal{H}_j$, we can write,

$$\forall \psi \in D(A) : \psi = \sum_j P_j \psi$$

Since the projectors add up to $id_{\mathcal{H}}$, we have

$$D(A) = \bigoplus_j P_j D(A)$$

Now for any $\psi \in P_j D(A)$, we have $\psi = P_j \psi$. Hence,

$$A\psi = AP_j\psi = P_j A\psi \in \mathcal{H}_j$$

Hence define,

$$A_j : P_j D(A) \rightarrow \mathcal{H}_j$$

as,

$$A_j\psi := P_j A\psi$$

This by definition satisfy $A_j\psi = A\psi$ for all $\psi \in P_j D(A)$ since A is reduced by $\mathcal{H}_j$. Hence we are done. $\blacksquare$

Now for any $\psi \in D(f)$, define,

Definition 4.1.13. $\mathcal{H}_\psi := \left\{ \left(\int_{\mathbb{R}} g dP \right) \psi \mid g \in L^2(\mathbb{R}, \mu_\psi) \right\}$

$\mathcal{H}_\psi$ is closed since $L^2(\mathbb{R}, \mu_\psi)$ is closed.

Lemma 4.1.14. $\mathcal{H}_\psi$ reduces $\int_{\mathbb{R}} f dP$ for any f . i.e,

$$P_\psi \circ \left(\int_{\mathbb{R}} f dP \right) \subseteq \left(\int_{\mathbb{R}} f dP \right) \circ P_\psi$$

where P_ψ is the projection to $\mathcal{H}_\psi$

Proof. First, choose f to be bounded. Then for any $\phi \in \mathcal{H}$, we can write,

$$\phi = \left(\int_{\mathbb{R}} g dP \right) \psi + \phi^\perp$$

Where $\phi^\perp \perp \mathcal{H}_\psi$. We can do so because $\mathcal{H}_\psi$ is closed. Now for any $h \in B(\mathbb{R})$, we have,

$$\left(\left(\int_{\mathbb{R}} h dP \right) \psi, \left(\int_{\mathbb{R}} f dP \right) \phi^\perp \right) = \left(\left(\int_{\mathbb{R}} f^* h dP \right) \psi, \phi^\perp \right) = 0$$

This result can be extended to unbounded h also. Hence $\left(\int_{\mathbb{R}} f dP \right) \phi^\perp \in \mathcal{H}_\psi^\perp$. Therefore,

$$\begin{aligned} P_\psi \circ \left(\int_{\mathbb{R}} f dP \right) \phi &= P_\psi \circ \left(\int_{\mathbb{R}} f dP \right) \circ \left[\left(\int_{\mathbb{R}} g dP \right) \psi + \phi^\perp \right] \\ &= P_\psi \circ \left(\int_{\mathbb{R}} f \cdot g dP \right) \psi + 0 \\ &= \left(\int_{\mathbb{R}} f \cdot g dP \right) \psi \end{aligned}$$

Likewise,

$$\left(\int_{\mathbb{R}} f dP \right) \circ P_\psi \phi = \left(\int_{\mathbb{R}} f dP \right) \circ P_\psi \left[\left(\int_{\mathbb{R}} g dP \right) \psi + \phi^\perp \right] = \left(\int_{\mathbb{R}} f \cdot g dP \right) \psi$$

Hence we have proved the result for bounded functions. For unbounded f , use $f_n := \chi_{\Omega_n} f$ as we defined previously. Then,

$$\forall \phi \in \mathcal{H} : P_\psi \circ \left(\int_{\mathbb{R}} f_n dP \right) \phi = \left(\int_{\mathbb{R}} f_n dP \right) \circ P_\psi \phi$$

Write $\int_{\mathbb{R}} f_n dP = \left(\int_{\mathbb{R}} f dP \right) \circ P(\Omega_n)$. Then take limits. The limit for LHS exists only when $\phi \in D(f)$. In that case, $P(\Omega_n)P_\psi \phi \rightarrow P_\psi \phi \in D(f)$, because $P(\Omega_n)P_\psi \phi \in D(f)$ (we have seen this while proving the density of $D(f)$) and $D(f)$ is closed $((\int_{\mathbb{R}} f dP)^{**} = (\int_{\mathbb{R}} f^{**} dP) = (\int_{\mathbb{R}} f dP))$. Hence,

$$\forall \phi \in D(f) : P_\psi \circ \left(\int_{\mathbb{R}} f dP \right) \phi = \left(\int_{\mathbb{R}} f dP \right) \circ P_\psi \phi$$

Since the domain of RHS is $\{\phi \in \mathcal{H} \mid P_\psi \phi \in D(f)\} \supseteq D(f)$, we have our result. ■

This means that we can decompose $\int_{\mathbb{R}} f dP$ as,

$$\int_{\mathbb{R}} f dP = \int_{\mathbb{R}} f dP|_{\mathcal{H}_\psi} + \int_{\mathbb{R}} f dP|_{\mathcal{H}_\psi^\perp}$$

Notice that $P_\psi D(f) = D(f) \cap \mathcal{H}_\psi$, i.e., $\phi \in P_\psi D(f)$ if it can be written as $(\int_{\mathbb{R}} g dP)\psi$ and $\int_{\mathbb{R}} |f|^2 d\mu_{(\int_{\mathbb{R}} g dP)\psi} < \infty$. Now, for a bounded g ,

$$\begin{aligned} \int_{\mathbb{R}} |f|^2 d\mu_{(\int_{\mathbb{R}} g dP)\psi} &= \left(\left(\int_{\mathbb{R}} g dP \right) \psi, \left(\int_{\mathbb{R}} |f|^2 dP \right) \left(\int_{\mathbb{R}} g dP \right) \psi \right) \\ &= \left(\psi, \left(\int_{\mathbb{R}} |fg|^2 dP \right) \psi \right) \\ &= \int_{\mathbb{R}} |fg|^2 d\mu_\psi \end{aligned}$$

Hence,

$$P_\psi D(f) = \left\{ \left(\int_{\mathbb{R}} g dP \right) \psi \mid g, fg \in L^2(\mathbb{R}, \mu_\psi) \right\}$$

Also, in this domain,

$$\left(\int_{\mathbb{R}} f dP \right) \circ \left(\int_{\mathbb{R}} g dP \right) = \int_{\mathbb{R}} fg dP$$

Now define,

$$\mathcal{U}_\psi : \mathcal{H}_\psi \rightarrow L^2(\mathbb{R}, \mu_\psi)$$

as,

$$\mathcal{U}_\psi \left(\left(\int_{\mathbb{R}} g dP \right) \psi \right) := g$$

Now,

$$\|\mathcal{U}_\psi\| = \sup_{\mathcal{H}_\psi} \frac{\|\mathcal{U}_\psi((\int_{\mathbb{R}} g dP)\psi)\|}{\|(\int_{\mathbb{R}} g dP)\psi\|} = \sup_{\mathcal{H}_\psi} \frac{\|f\|_{L^2}}{(\int_{\mathbb{R}} |f|^2 d\mu_\psi)^{1/2}} = 1$$

The map $\mathcal{U}_\psi$ is a bijection as $\mathcal{U}_\psi^{-1}g = (\int_{\mathbb{R}} g dP)\psi$ is uniquely defined. Hence $\mathcal{U}_\psi$ is a unitary map. Now,

$$\begin{aligned} \mathcal{U}_\psi \circ \int_{\mathbb{R}} f dP \Big|_{\mathcal{H}_\psi} \left(\left(\int_{\mathbb{R}} g dP \right) \psi \right) &= \mathcal{U}_\psi \left(\left(\int_{\mathbb{R}} f g dP \right) \psi \right) \\ &= f g \\ &= f \mathcal{U}_\psi \left(\left(\int_{\mathbb{R}} g dP \right) \psi \right) \end{aligned}$$

Hence,

$$\mathcal{U}_\psi \circ \int_{\mathbb{R}} f dP \Big|_{\mathcal{H}_\psi} = f \mathcal{U}_\psi \quad \text{or} \quad \mathcal{U}_\psi \circ \int_{\mathbb{R}} f dP \Big|_{\mathcal{H}_\psi} \circ \mathcal{U}_\psi^{-1} = f$$

The Spectral Decomposition of a PVM

Definition 4.1.14. ψ is called a cyclic vector if $\mathcal{H}_\psi = \mathcal{H}$

Definition 4.1.15. $\{\psi_j \in \mathcal{H} \mid j \in J\}$ is called a set of spectral vectors if $\forall j \in J : \|\psi_j\| = 1$ and $\mathcal{H}_{\psi_j} \perp \mathcal{H}_{\psi_k} \forall j \neq k$.

Definition 4.1.16. $\{\psi_j \in \mathcal{H} \mid j \in J\}$ is called a spectral basis if it is a spectral set with $\mathcal{H} = \bigoplus_{j \in J} \mathcal{H}_{\psi_j}$

Theorem 4.1.15. For a PVM, P , there is always a countable spectral basis.

Proof. Let $\{\tilde{\psi}_j \in \mathcal{H} \mid j \in J\}$ be a spectral set. Take $\psi \in \mathcal{H}$ such that $\psi \perp \mathcal{H}_{\psi_j}$ for all $j \in J$. Then for bounded f, g ,

$$\left(\left(\int_{\mathbb{R}} g dP \right) \psi_j, \left(\int_{\mathbb{R}} f dP \right) \psi \right) = \left(\left(\int_{\mathbb{R}} f^* g dP \right) \psi_j, \psi \right) = 0$$

This result is easily extended for all f, g . Hence $(\int_{\mathbb{R}} f dP)\psi \perp \mathcal{H}_{\psi_j}$ or $\mathcal{H}_\psi \perp \mathcal{H}_{\psi_j}$. Using this procedure, construct the maximal spectral set. Then choose $\psi_1 = \tilde{\psi}_1$ and normalise it. Project $\tilde{\psi}_2$ to $\mathcal{H}_{\psi_1}^\perp$ and normalise it to get ψ_2 and so on. Hence,

$$\text{span} \left(\left\{ \psi_j \in \mathcal{H} \mid j \in J \right\} \right) \subseteq \bigoplus_{j \in J} \mathcal{H}_{\psi_j}$$

and since we have not left out any vector in the construction, we have at least,

$$\mathcal{H} = \overline{\text{span}\left(\left\{\psi_j \in \mathcal{H} \mid j \in J\right\}\right)} \subseteq \bigoplus_{j \in J} \overline{\mathcal{H}_{\psi_j}} = \bigoplus_{j \in J} \mathcal{H}_{\psi_j}$$

The basis must be countable since we are dealing with separable Hilbert spaces. ■

Remark. The cardinality of the spectral basis is not well defined.

Theorem 4.1.16. Spectral Theorem: *Any self adjoint operator A can be diagonalised.*

Proof. By the first version of the theorem, we have $A = \int_{\mathbb{R}} id_{\mathbb{R}} dP_A$. The PVM P_A has a countable spectral basis. We have already shown that, there is $\mathcal{U}_{\psi}$ for each ψ in the spectral basis such that,

$$\mathcal{U}_{\psi} \circ \left(\int_{\mathbb{R}} id_{\mathbb{R}} dP_A \right) \Big|_{\mathcal{H}_{\psi}} = id_{\mathbb{R}} \mathcal{U}_{\psi} \quad \text{or} \quad \mathcal{U}_{\psi} \circ \left(\int_{\mathbb{R}} id_{\mathbb{R}} dP_A \right) \Big|_{\mathcal{H}_{\psi}} \circ \mathcal{U}_{\psi}^{-1} = id_{\mathbb{R}}$$

Now this tells us that,

$$\mathcal{U}_{\psi} \circ A \Big|_{\mathcal{H}_{\psi}} \circ \mathcal{U}_{\psi}^{-1} = id_{\mathbb{R}}$$

Hence A is proportional to the identity in each of the subspaces generated by elements from the spectral basis. The proportionality constant maybe different from subspace to subspace. This along with the fact that a spectral basis completely span the entire Hilbert space tells us that A is diagonalisable. ■

Example 4.1.16.1. For the Harmonic Oscillator, it can be shown¹ that the spectrum is

$$\sigma(H) = \left\{ \omega\left(n + \frac{1}{2}\right) \mid n \in \mathbb{N}_0 \right\}$$

all of the spectrum consists of eigenvalues and the corresponding eigenvectors are orthonormal,

$$(\psi_n, \psi_m) = \delta_{mn}$$

Now define a projection,

$$P_n(\cdot) := (\psi_n, \cdot) \psi_n$$

¹shown in chapter 5

This operator is bounded. Hence we can use the operator norm in $\mathcal{L}(\mathcal{H})$ to decide on the convergence of sequences of these projections.

$$\sum_{n=0}^{\infty} P_n = id_{\mathcal{H}}$$

The proof is straightforward because $\{\psi_n \mid n \in \mathbb{N}_0\}$ is a complete system. Now define, for $\Omega \in \sigma(\mathcal{O}_{\mathbb{R}})$

$$P_H(\Omega) = \sum_{n \text{ for } E_n \in \Omega} P_n$$

This is the PVM for the Hamiltonian of the Harmonic oscillator since $\{\psi_n \mid n \in \mathbb{N}_0\}$ is a spectral basis.

$$H = \int_{\mathbb{R}} \lambda P_H(d\lambda) = \sum_{n=0}^{\infty} E_n P_n$$

4.2 Some Applications of the Spectral Theorem

[1, 2]

Spectrum of a Self Adjoint Operator

Lemma 4.2.1. *If there is a sequence $\psi_n \in D(A)$ such that $\|\psi_n\| = 1$ and $\|(A - z)\psi_n\| \rightarrow 0$, then $z \in \sigma(A)$,*

Remark. Such a sequence is called a Weyl sequence

Proof. If $z \in \rho(A)$, then

$$1 = \|\psi_n\| = \|R_A(z)(A - z)\psi_n\| \leq \|R_A(z)\| \cdot \|(A - z)\psi_n\| \rightarrow 0$$

which is a contradiction, and hence the proof. ■

Theorem 4.2.2. *For a self adjoint A , the spectrum is given by,*

$$\sigma(A) = \left\{ \lambda \in \mathbb{R} \mid \forall \epsilon > 0 : P_A(\lambda - \epsilon, \lambda + \epsilon) \neq 0 \right\}$$

Proof. Let $\lambda_0 \in \text{RHS}$. Then there is a $\Omega_n = (\lambda_0 - \frac{1}{n}, \lambda_0 + \frac{1}{n})$ and $P(\Omega_n) \neq 0$. For $\psi \in D(A)$, define $\psi_n := P(\Omega_n)\psi$ with $\|\psi_n\| = 1$. Then,

$$\|(A - \lambda_0)\psi_n\|^2 = \|(A - \lambda_0)P(\Omega_n)\psi\|^2 = \int_{\mathbb{R}} (\lambda - \lambda_0)^2 \chi_{\Omega_n}(\lambda) \mu_{\psi_n}(d\lambda) \leq \frac{1}{n^2}$$

Hence the above sequence is a Weyl sequence, which gives us $\lambda_0 \in \sigma(A)$. Now to prove the other inclusion, we prove the contrapositive. If $P(\lambda_0 - \epsilon, \lambda_0 + \epsilon) = 0$, define,

$$f_\epsilon(\lambda) := \chi_{\mathbb{R} \setminus (\lambda_0 - \epsilon, \lambda_0 + \epsilon)}(\lambda)(\lambda - \lambda_0)^{-1}$$

Now,

$$\begin{aligned} (A - \lambda_0) \int_{\mathbb{R}} f_\epsilon dP &= \left(\int_{\mathbb{R}} \lambda P(d\lambda) - \int_{\mathbb{R}} \lambda_0 \chi_{\mathbb{R}} P(d\lambda) \right) \circ \int_{\mathbb{R}} f_\epsilon(\lambda) P(d\lambda) \\ &= \int_{\mathbb{R}} (\lambda - \lambda_0) f_\epsilon(\lambda) P(d\lambda) \\ &= \int_{\mathbb{R}} \chi_{\mathbb{R} \setminus (\lambda_0 - \epsilon, \lambda_0 + \epsilon)} P(d\lambda) \\ &= P(\chi_{\mathbb{R} \setminus (\lambda_0 - \epsilon, \lambda_0 + \epsilon)}) \\ &= id_{\mathcal{H}} \end{aligned}$$

Likewise,

$$\left(\int_{\mathbb{R}} f_\epsilon dP \right) \circ (A - \lambda_0) = id_{D(A)}$$

Hence $A - \lambda_0$ is invertible, which gives $\lambda_0 \notin \sigma(A)$ ■

Commuting Operators

First start with two bounded operators A and B .

Definition 4.2.1. A and B are said to commute if,

$$[A, B] := A \circ B - B \circ A = 0$$

However, if either of the operators is unbounded, then the above definition may end up giving us zero information. We illustrate this using the example below.

Example 4.2.1.1. Let $A : D(A) \rightarrow \mathcal{H}$ be unbounded. Then define, for some fixed $\phi \in \mathcal{H}$,

$$B_\phi(\cdot) := (\phi, \cdot) \psi$$

where $\psi \notin D(A)$ is also fixed. B_ϕ is a bounded operator. If we define the commutator as above, then,

$$D([B_\phi, A]) = \text{ran } B_\phi \cap D(A) = \{0\}$$

We have to use a bounded operator that can be uniquely constructed from an unbounded operator to resolve this issue.

Definition 4.2.2. If A is self adjoint and B is bounded, A, B are said to commute if

$$\forall \Omega \in \sigma(\mathcal{O}_{\mathbb{R}}) : [P_A(\Omega), B] = 0$$

Lemma 4.2.3. If A is self adjoint and B is bounded such that they commute, then $[f(A), B] = 0$ for any $f \in B(\mathbb{R})$ and vice versa.

Proof. For simple f ,

$$f(A) = \sum_{n=1}^N f_n P_A(\Omega_n)$$

Then $f(A)B = Bf(A)$ is trivial. Now, $\text{Sim}(\mathbb{R}) \subseteq B(\mathbb{R})$ is dense, which gives us the only if part.

Now if $[f(A), B] = 0$ for any $f \in B(\mathbb{R})$, then it is true for $f = \chi_{\Omega}$ for any $\Omega \in \sigma(\mathcal{O}_{\mathbb{R}})$, which gives us the if part. ■

This lemma gives us equivalent definitions for commutativity used in the literature.

Definition 4.2.3. A self adjoint and B bounded are said to commute if,

1. $\forall \Omega \in \sigma(\mathcal{O}_{\mathbb{R}}) : [P_A(\Omega), B] = 0$
2. for some $z \in \rho(A) : [R_A(z), B] = 0$
3. for some $t \in \mathbb{R} : [e^{-itA}, B] = 0$

4.3 The Stone's Theorem [1, 2]

Stone's theorem is responsible for the unique definition of self adjoint observables. It tells us how to construct a self adjoint operator from certain symmetries.

Definition 4.3.1. A one parameter group is a group whose underlying set is parameterised by a real parameter,

$$G := \left\{ u(t) \middle| t \in \mathbb{R} \right\}$$

and,

$$u(t) \circ u(s) = u(f(t, s))$$

where f is some real valued function.

Remark. If the group is abelian, then the function f is symmetric. In that case, it is always possible to reparameterise the group to get $f(t, s) = t + s$, and $u(0) = 1$.

We are interested in Unitary, abelian one parameter groups(UAOPG), which are strongly continuous. Which means that,

$$G := \left\{ u(t) \mid t \in \mathbb{R}, u(t)^* u(t) = id_{\mathcal{H}} \right\}$$

with,

$$\forall \psi \in \mathcal{H} : \lim_{t \rightarrow t_0} u(t)\psi = u(t_0)\psi$$

Definition 4.3.2. Let $u(\cdot) : \mathbb{R} \rightarrow \mathcal{L}(\mathcal{H})$ be a UAOPG. Then a linear map, $A : D^s(A) \rightarrow \mathcal{H}$ defined by,

$$A\psi := \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} [u(\epsilon)\psi - \psi] \equiv i[u(\cdot)\psi]'(0)$$

where

$$D^s(A) := \left\{ \psi \in \mathcal{H} \mid \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} [u(\epsilon)\psi - \psi] \text{ exists} \right\}$$

is called the generator of the UAOPG. The domain is called the Stone domain.

Lemma 4.3.1. For a self adjoint A , $u(t) := e^{-itA}$ is UAOPG

Proof.

$$\begin{aligned} u(t) \circ u(s) &= e^{-itA} \circ e^{-isA} \\ &= \int_{\mathbb{R}} e^{-it\lambda} P_A(d\lambda) \circ \int_{\mathbb{R}} e^{-is\lambda} P_A(d\lambda) \\ &= \int_{\mathbb{R}} e^{-i(t+s)\lambda} P_A(d\lambda) \\ &= e^{-i(t+s)A} \\ &= u(t+s) \end{aligned}$$

and,

$$u(0) = e^0 = P_A(\mathbb{R}) = id_{\mathcal{H}}$$

Now,

$$u^*(t) = \left(\int_{\mathbb{R}} e^{-it\lambda} P_A(d\lambda) \right)^* = \int_{\mathbb{R}} e^{+it\lambda} P_A(d\lambda) = u(-t)$$

Hence,

$$u^*(t) \circ u(t) = id_{\mathcal{H}}$$

$$\|u^*(t)u(t)\| = 1 \Rightarrow \|u(t)\|^2 = 1 \Rightarrow \|u(t)\| = 1$$

■

Theorem 4.3.2. Stone's Theorem: Let $u(\cdot) : \mathbb{R} \rightarrow \mathcal{L}(\mathcal{H})$ be a UAOPG. Then its generator $A : D^s(A) \rightarrow \mathcal{H}$ is self adjoint on the stone domain and,

$$u(t) = e^{-itA}$$

Proof. First we show that the stone domain is dense. Take $\psi \in \mathcal{H}$ and define,

$$\psi_\tau := \int_0^\tau dr u(r)\psi$$

Then,

$$\left. \frac{d\psi_\tau}{d\tau} \right|_{\tau=0} = \lim_{\tau \rightarrow 0} \frac{\psi_\tau - \psi_0}{\tau} = \lim_{\tau \rightarrow 0} \frac{\psi_\tau}{\tau} = \psi$$

Where the last step is obtained via the application of L'Hospital's rule. This tells us that there is a ψ_{τ_0} in the neighbor hood of ψ . Now, what is left to show is that this ψ_{τ_0} is in the stone domain. For that, we have to show that the limit in the definition of the generator exists.

$$\begin{aligned} u(\epsilon)\psi_\tau - \psi_\tau &= u(\epsilon) \int_0^\tau dr u(r)\psi - \int_0^\tau dr u(r)\psi \\ &= \int_0^\tau dr u(r+\epsilon)\psi - \int_0^\tau dr u(r)\psi \\ &= \int_\epsilon^{\tau+\epsilon} dr u(r)\psi - \int_0^\tau dr u(r)\psi \\ &= \int_\tau^{\tau+\epsilon} dr u(r)\psi + \int_\epsilon^\tau dr u(r)\psi - \int_0^\tau dr u(r)\psi \\ &= \int_\tau^{\tau+\epsilon} dr u(r)\psi - \int_0^\epsilon dr u(r)\psi \\ &= \int_0^\epsilon dr u(r+\tau)\psi - \int_0^\epsilon dr u(r)\psi \\ &= u(\tau)\psi_\epsilon - \psi_\epsilon \end{aligned}$$

Hence,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [u(\epsilon)\psi_\tau - \psi_\tau] &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} [u(\tau) - id_{\mathcal{H}}]\psi_\epsilon \\ &= [u(\tau) - id_{\mathcal{H}}] \lim_{\epsilon \rightarrow 0} \frac{\psi_\epsilon}{\epsilon} = [u(\tau) - id_{\mathcal{H}}]\psi \in \mathcal{H} \end{aligned}$$

Hence all $\psi_\tau \in D^s(A)$. Which proves the stone domain is dense.

Next, we prove that the generator A is symmetric. Let $\psi, \phi \in D^s(A)$. Then,

$$\begin{aligned} (\psi, A\phi) &= \left(\psi, \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} [u(\epsilon)\phi - \phi] \right) \\ &= \left(\lim_{\epsilon \rightarrow 0} \frac{-i}{\epsilon} [u^*(\epsilon) - id_{\mathcal{H}}]\psi, \phi \right) \\ &= \left(\lim_{\epsilon \rightarrow 0} \frac{i}{-\epsilon} [u(-\epsilon) - id_{\mathcal{H}}]\psi, \phi \right) \\ &= (A\psi, \phi) \end{aligned}$$

Hence A is symmetric.

Now to check the essential self adjointness of A , we need to look at $\ker(A^* \mp i)$. Let $\phi \in \ker(A^* + i)$. Then we have, $A^*\phi = -i\phi$. Now for any $\psi \in D^s(A)$,

$$\begin{aligned} \frac{d}{dt} (\phi, u(t)\psi) &= (\phi, -iAu(t)\psi) \\ &= -i (A^*\phi, u(t)\psi) \\ &= -(\phi, u(t)\psi) \end{aligned}$$

Hence,

$$(\phi, u(t)\psi) = e^{-t} (\phi, \psi)$$

Since the LHS is bounded for all $t \in \mathbb{R}$, RHS must also be bounded, which is possible only if,

$$(\phi, \psi) = 0$$

Since $D^s(A)$ is dense, we must have $\phi = 0$. This concludes that $\ker(A^* + i) = \{0\}$.

Hence A is essentially self adjoint.

Now construct,

$$\tilde{u}(t) := e^{-itA^{**}}$$

Let $\psi \in D^s(A) \subseteq D(A^{**})$, and,

$$\psi(t) := (e^{-itA^{**}} - u(t))\psi$$

$$\frac{d\psi(t)}{dt} = [-itA^{**}e^{-itA^{**}} + iA^{**}u(t)]\psi = -iA^{**}\psi(t)$$

Where we have used $A^{**} = A$ in the common domain. Now,

$$\frac{d}{dt}\|\psi(t)\|^2 = 2\operatorname{Re}\{(\psi(t), \psi'(t))\} = 2\operatorname{Re}\{(\psi(t), -iA^{**}\psi(t))\} = 0$$

Hence,

$$\|\psi(t)\| = \|\psi(0)\| = 0$$

Hence, $\psi(t) = 0$, which gives us $A = A^{**}$. Thus A is self adjoint.

This concludes the proof of Stone's theorem. ■

Corollary 4.3.2.1. *Let $D \subseteq D^s(A)$ be dense and invariant under $u(t)$. Then A is essentially self adjoint on D*

Proof. Replace $D^s(A)$ with D in the second part of the proof, where we showed essential self adjointness. ■

Example 4.3.2.1. Position Operator: Define a UAOPG,

$$u^j(\cdot) : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$$

as,

$$(u^j(t)\psi)(x) := \psi(x)e^{-itx^j}$$

Notice this is translation in the momentum space in the j th direction. The corresponding generators are,

$$Q^j : D^s(Q^j) \rightarrow L^2(\mathbb{R}^3)$$

$$\begin{aligned} (Q^j\psi)(x) &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (u^j(\epsilon)\psi - \psi)(x) \\ &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (e^{-i\epsilon x^j} - 1)\psi(x) \\ &= x^j\psi(x) \end{aligned}$$

Example 4.3.2.2. Momentum Operator: Define a UAOPG,

$$u_j(\cdot) : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$$

as,

$$(u_j(t)\psi)(x) := \psi(x^1, \dots, x^j - t, \dots)$$

This is a translation in the position space. The generator,

$$P_j j : D^s(P_j) \rightarrow L^2(\mathbb{R}^3)$$

$$\begin{aligned} (P_j \psi)(x) &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (u^j(\epsilon) \psi - \psi)(x) \\ &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (\psi(x^1, \dots, x^j - \epsilon, \dots) - \psi(x^1, \dots, x^j, \dots)) \\ &= -i \partial_j \psi(x) \end{aligned}$$

Example 4.3.2.3. Angular Momentum Operator: Define a UAOPG,

$$u_j(\cdot) : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}^3)$$

as,

$$(u_j(t) \psi)(x) := \psi(D_j(\epsilon)x)$$

Where $D_j(t)$ is the rotation about the j th axis by an angle ϵ . For infinitesimal rotation about say, the 3 axis, the generator is,

$$\begin{aligned} (L_3 \psi)(x) &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (\psi(\cos \epsilon x^1 + \sin \epsilon x^2, -\sin \epsilon x^1 + \cos \epsilon x^2, x^3) - \psi(x^1, x^2, x^3)) \\ &= \lim_{\epsilon \rightarrow 0} \frac{i}{\epsilon} (\psi(x^1 + \epsilon x^2, -\epsilon x^1 + x^2, x^3) - \psi(x^1, x^2, x^3)) \\ &= -i(x^1 \partial_2 \psi(x) - x^2 \partial_1 \psi(x)) \end{aligned}$$

4.4 The Schwartz Space

When we do quantum mechanics, we use more than one operator at a time to do calculations. But there is a major problem in that case. Which is,

$$D^s(Q) \neq D^s(P) \neq D^s(L)$$

Even worse,

$$D^s(Q) \neq D^s(Q^2) \neq D^s(Q^3) \dots$$

To resolve this issue, we introduce the Schwartz space.

Definition 4.4.1. The Schwartz space on $\mathbb{R}^d$, denoted by $S(\mathbb{R}^d)$ is,

$$S(\mathbb{R}^d) := \left\{ \psi \in C^\infty(\mathbb{R}^d) \mid \sup_{x \in \mathbb{R}^d} |x^\alpha \partial_\beta \psi(x)| < \infty, \forall \alpha, \beta \in \mathbb{N}_0^{\times d} \right\}$$

Here,

$$\mathbb{N}_0^{\times d} = \mathbb{N}_0 \times \mathbb{N}_0 \times \cdots \times \mathbb{N}_0 \quad d \text{ times}$$

and,

$$x^\alpha = (x^1)^{\alpha_1} (x^2)^{\alpha_2} \cdots (x^d)^{\alpha_d}$$

$$\partial_\beta = (\partial_1)^{\beta_1} (\partial_2)^{\beta_2} \cdots (\partial_d)^{\beta_d}$$

The Schwartz space is a vector space.

Lemma 4.4.1. *If $\psi, \phi \in S(\mathbb{R}^d)$, then $\psi\phi \in S(\mathbb{R}^d)$*

Proof.

$$\begin{aligned} \sup_{x \in \mathbb{R}^d} |x^\alpha \partial_\beta(\psi\phi)| &= \sup_{x \in \mathbb{R}^d} |x^\alpha (\partial_\beta \psi)(x) \phi(x) + x^\alpha (\partial_\beta \phi)(x) \psi(x)| \\ &\leq \sup_{x \in \mathbb{R}^d} |x^\alpha (\partial_\beta \psi)(x) \phi(x)| + \sup_{x \in \mathbb{R}^d} |x^\alpha (\partial_\beta \phi)(x) \psi(x)| \end{aligned}$$

■

From the definition of Schwartz space, we can see that,

$$S(\mathbb{R}^d) \subseteq D^s(Q), D^s(P)$$

Also,

$$Q : S(\mathbb{R}^d) \rightarrow S(\mathbb{R}^d) \quad \text{and} \quad P : S(\mathbb{R}^d) \rightarrow S(\mathbb{R}^d)$$

It can be shown that $S(\mathbb{R}^d)$ is dense. Hence the Schwartz space can be used as the domain while we do quantum mechanical problems that involve more than one operator!

Chapter 5

Applications

5.1 Particle in a “Box” [1, 2]

We will work out the problem of particle in a box using all the machinery we have developed till now in this section.

Momentum on a Line

Consider the interval, $I = [0, 2\pi]$. The momentum operator is given by the Stone’s theorem as,

$$P : D(P) \rightarrow L^2(I)$$

with,

$$P\psi = -i\psi'$$

We must choose $D(P)$ in such a way that the momentum is at least essentially self adjoint. Guess;

$$D(P) = \left\{ \psi \in C^1(I) \mid \psi(0) = 0 = \psi(2\pi) \right\}$$

We are not using the Schwartz space here, but a larger set to start with. First check if P is symmetric. Let $\psi, \phi \in D(P)$. Then,

$$\begin{aligned} (\psi, P\phi) &= \int_0^{2\pi} dx \psi^*(x)(-i)\phi'(x) \\ &= - \int_0^{2\pi} dx (-i)(\psi'(x))^* \phi(x) + (-i)[\psi^*(x)\phi(x)]_0^{2\pi} \\ &= (P\psi, \phi) \end{aligned}$$

Hence P is symmetric. Now we check for self adjointness. Since P is symmetric, we have $P \subseteq P^*$. Hence P^* behaves exactly as P does in the smaller domain. So we propose,

$$P^* : D(P^*) \rightarrow L^2(I)$$

with,

$$P^*\psi = -i\psi'$$

Now for $\psi \in D(P^*)$, $\exists \eta \in L^2(I) : \forall \phi \in D(P) : (\psi, P\phi) = (\eta, \phi)$. i.e,

$$\int_0^{2\pi} dx \psi^*(x)(-i)\phi'(x) = \int_0^{2\pi} dx \eta^*(x)\phi(x)$$

Let $N : I \rightarrow \mathbb{C}$ such that $N' =_{a.e.} \eta$. Then after integration by parts,

$$\int_0^{2\pi} dx \psi^*(x)(-i)\phi'(x) = - \int_0^{2\pi} dx N^*(x)\phi'(x) + [N^*(x)\phi(x)]_0^{2\pi}$$

Rearrange to get,

$$\int_0^{2\pi} dx (\psi(x) - iN(x))^* \phi'(x) = 0 \iff (\psi - iN, \phi') = 0$$

i.e, $\psi - iN \in \left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}^\perp$. To simplify things, we prove the following lemma.

Lemma 5.1.1. $\left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\} = \left\{ \xi \in C^0(I) \mid \int_0^{2\pi} dx \xi(x) = 0 \right\}$

Proof. Let $\phi' \in \left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}$. Then, $\int_0^{2\pi} dx \phi'(x) = \phi(x) \Big|_0^{2\pi} = 0$. Hence $\phi' \in \left\{ \xi \in C^0(I) \mid \int_0^{2\pi} dx \xi(x) = 0 \right\}$.

Now if $\xi \in \left\{ \xi \in C^0(I) \mid \int_0^{2\pi} dx \xi(x) = 0 \right\}$, let $\phi(x) := \int_0^x dy \xi(y)$. $\phi \in C^1(I)$ and $\phi(0) = 0 = \phi(2\pi)$ which gives $\phi \in D(P)$. Hence $\phi' \in \left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}$ ■

Lemma 5.1.2. $\overline{\left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}} = \{1\}^\perp$

Proof.

$$\begin{aligned} \overline{\left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}} &= \left\{ \xi \in \overline{C^0(I)} \mid (1, \xi) = 0 \right\} \\ &= \left\{ \xi \in L^2(I) \mid (1, \xi) = 0 \right\} \\ &= \{1\}^\perp \end{aligned}$$

■

Now, this gives us,

$$\begin{aligned} \psi - iN &\in \left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}^\perp \\ &= \overline{\left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}}^\perp \\ &= \left\{ \phi' \in C^0(I) \mid \phi \in D(P) \right\}^{\perp\perp\perp} = \overline{\{1\}} = \text{span}(\{1\}) \end{aligned}$$

Hence $\psi - iN = \text{constant}$. Which gives,

$$\psi(x) = \psi(0) + i \int_0^x dy \eta(y)$$

A function which can be written as above is called Absolutely continuous. Hence,

$\psi \in AC(I)$. Since we want the derivative of ψ also square integrable, $\psi \in \left\{ \phi \in AC(I) \mid \phi' \in L^2(I) \right\} = H^1(I)$. Hence,

$$D(P^*) = H^1(I)$$

This tells us that P is not self adjoint. Now we check for essential self adjointness.

$$\forall \psi \in D(P^{**}), \forall \phi \in D(P^*), (\psi, P^* \phi) = (P^{**} \psi, \phi)$$

But since $P \subseteq P^{**} \subseteq P^*$, we have $(\psi, P^* \phi) = (P^* \psi, \phi)$. This gives us after integrating by parts,

$$\int_0^{2\pi} dx \psi^*(x) (-i) \phi'(x) = \int_0^{2\pi} dx \psi^*(x) (-i) \phi'(x) + i [\psi(x) \phi(x)]_0^{2\pi}$$

This gives,

$$\psi(2\pi) \phi(2\pi) - \psi(0) \phi(0) = 0$$

Since there are no restriction on the boundaries of $D(P^*)$, we must choose,

$$\psi(2\pi) = 0 = \psi(0)$$

Hence, we can extend the domain of the closure to,

$$D(P^{**}) = \left\{ \psi \in H^1(I) \mid \psi(0) = 0 = \psi(2\pi) \right\}$$

Hence, P is not even essentially self adjoint!!

Particle on a Circle

For this setup, we choose the domain,

$$D(P_c) = \left\{ \psi \in C^1(I) \mid \psi(0) = \psi(2\pi) \right\}$$

It is clear that $D(P_I) \subseteq D(P_c)$. We first check for symmetry. The exact same calculation as we had for the interval will tell us that P_c is symmetric. Here the boundary terms are both equal and get cancelled instead of both being zero. To check self adjointness, we use the property, $P_I \subseteq P_c$. This gives us $P_c^* \subseteq P_I^*$. For $\psi \in D(P_c^*)$, and for any $\phi \in D(P_c)$, we have,

$$(\psi, P_c \phi) = (P_c^* \psi, \phi) = (P_I^* \psi, \phi)$$

This gives,

$$\int_0^{2\pi} dx \psi^*(x)(-i)\phi'(x) = \int_0^{2\pi} dx \psi^*(x)(-i)\phi'(x) - i[\psi^*(x)\phi(x)]_0^{2\pi}$$

which imply,

$$(\psi^*(2\pi) - \psi^*(0))\phi(0) = 0 \Rightarrow \psi(2\pi) = \psi(0)$$

$D(P_c)$ can be at most extended to $D(P_I)$, but obeying the above condition. Hence,

$$D(P_c^*) = \left\{ \psi \in H^1(I) \mid \psi(0) = \psi(2\pi) \right\}$$

Hence not self adjoint.

Now check for essential self adjointness. For $\psi \in D(P_c^{**})$, and for any $\phi \in D(P_c^*)$, we have,

$$(\psi, P_c^* \phi) = (P_c^{**} \psi, \phi)$$

Using $P_c \subseteq P_c^{**} \subseteq P_c^*$, and the same integration by parts technique,

$$(\psi^*(2\pi) - \psi^*(0))\phi(0) = 0 \Rightarrow \psi(2\pi) = \psi(0)$$

Hence,

$$D(P_c^{**}) = \left\{ \psi \in H^1(I) \mid \psi(0) = \psi(2\pi) \right\} = D(P_c^*)$$

Which tells us that P_c is essentially self adjoint!

5.2 Classifying Self Adjoint Extensions [2]

Definition 5.2.1. Let $A : D(A) \rightarrow \mathcal{H}$ be symmetric. Then a map $V : \text{ran}(A+i) \rightarrow \text{ran}(A-i)$ defined by,

$$V := (A-i)(A+i)^{-1}$$

is called the **Cayley Transform**

Definition 5.2.2. An operator $B : D(B) \rightarrow \mathcal{H}$ is called an **isometry** if $\forall \phi \in D(B) : \|B\phi\| = \|\phi\|$

Remark. An isometry is not unitary because $D(B)$ is not the entire Hilbert space.

Theorem 5.2.1. *There is a bijection between the set of all symmetric operators A and the set of all isometries V with $\text{ran}(1-V)$ is dense, such that A and V are Cayley transforms of each other.*

Proof. Let A be symmetric. We need to show that the Cayley transform V of A is an isometry and $\text{ran}(1-V)$ is dense. Since A is symmetric,

$$\forall \psi \in D(A) : \|(A \pm i)\psi\| = \|A\psi\| + \|\psi\|$$

Now for $\phi \in D(V) = \text{ran}(A+i)$, there is a $\psi \in D(A)$ such that $\phi = (A+i)\psi$. Hence,

$$\|V\phi\| = \|(A-i)\psi\| = \|A\psi\| + \|\psi\| = \|\phi\|$$

Hence, V is an isometry. Now,

$$1 \pm V = [(A+i) \pm (A-i)](A+i)^{-1} = \begin{cases} 2A(A+i)^{-1} \\ 2i(A+i)^{-1} \end{cases}$$

Now,

$$\operatorname{ran}(1 - V) = (1 - V)D(V) = 2i(A + i)^{-1}\operatorname{ran}(A + i) = D(A)$$

Hence $\operatorname{ran}(1 - V)$ is dense.

Now assume V is an isometry with $\operatorname{ran}(1 - V)$ dense. Define,

$$A_V := i(1 + V)(1 - V)^{-1}$$

with $\operatorname{ran}(1 - V)$ and $\operatorname{ran}(1 + V)$ and domain and range. The domain is dense by definition. Now,

$$\forall \phi \in D(V) : ((1 \pm V)\phi, (1 \mp V)\phi) = \|\phi\|^2 \pm (V\phi, \phi) \mp (\phi, V\phi) - \|V\phi\|^2 = \pm 2i \operatorname{Im}\{(V\phi, \phi)\}$$

For $\psi \in D(A) = \operatorname{ran}(1 - V)$, we have $\phi \in D(V) : \psi = (1 - V)\phi$. Then,

$$(A\psi, \psi) = -i((1 + V)\phi, (1 - V)\phi) = -i(\pm 2i \operatorname{Im}\{(V\phi, \phi)\}) = -i((1 - V)\phi, (1 + V)\phi) = (\psi, A\psi)$$

This gives us A is symmetric.

$$A \pm i = i[(1 + V) \pm (1 - V)](1 - V)^{-1} = \begin{cases} 2i(1 - V)^{-1} \\ 2iV(1 - V)^{-1} \end{cases}$$

Hence, V would be A 's Cayley transform ■

Corollary 5.2.1.1. *A is self adjoint iff V is unitary.*

Proof. If A is self adjoint, we have $\operatorname{ran}(A \pm i) = \mathcal{H}$. Hence V is unitary. Now if V is unitary, then $D(V) = \mathcal{H} = \operatorname{ran}(V)$, hence $\operatorname{ran}(A \pm i) = \mathcal{H}$ which gives A is self adjoint. ■

Remark. Finding the self adjoint extension of a symmetric operator A is then equivalent to finding unitary extension to the corresponding Cayley transform V . For that, we need to find a bijection from $\mathcal{H} \setminus D(V) = D(V)^\perp$ to $\mathcal{H} \setminus (\operatorname{ran}(V)) = \operatorname{ran}(V)^\perp$. This is possible only if,

$$\dim(D(V)^\perp) = \dim(\operatorname{ran} V^\perp)$$

But

$$\begin{aligned} D(V)^\perp &= \operatorname{ran}(A + i)^\perp = \ker(A^* - i) \\ \operatorname{ran}(V)^\perp &= \operatorname{ran}(A - i)^\perp = \ker(A^* + i) \end{aligned}$$

Definition 5.2.3. Let A be symmetric. Then the defect indices are defined as,

$$d_{\pm} := \dim(\ker(A^* \mp i))$$

Theorem 5.2.2. *A symmetric operator is at least one self adjoint extension (may be more than one!) if the defect indices are equal*

Proof. Developed in the last remark. ■

Theorem 5.2.3. *If the defect indices both equal zero, then the symmetric operator has exactly one self adjoint extension*

Proof. Proof was already done in a previous section. ■

Momentum on an Interval Again!

To calculate the defect indices of the momentum operator, solve for

$$(P^* \pm i)\psi = 0 \Rightarrow -i\psi'(x) \pm i\psi(x) = 0$$

The solution in the corresponding domain is,

$$\psi(x) = a_{\pm} e^{\mp x}$$

Hence $d_+ = 1 = d_-$. There might be several self adjoint extensions possible. Here,

$$\ker(P^* \pm i) = \text{span}(\{e^{\mp x}\})$$

To find the self adjoint extension, we need a unitary map from $\text{span}(\{e^{+x}\}) \rightarrow \text{span}(\{e^{-x}\})$. There are infinite maps possible,

$$V_{\theta} e^{2\pi-x} = e^{i\theta} e^x \quad \theta \in [0, 2\pi)$$

Let the corresponding self adjoint extension be P_{θ} . Then,

$$D(P_{\theta}) = \text{ran}(1 - V_{\theta}) \oplus D(P)$$

For $\psi_{\theta} \in D(P_{\theta})$, we have,

$$\psi_{\theta}(x) = \psi(x) + \alpha(e^{2\pi-x} - e^{i\theta} e^x)$$

where $\psi \in D(P)$ and $\alpha \in \mathbb{C}$. Now,

$$\psi_\theta(2\pi) = \alpha(1 - e^{i\theta}e^{2\pi})$$

and

$$\psi_\theta(0) = \alpha(e^{2\pi} - e^{i\theta})$$

Then,

$$\psi_\theta(2\pi) = e^{i\tilde{\theta}}\psi_\theta(0)$$

where,

$$e^{i\tilde{\theta}} = \frac{(1 - e^{i\theta}e^{2\pi})}{e^{2\pi} - e^{i\theta}}$$

Hence, P has infinite possible self adjoint extensions with domains,

$$D(P_\theta) := \left\{ \psi \in H^1(I) \mid \psi(0)e^{i\tilde{\theta}} = \psi(2\pi) \right\}$$

Remark. For no values of θ we get the particle in a box boundary condition back. This means that it is not possible to define a self adjoint momentum for particle in a box problem! However, $\tilde{\theta} = 0$ gives us the particle in a circle problem, which we already showed that a self adjoint extension exists.

5.3 The Quantum Harmonic Oscillator [1]

Let the Hilbert space under consideration be $L^2(\mathbb{R})$. We propose the Hamiltonian as,

$$H := \frac{P \circ P}{2m} + \frac{m\omega^2}{2}Q \circ Q$$

Where the domain of Q, P is taken to be the Schwartz space, and obeys the commutation relation $[Q, P] = i \cdot id_{S(\mathbb{R})}$. This only makes these operators essentially self adjoint. We will show that H is also essentially self adjoint using the following theorem.

Theorem 5.3.1. *If a symmetric operator has an orthonormal basis, then it is essentially self adjoint.*

Proof. Let A be symmetric and $\{\psi_n\}$ be its ON basis. We have,

$$A \subseteq A^*$$

Now, we need to find $\ker(A^* \pm i)$. Choose any ψ in the domain of A^* . Using the ON basis,

$$\psi = \sum_n c_n \psi_n$$

Now,

$$(A^* + i)\psi = 0 \Rightarrow \sum_n c_n (A^* + i)\psi_n = 0$$

Hence,

$$\sum_n c_n (A + i)\psi_n = 0$$

Taking an inner product with another vector ψ_m ,

$$\sum_n c_n ((\psi_m, A\psi_n) + i\delta_{mn}) = 0$$

Since all c_n are not zero by assumption, we need,

$$(\psi_m, A\psi_n) = -i\delta_{mn}$$

or,

$$(\psi_n, A\psi_n) = -i$$

Since LHS is real, this is not possible, which tells us that $c_n = 0$ for all n . This gives $\ker(A^* \pm i) = \{0\}$. Hence A is essentially self adjoint. $\blacksquare$

Now, we just have to show that H has an ortho-normal basis. To simplify things, we define,

Definition 5.3.1. Define $a_{\pm} : S(\mathbb{R}) \rightarrow S(\mathbb{R})$ as,

$$a_{\pm} := \sqrt{\frac{m\omega}{2}}Q \mp \frac{i}{\sqrt{2m\omega}}P$$

Using this new definition, we can rewrite the Hamiltonian as,

$$H = \omega \left(a_+ a_- + \frac{1}{2} id_{S(\mathbb{R})} \right)$$

Once we have this, we can compute,

$$[a_-, a_+] = id_{S(\mathbb{R})} \quad [H, a_{\pm}] = \pm \omega a_{\pm}$$

Let ψ be an eigenvector of H with eigenvalue E . Then Using these commutation relations, we can see that,

$$H(a_{\pm}\psi) = (E \pm \omega)\psi$$

This tells us that the eigenvalues of H are of the form $\{E \pm n\omega\}$, where $\{(a_{\pm})^n\psi\}$ are the corresponding eigenvectors.

Now,

$$\|a_+\psi\|^2 = (\psi, a_-a_+\psi) = (\psi, a_+a_-\psi) + (\psi, \psi) = (a_-\psi, a_-\psi) + (\psi, \psi) \geq \|\psi\|^2$$

and,

$$E\|\psi\|^2 = (\psi, H\psi) = \omega((\psi, a_+a_-\psi) + (\psi, \psi)) \geq \frac{\omega}{2}\|\psi\|^2$$

As $E \geq \frac{\omega}{2}$, a_- cannot lower the eigen values forever. It must terminate, say at $N \in \mathbb{N}_0$. i.e, there is a non vanishing $\psi_0 = (a_-)^N\psi$ such that $a_-\psi_0 = 0$. The energy of this state is the lowest possible,

$$H\psi_0 = \frac{\omega}{2}\psi_0$$

We can build all the eigen states of H by starting from ψ_0 and acting with a_+ repeatedly. Now putting back the differential operators in place, we get from $a_-\psi_0 = 0$,

$$\left(\sqrt{\frac{m\omega}{2}}x + \frac{i}{\sqrt{2m\omega}}(-i)\frac{d}{dx}\right)\psi_0(x) = 0$$

This can be solved to get,

$$\psi_0(x) = A \exp\left\{-\frac{m\omega}{2}x^2\right\}$$

Then the n th eigenstate is obtained in terms of the Hermite polynomials as,

$$\psi_n(x) \propto H_n(\sqrt{m\omega}x) \exp\left\{-\frac{m\omega}{2}x^2\right\}$$

Since there is only one eigenstate per eigenvalue, ψ_0 is unique. We quote a result without proof,

Lemma 5.3.2. *Hermite polynomials form an Ortho normal basis for $L^2(\mathbb{R})$*

This result, along with the theorem we proved in the begining of the section tells us that H is essentially self adjoint.

Chapter 6

Algebraic Formulation of Quantum Field Theory

6.1 Introduction

Haag and Castler, in 1964 introduced a new approach to quantum field theory using the formalism of a C^* -algebra. The main reason to have a new formulation despite the success of quantum field theory in predicting experiments was that the mathematical foundation of quantum field theory is a bit shaky. People have shown that the formalism of QFT using Feynman diagrams is not mathematically rigorous. Through an algebraic approach, we hope that many of the problems if not all can be remedied.

In the next section, we will discuss the usual formalism of QFT, in Fock space, and some issues that pops up. Then we will present a motivation for an algebraic formulation(which is anecdotal). And then we will discuss how do we reformulate quantum mechanics in the language of algebra.

6.2 Issues With the Fock Space

We discuss the Fock space formalism and certain issues with it using a specific model of QFT, the van-Hove model.

Constructing the Fock Space [4]

In Quantum Field Theory, the number of particles is not a sacred quantity. Hence we need a formalism in which we can deal with any number of particles at the same time. This is where the Fock space comes in. Denote the one particle Hilbert space for particle type s_i by $\mathcal{H}_{s_i}^{(1)}$. First, we construct an N -particle Hilbert space. Denote the elements of the one particle Hilbert space by $\psi_{s_i}^{(1)}$. Now define,

$$\psi_{(s_1, \dots, s_N)}^{(N)} := \psi_{s_1}^{(1)} \otimes \psi_{s_2}^{(1)} \otimes \dots \otimes \psi_{s_N}^{(1)}$$

We define a scalar product between two such constructions as,

$$\left(\phi_{(s_1, \dots, s_N)}^{(N)}, \psi_{(s_1, \dots, s_N)}^{(N)} \right) := \prod_{k=1}^N \left(\phi_{s_k}^{(1)}, \psi_{s_k}^{(1)} \right)$$

Define a vector space as,

$$\mathcal{H}_{(s_1, \dots, s_N)}^{(N)} := \left\{ \text{all linear combinations of possible } \psi_{(s_1, \dots, s_N)}^{(N)} \right\}$$

The scalar product defined can be extended to this new space via linearity. To make this a Hilbert space, we must complete it. Hence the N -particle Hilbert space finally is $\overline{\mathcal{H}}_{(s_1, \dots, s_N)}^{(N)}$

Identical Particles

In case we have identical particles, the above constructed Hilbert space will be more than what is needed. Say we are looking at N identical bosons. In that case, we need states which are symmetric. We construct a Hilbert space for it as follows. Consider $\psi_{(1, \dots, N)}^{(N)} \in \mathcal{H}_{(1, \dots, N)}^{(N)}$ such that $\psi_{(1, \dots, N)}^{(N)}$ can be written as a direct product. i.e,

$$\psi_{(1, \dots, N)}^{(N)} = \psi_1^{(1)} \otimes \psi_2^{(1)} \otimes \dots \otimes \psi_N^{(1)}$$

Define a “symmetrizing map” S such that,

$$S\left(\psi_{(1, \dots, N)}^{(N)}\right) = \frac{1}{\sqrt{N!}} \sum_{P \in \tau(N)} \psi_{P(1, \dots, N)}^{(N)} =: \psi_{S(1, \dots, N)}^{(N)}$$

Where τ is the permutation group. This map can be extended to $\mathcal{H}_{(1, \dots, N)}^{(N)}$ by linearity.

Proposition 6.2.1. $S : \mathcal{H}_{(1,\dots,N)}^{(N)} \rightarrow \mathcal{H}_{(1,\dots,N)}^{(N)}$ is bounded

Proof. Consider $\psi_{(1,\dots,N)}^{(N)}$ with a norm 1. Using it,

$$\begin{aligned} \|S\| &= \sup \left\| S\psi_{(1,\dots,N)}^{(N)} \right\| \\ &= \sup \left\| \frac{1}{\sqrt{N!}} \sum_{P \in \tau(N)} \psi_{P(1,\dots,N)}^{(N)} \right\| \\ &\leq \sup \frac{1}{\sqrt{N!}} \sum_{P \in \tau(N)} \left\| \psi_{P(1,\dots,N)}^{(N)} \right\| \\ &= \sup \frac{1}{\sqrt{N!}} \sum_{P \in \tau(N)} 1 \\ &= \sqrt{N!} \end{aligned}$$

■

Hence by BLT theorem, we can extend this to the entire space, $\overline{\mathcal{H}}_{(1,\dots,N)}^{(N)}$. The range of the map S is the symmetrized space that we need.

Fock Space

Define,

$$\mathcal{H} := \bigoplus_{N=0}^{\infty} \overline{\mathcal{H}}^{(N)}$$

With $\overline{\mathcal{H}}^{(0)} := \mathbb{C}$ Any $\psi \in \mathcal{H}$ is of the form,

$$\psi = \{\psi^{(0)}, \psi^{(1)}, \dots\}$$

Define a sum and s-multiplication as,

$$(\lambda \cdot \psi + \mu \cdot \phi)^{(N)} := \lambda \cdot \psi^{(N)} + \mu \cdot \phi^{(N)}$$

Also, define the scalar product as,

$$(\psi, \phi) := \sum_{N=0}^{\infty} (\psi^{(N)}, \phi^{(N)})$$

The closure of $\mathcal{H}$ will give us the Fock space, $\mathcal{F}$.

Remark. If the direct sum were finite, then we will have a Hilbert space even before the closure.

The Creation and Annihilation Operators

As an example, we take $\overline{\mathcal{H}}^{(1)} = L^2(\mathbb{R}^3)$. And assume the particles obey bosonic statistics. Then we take the symmetric part of the pre-Fock space, $\mathcal{H}$, and on it define the creation and annihilation operators as,

$$\begin{aligned} (a(f)\psi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) &:= \sqrt{N+1} \int d^3x f^*(\vec{x}) \psi^{(N+1)}(\vec{x}, \vec{x}_1, \dots, \vec{x}_N) \\ (a^*(f)\psi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) &:= \frac{1}{\sqrt{N}} \sum_{k=1}^N f(\vec{x}_k) \psi^{(N-1)}(\vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N) \end{aligned}$$

Where $\hat{\vec{x}}_k$ means that argument is omitted. Now,

$$\begin{aligned} \left\| (a^*(f)\psi)^{(N)} \right\|^2 &= \frac{1}{N} \left\| \sum_{k=1}^N f \cdot \psi^{(N-1)} \right\|^2 \\ &\leq \frac{N^2}{N} \|f\|^2 \cdot \|\psi^{(N-1)}\|^2 \end{aligned}$$

Hence $a^*(f) : \overline{\mathcal{H}}^{(N)} \rightarrow \overline{\mathcal{H}}^{(N+1)}$ is bounded by $\sqrt{N+1}\|f\|$. Likewise,

$$\begin{aligned} \left\| (a(f)\psi)^{(N)} \right\|^2 &= (N+1) \left\| \int d^3x f^* \cdot \psi^{(N+1)} \right\|^2 \\ &\leq (N+1) \|f\|^2 \cdot \|\psi^{(N+1)}\|^2 \end{aligned}$$

Hence $a(f) : \overline{\mathcal{H}}^{(N)} \rightarrow \overline{\mathcal{H}}^{(N-1)}$ is bounded by $\sqrt{N}\|f\|$. But these operators are unbounded in $\mathcal{H}$.

Proposition 6.2.2.

$$a(f)^* = a^*(f)$$

Proof.

$$\begin{aligned} (\psi^{(N)}, (a(f)\phi)^{(N)}) &= \int d^3x_1 \dots d^3x_N \psi^{*(N)}(\vec{x}_1, \dots, \vec{x}_N) (a(f)\phi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) \\ &= \int d^3x_1 \dots d^3x_N \int d^3x \psi^{*(N)}(\vec{x}_1, \dots, \vec{x}_N) \sqrt{N+1} f^*(\vec{x}) \phi^{(N+1)}(\vec{x}, \vec{x}_1, \dots, \vec{x}_N) \\ &= \sqrt{N+1} (f \otimes \psi^{(N)}, \phi^{(N+1)}) \end{aligned}$$

Now,

$$\begin{aligned}
((a^*(f)\psi)^{(N)}, \phi^{(N)}) &= \int d^3x_1 \cdots d^3x_N \left(\frac{1}{\sqrt{N}} \sum_{k=1}^N f(\vec{x}_k) \psi^{(N-1)}(\vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N) \right)^* \phi^{(N)}(\vec{x}_1, \dots, \vec{x}_N) \\
&= \frac{1}{\sqrt{N}} \sum_{k=1}^N \int d^3x_1 \cdots d^3x_N f^*(\vec{x}_k) \psi^{*(N-1)}(\vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N) \\
&= \frac{1}{\sqrt{N}} \sum_{k=1}^N (f \otimes \psi^{(N-1)}, \phi^{(N)}) \\
&= \sqrt{N} (f \otimes \psi^{(N-1)}, \phi^{(N)})
\end{aligned}$$

From this, we have our result. ■

Proposition 6.2.3.

$$a^*(\lambda f + \mu g) = \lambda a^*(f) + \mu a^*(g)$$

Proof. Immediate from the definition. ■

Proposition 6.2.4.

$$[a(f), a^*(g)] = (f, g)$$

Proof.

$$\begin{aligned}
(a(f)a^*(g)\psi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) &= \sqrt{N+1} \int d^3x_0 f^*(\vec{x}_0) (a^*(g)\psi)^{(N+1)}(\vec{x}_0, \vec{x}_1, \dots, \vec{x}_N) \\
&= \int d^3x_0 f^*(\vec{x}_0) \sum_{k=0}^N g(\vec{x}_k) \psi^{(N)}(\vec{x}_0, \vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N)
\end{aligned}$$

Similarly,

$$\begin{aligned}
(a^*(g)a(f)\psi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) &= \frac{1}{\sqrt{N}} \sum_{k=1}^N g(\vec{x}_k) (a(f)\psi)^{(N-1)}(\vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N) \\
&= \sum_{k=1}^N g(\vec{x}_k) \int d^3x_0 f^*(\vec{x}_0) \psi^{(N)}(\vec{x}_0, \vec{x}_1, \dots, \hat{\vec{x}}_k, \dots, \vec{x}_N)
\end{aligned}$$

Hence,

$$([a(f), a^*(g)]\psi)^{(N)}(\vec{x}_1, \dots, \vec{x}_N) = \int d^3x_0 f^*(\vec{x}_0) g(\vec{x}_0) \psi^{(N)}(\vec{x}_1, \dots, \vec{x}_N)$$

Hence,

$$[a(f), a^*(g)] = (f, g)$$

■

Proposition 6.2.5.

$$[a(f), a(g)] = 0 = [a^*(f), a^*(g)]$$

Proof. Proceed the same way as the previous. ■

The last two propositions are called the Canonical Commutation Relations. Now in general for any one particle Hilbert space $\mathcal{H}^{(1)}$, we can define,

$$\begin{aligned} a(f)\phi^{(N)} &:= \frac{1}{\sqrt{(N-1)!}} \sum_{p \in \tau(N)} \left(f, \phi_{p(1)}^{(1)} \right) \phi_{p(2)}^{(1)} \otimes \phi_{p(3)}^{(1)} \otimes \cdots \otimes \phi_{p(N)}^{(1)} \\ a^*(f)\phi^{(N)} &:= \frac{1}{\sqrt{(N+1)!}} \sum_{p \in \tau(N+1)} \phi_{p(0)}^{(1)} \otimes \phi_{p(1)}^{(1)} \otimes \cdots \otimes \phi_{p(N)}^{(1)} \end{aligned}$$

where $\phi_{p(0)}^{(1)} = f$. All the calculations we did above goes through.

Second Quantized Form

Given an observable in $\mathcal{H}^{(1)}$, we would like to define its corresponding operator in the Fock space. For that, first we do the following general construction; Consider $A_i : D(A_i) \rightarrow \mathcal{H}_i^{(N)}$ for $i = 1, \dots, N$ with all operators closed and densely defined. We want to define $A_1 \otimes \cdots \otimes A_N$ such that it is closed. To circumvent the burden of proving the operator is closed and so on, we define it in a rather roundabout manner. First, since A_i is closed, $D(A_i^*)$ is dense. Hence,

$$D_* := \text{span}\{D(A_1^*) \times \cdots \times D(A_N^*)\}$$

is dense in $\overline{\mathcal{H}}^{(N)}$. Then define, $A_* : D_* \rightarrow \overline{\mathcal{H}}^{(N)}$,

$$A_*(\phi_1^{(1)} \otimes \cdots \otimes \phi_N^{(1)}) := A_1^* \phi_1^{(1)} \otimes \cdots \otimes A_N^* \phi_N^{(1)}$$

where $\phi_i^{(1)} \in D(A_i^*)$. Then define,

$$A_1 \otimes \cdots \otimes A_N := A_*^*$$

Since $D(A_*^*) \supseteq D(A_1) \times \cdots \times D(A_N)$, hence $A_1 \otimes \cdots \otimes A_N$ is densely defined. The adjoint,

$$(A_1 \otimes \cdots \otimes A_N)^* = A_*^{**}$$

is closed, hence $A_1 \otimes \cdots \otimes A_N$ is closable. Now, we concentrate onto the particular case we are interested in. Let $\mathcal{H}_i^{(1)} = \mathcal{H}^{(1)}$. For a densely defined closed $A : D(A) \rightarrow \mathcal{H}^{(1)}$, define,

$$\begin{aligned} A_1^{(N)} &:= A \otimes id \otimes id \otimes \cdots \otimes id \\ A_2^{(N)} &:= id \otimes A \otimes id \otimes \cdots \otimes id \\ &\vdots \\ A_N^{(N)} &:= id \otimes id \otimes id \otimes \cdots \otimes A \end{aligned}$$

All of them are closed. Now define, $\sum_{i=1}^N A_i^{(N)}$ on the domain $D(A_1^{(N)}) \cap \cdots \cap D(A_N^{(N)})$. This domain atleast contain $D(A) \times D(A) \times \cdots \times D(A)$, hence dense. Let $A^{(N)}$ be the minimal closed extension of $\sum_{i=1}^N A_i^{(N)}$. Now define,

$$D(\Omega(A)) := \left\{ \psi \in \mathcal{F} \mid \psi^{(N)} \in D(A^{(N)}) : \sum_N \|A^{(N)} \psi^{(N)}\|^2 < \infty \right\}$$

Then, $\Omega(A) : D(\Omega(A)) \rightarrow \mathcal{F}$

$$\Omega(A)\psi := \sum_N A^{(N)} \psi^{(N)}$$

This operator is also closed since each of $A^{(N)}$ is closed. If A were self adjoint to start with, $\Omega(A)$ also would be so. This operator is called the second quantized form of a single particle observable.

Theorem 6.2.1. *Let H be self adjoint and A closed. Then,*

1. $\Omega(e^{iHt} A e^{-iHt}) = e^{i\Omega(H)t} \Omega(A) e^{-i\Omega(H)t}$
2. $a(e^{-iHt} f) = e^{-i\Omega(H)t} a(f) e^{i\Omega(H)t}$
3. $a^*(e^{-iHt} f) = e^{-i\Omega(H)t} a^*(f) e^{i\Omega(H)t}$

Proof. . ■

Remark. This theorem tells us that if H is the one particle time evolution generator, then $\Omega(H)$ is the corresponding time evolution generator in the Fock space.

6.2.1 The Free Scalar Field [4]

Here we will illustrate how do we construct a field theory for a free scalar particle with the formalism developed in the last section. The one particle space of mass m and spin 0 particles is taken to be square integrable functions on the Minkowski space with respect to the Lorentz invariant measure $\frac{d^3\vec{k}}{\omega_{\vec{k}}}$. Denote,

$$\mathcal{H}^{(1)} := L_{\mu}^2(\mathbb{R}^3)$$

now for $\phi \in \mathcal{H}^{(1)} : \exists \tilde{\phi}(\vec{k})$ such that,

$$\|\phi\|_{\mu}^2 := \int \frac{d^3\vec{k}}{\omega_{\vec{k}}} |\tilde{\phi}(\vec{k})|^2 < \infty$$

Define the inner-product,

$$(\phi, \psi)_{\mu} := \int \frac{d^3\vec{k}}{\omega_{\vec{k}}} \tilde{\phi}^*(\vec{k}) \tilde{\psi}(\vec{k})$$

where $\omega_{\vec{k}} := \sqrt{\vec{k}^2 + m^2}$.

We now define the one particle Hamiltonian as,

$$\left(\widetilde{H_0^{(1)} \phi} \right) (\vec{k}) := \omega_{\vec{k}} \tilde{\phi}(\vec{k})$$

Where the domain is,

$$D(H_0^{(1)}) := \left\{ \phi \in L_{\mu}^2(\mathbb{R}^3) \mid \left\| H_0^{(1)} \phi \right\|_{\mu}^2 < \infty \right\}$$

Example 6.2.0.1. .

We use the symmetrised version of the Fock space for this case. We define, $a(f)$, $a^*(f)$ and $H_0 = \Omega(H_0^{(1)})$. Then construct,

$$F(f) := \frac{1}{\sqrt{2}} [a^*(f) + a(f)]$$

F is self adjoint by construction. Now, the time evolution of F is given by,

$$F_t(f) = e^{iH_0 t} F(f) e^{-iH_0 t} = F(e^{iH_0^{(1)} t} f)$$

Define,

$$P_t(f) := \frac{d}{dt} F_t(f)$$

From the definition of creation and annihilation operators, it directly follows that,

$$\frac{d}{dt}a(e^{iH_0^{(1)}t}f) = a(e^{iH_0^{(1)}t}iH_0^{(1)}f)$$

Hence,

$$P_t(f) = F(e^{iH_0^{(1)}t}iH_0^{(1)}f) = F_t(iH_0^{(1)}f)$$

Also,

$$P(f) := P_0(f) = \frac{i}{\sqrt{2}} \left[a^*(H_0^{(1)}f) - a(H_0^{(1)}f) \right]$$

and,

$$\frac{d}{dt}P_t(f) = F_t\left(-\left(H_0^{(1)}\right)^2 f\right)$$

Now the commutation relations can be written as,

$$\begin{aligned} [F(f), F(g)] &= \frac{1}{2} \left[(f, g)_\mu - (g, f)_\mu \right] \\ [P(f), P(g)] &= -\frac{1}{2} \left[\left(H_0^{(1)}f, H_0^{(1)}g \right)_\mu - \left(H_0^{(1)}g, H_0^{(1)}f \right)_\mu \right] \\ [F(f), P(g)] &= \frac{i}{2} \left[\left(f, H_0^{(1)}g \right)_\mu - \left(H_0^{(1)}g, f \right)_\mu \right] \end{aligned}$$

Now,

$$\left(f, H_0^{(1)}g \right)_\mu = \int d^3\vec{k} \tilde{f}^*(\vec{k}) \tilde{g}(\vec{k}) = (f, g)$$

Then for real f and g ,

$$\begin{aligned} [F(f), F(g)] &= 0 \\ [P(f), P(g)] &= 0 \\ [F(f), P(g)] &= i(f, g) \end{aligned}$$

This conclude the mathematical formalism of the free scalar field.

6.2.2 The van-Hove Model [4]

In this section, we discuss the van-Hove model and through it, illustrate certain weaknesses of the Fock space formalism. van-Hove model is the theory of a scalar field with a source. We first investigate the changes in the spectra compared to the free particle. Let the time evolution operator in the Fock space be,

$$H = H_0 + F(\rho)$$

where ρ is a source. Let D be the domain of the Hamiltonian in the Fock space.

$$D := \left\{ \phi \in \mathcal{F} \mid \sum_N \left\| H_0^{(N)} \phi^{(N)} \right\|_\mu^2 < \infty \right\}$$

We now verify that H is also well defined and self adjoint on this domain. Take $\psi \in D$. We have,

$$\infty > \|H_0 \psi\|_\mu^2 = \sum_N \left\| H_0^{(N)} \psi^{(N)} \right\|_\mu^2 \geq m^2 \sum_N N^2 \left\| \psi^{(N)} \right\|_\mu^2$$

The last inequality follows because $H_0^{(1)}$ is bounded below by m . Now, we have,

$$\begin{aligned} \sum_N N^2 \left\| \psi^{(N)} \right\|_\mu^2 &< \infty \\ \Rightarrow \sum_N N \left\| \psi^{(N)} \right\|_\mu^2 &< \infty \\ \Rightarrow \sum_N (N+1) \left\| \psi^{(N)} \right\|_\mu^2 &< \infty \end{aligned}$$

Now,

$$\begin{aligned} \|F(\rho)\psi\|_\mu^2 &\leq \|a(\rho)\psi\|_\mu^2 + \|a^*(\rho)\psi\|_\mu^2 \\ &\leq \sum_N \|\rho\|_\mu^2 N \left\| \psi^{(N)} \right\|_\mu^2 + \sum_N \|\rho\|_\mu^2 (N+1) \left\| \psi^{(N)} \right\|_\mu^2 \\ &= \|\rho\|_\mu^2 \sum_N (2N+1) \left\| \psi^{(N)} \right\|_\mu^2 < \infty \end{aligned}$$

Hence $F(\rho)$ exists in D . We now use the following theorem,

Theorem 6.2.2. Kato-Rellich Theorem: *If $A : D \rightarrow \mathcal{H}$ is self adjoint and $B : D \rightarrow \mathcal{H}$ is symmetric and*

$$\forall \psi \in D : \|B\psi\| \leq a\|A\psi\| + b\|\psi\|$$

for $0 < a < 1$ and $0 \leq b < \infty$, then $A + B$ is self adjoint on D

Hence H will be self adjoint on D if we can find a, b such that,

$$\|F(\rho)\psi\|_\mu^2 \leq a\|H_0\psi\|_\mu^2 + b\|\psi\|_\mu^2$$

which gives,

$$a^2 m^2 \sum_N N^2 \|\psi\|_\mu^2 + b^2 \sum_N \|\psi\|_\mu^2 - \|\rho\|_\mu^2 \sum_N (2N+1) \|\psi\|_\mu^2 \geq 0$$

which gives,

$$\forall N \in \mathbb{N} : a^2 m^2 N^2 - 2 \|\rho\|_\mu^2 N + (b^2 - \|\rho\|_\mu^2) \geq 0$$

The choice for a and b are,

$$a = \frac{\|\rho\|_\mu}{\lambda m} \quad \text{and} \quad b = \|\rho\|_\mu \gamma$$

where $\lambda > \frac{\|\rho\|_\mu}{m}$ and $\gamma > \lambda + 1$. Hence H is self adjoint on D .

Spectrum of H

We now investigate how does the spectrum gets modified from that of the free field by the introduction of the source term. Since $P(f)$ which is self adjoint is defined on D , it can generate a Unitary abelian one parameter group,

$$V_\lambda(f) := \exp\{-iP(f)\lambda\}$$

Let $V \equiv V_1((H_0^{(1)})^{-2}\rho)$. Now,

$$V^{-1}H_0V = H_0 + i\left[P((H_0^{(1)})^{-2}\rho), H_0\right] - \frac{1}{2}\left[P(H_0^{(1)})^{-2}\rho, \left[P(H_0^{(1)})^{-2}\rho, H_0\right]\right] + \dots$$

Now using,

$$\frac{d}{dt}P_t(f) = i[H_0, P_t(f)]$$

$$\begin{aligned} i\left[P((H_0^{(1)})^{-2}\rho), H_0\right] &= -\frac{d}{dt}P_t((H_0^{(1)})^{-2}\rho)\Big|_{t=0} \\ &= -F_t(-(H_0^{(1)})^2(H_0^{(1)})^{-2}\rho)\Big|_{t=0} \\ &= F(\rho) \end{aligned}$$

Then the second term becomes,

$$\left[P((H_0^{(1)})^{-2}\rho), -iF(\rho)\right] = -\left(\rho, (H_0^{(1)})^{-2}\rho\right) = -\left\|(H_0^{(1)})^{-1}\rho\right\|^2$$

Hence the rest of the terms in the series are zero. Hence,

$$V^{-1}H_0V = H_0 + F(\rho) + \frac{1}{2}\left\|(H_0^{(1)})^{-1}\rho\right\|^2 = H + \frac{1}{2}\left\|(H_0^{(1)})^{-1}\rho\right\|^2$$

Rewrite this as,

$$VHV^{-1} = H_0 + W$$

Where $W = -\frac{1}{2} \left\| (H_0^{(1)})^{-1} \rho \right\|^2$. Since the spectrum is invariant under unitary transformations, the spectrum of H is the shifted version of the spectrum of H_0 , by a value W .

$$W = -\frac{1}{2} \int d^3k \frac{1}{\omega_{\mathbf{k}}^2} \tilde{\rho}^*(\mathbf{k}) \tilde{\rho}(\mathbf{k})$$

Since $\frac{1}{\omega_{\mathbf{k}}^2}$ is the Fourier transform of $Y(x) = \frac{e^{-mx}}{4\pi x}$, we can write,

$$W = \frac{1}{2} \int dx dy Y(x-y) \rho^*(x) \rho(y)$$

Which tells that W is the consequence of the sources interacting via the Yukawa potential. Since a constant in the Hamiltonian do not play any role in the time evolution, we define a “renormalized” Hamiltonian,

$$\hat{H} = H - W \cdot id$$

Since $\hat{H}$ and H_0 are unitarily equivalent, they have the same spectrum.

Scattering

Since $\hat{H}$ is unitarily equivalent to the free Hamiltonian, we might immediately say that the S matrix, $S = id$. But to say this, we have to make sure that it is possible to set up a scattering situation in this model, by which we mean the fields satisfy the asymptotic conditions. The candidates for F^{in} and F^{out} are the free fields themselves.

$$F_t(f) = e^{iH_0 t} F(f) e^{-iH_0 t} = F\left(e^{iH_0^{(1)} t} f\right)$$

But the correct interpolating field is given by,

$$\hat{F}_t(f) = V^{-1} F_t(f) V = \hat{F}\left(e^{iH_0^{(1)} t} f\right)$$

Now,

$$\begin{aligned} \hat{F}_t(f) &= V^{-1} e^{iH_0 t} F(f) e^{-iH_0 t} V \\ &= e^{i\hat{H} t} V^{-1} F(f) V e^{-i\hat{H} t} \\ &= e^{i\hat{H} t} \hat{F}(f) e^{-i\hat{H} t} \end{aligned}$$

The evolution of $\hat{F}$ is governed by $\hat{H}$, as needed for the scattering formalism. Now,

$$\begin{aligned}\hat{F}(f) &= V^{-1}F(f)V \\ &= F(f) + i\left[P\left((H_0^{(1)})^{-2}\rho\right), F(f)\right] - \frac{1}{2}\left[P\left((H_0^{(1)})^{-2}\rho\right), \left[P\left((H_0^{(1)})^{-2}\rho\right), F(f)\right]\right] + \cdots \\ &= F(f) + \left(\rho, \left(H_0^{(1)}\right)^{-2}f\right)\end{aligned}$$

Let,

$$c(f, \rho) := \frac{1}{2}\left[\left(\rho, \left(H_0^{(1)}\right)^{-2}f\right) + \text{c.c.}\right]$$

Hence,

$$\hat{F}_t(f) = \hat{F}\left(e^{iH_0^{(1)}t}f\right) = F\left(e^{iH_0^{(1)}t}f\right) + c\left(e^{iH_0^{(1)}t}f, \rho\right)$$

For any state ϕ , let Φ be the density matrix. Then the asymptotic condition can be written as,

$$\lim_{t \rightarrow \pm\infty} \left\langle \Phi : \hat{F}_t(f) - F_t(f) \right\rangle = 0$$

Which boils down to,

$$\lim_{t \rightarrow \pm\infty} c\left(e^{iH_0^{(1)}t}f, \rho\right) = 0$$

For $f, \rho \in \mathcal{H}^{(1)}$, the above condition can be satisfied always. Hence we have a scattering scenario possible in the model and $S = id$.

Vacuum: Denote by ψ_0 the bare vacuum. Which means,

$$a(f)\psi_0 = 0 \quad \text{and} \quad U_t^0\psi_0 = \psi_0$$

where $U_t^0 := e^{iH_0 t}$. Now define, the “physical” or “dressed” vacuum as,

$$\hat{\psi}_0 := V^{-1}\psi_0$$

This satisfies,

$$\hat{a}(f)\hat{\psi}_0 = 0 \quad \text{and} \quad \hat{U}_t\hat{\psi}_0 = \hat{\psi}_0$$

where $\hat{U}_t := e^{i\hat{H}t}$. We can reconstruct the entire Fock space by application of $\hat{a}$ and $\hat{a}^*$ on the vacuum using all possible polynomials and finally taking a closure. Till this point, everything neatly fitted in the Fock space formalism. Now, we take

a particular case, in which the source is concentrated at the origin. This means $\tilde{\rho}(\vec{\mathbf{k}}) \rightarrow \text{constant}(=1)$. Now take $\psi \in D$. Then,

$$\|F(\rho)\psi\|^2 \leq \|\rho\|_\mu^2 \sum_{N=0}^{\infty} (2N+1) \|\psi^{(N)}\|^2$$

But,

$$\|\rho\|_\mu^2 = \int_{\mathbb{R}^3} \frac{d^3k}{\omega_{\vec{\mathbf{k}}}} \tilde{\rho}^*(\vec{\mathbf{k}}) \tilde{\rho}(\vec{\mathbf{k}}) = \int_{\mathbb{R}^3} \frac{d^3k}{\omega_{\vec{\mathbf{k}}}} \rightarrow \infty \quad (\text{diverge for large } k)$$

This means $H = H_0 + F(\rho)$ loses its meaning as an operator acting on the Fock space. Also, the constant W also becomes infinite.

6.2.3 Motivation for an Algebraic Reformulation

Is there a way we can circumvent the issue when ρ is concentrated at the origin? The primary objects of the theory are fields and their expectation values. $F(f)$ is well defined on $\mathcal{H}$ and $F(f)\mathcal{H} = \mathcal{H}$. What happens on the entire Fock space is obtained by taking the closure. Take all possible $F(f)\Big|_{\mathcal{H}}$. Form all finite linear combinations and products of these. Along with the id , this forms an algebra, denoted by $\mathcal{U}$. The states then can be defined as normalised linear functionals, $\langle \Phi : \cdot \rangle : \mathcal{U} \rightarrow \mathbb{R}$. Now the time evolution of the elements in the algebra is also well defined. For the free fields,

$$F(f) \rightarrow F_t^0(f) = F\left(e^{iH_0^{(1)}t}f\right)$$

Now for the interpolating field,

$$\hat{F}(f) \rightarrow \hat{F}_t(f) = F\left(e^{iH_0^{(1)}t}f\right) + c\left(e^{iH_0^{(1)}t}f, \rho\right)$$

Since $F(f) = \hat{F}(f) - c(f, \rho)$, write,

$$F_t(f) = F\left(e^{iH_0^{(1)}t}f\right) + c\left(e^{iH_0^{(1)}t}f - f, \rho\right)$$

The above relation stays valid even in the $\tilde{\rho}(\vec{\mathbf{k}}) \rightarrow 1$ limit, since,

$$\frac{1}{2} \left[\int_{\mathbb{R}^3} \frac{d^3k}{\omega_{\vec{\mathbf{k}}}^2} \tilde{g}(\vec{\mathbf{k}}) + \text{c.c.} \right] \quad \text{is finite for } g \in \mathcal{H}^{(1)}$$

Hence if we abandon the idea of calculating the time evolution from the Hamiltonian and define the time evolution as an automorphism of the algebra, then we can potentially circumvent the problem.

But if we choose this view point, then we lose some of the features we had originally;

Lemma 6.2.3. *A vacuum for the dressed field $\hat{\psi}_0 \neq 0$ cannot exist.*

Proof. If it exists, then

$$\hat{a}(f)\hat{\psi}_0 = 0 \quad \text{and} \quad \hat{U}_t\hat{\psi}_0 = \hat{\psi}_0$$

Write,

$$\hat{\psi}_0 = \sum_N \hat{\psi}_0^{(N)}$$

Now,

$$\begin{aligned} \hat{a}(f) &= V^{-1}a(f)V \\ &= a(f) + \frac{1}{\sqrt{2}} \left[a^* \left((H_0^{(1)})^{-1} \rho \right), a(f) \right] + \text{rest of the terms are zero} \\ &= a(f) + \frac{1}{\sqrt{2}} \left((H_0^{(1)})^{-1} \rho, f \right)_\mu \\ &=: a(f) + d(f, \rho) \end{aligned}$$

Now,

$$\hat{a}(f)\hat{\psi}_0 = 0 \Rightarrow 0 = \sum_N \hat{a}(f)\hat{\psi}_0^{(N)} = \sum_N a(f)\hat{\psi}_0^{(N)} + \sum_N d(f, \rho)\hat{\psi}_0^{(N)}$$

This gives,

$$-d(f, \rho)\hat{\psi}_0^{(N)} = a(f)\hat{\psi}_0^{(N+1)}$$

Taking the norm,

$$|d(f, \rho)| \left\| \hat{\psi}_0^{(N)} \right\| \leq \sqrt{N+1} \|f\|_\mu \left\| \hat{\psi}_0^{(N+1)} \right\|$$

For this to satisfy, we need either $|d(f, \rho)| \leq \sqrt{N+1} \|f\|_\mu$ or $\left\| \hat{\psi}_0^{(N)} \right\| = 0$. If the first condition was true, the Riesz lemma will tell us that,

$$d(f, \rho) = (f, \eta)$$

But when $\tilde{\rho}(\vec{\mathbf{k}}) = 1$, $(H_0^{(1)})^{-1} \rho$ is not in the Hilbert space, hence $d(f, \rho)$ is not well defined. So, the second option is true, which gives $\hat{\psi}_0 = 0$ ■

Corollary 6.2.3.1. *The representations of $\{a(f), a^*(g)\}$ and $\{\hat{a}(f), \hat{a}^*(g)\}$ are unitarily inequivalent.*

Proof. If they were not, there will be a unitary operator V such that $\hat{a}(f) = V^{-1}a(f)V$. Then we will get a non zero dressed vacuum constructed from the bare vacuum via $V^{-1}\psi_0$. This is not possible. ■

In the next section, we will look at an algebraic reformulation of quantum mechanics.

6.3 Jordan Algebra of Quantum Mechanics [4]

We review the Jordan algebra structure of Quantum Mechanics in this section.

Postulate 1. For each observable A on a given physical system, there is a linear self-adjoint operator $\pi(A)$ acting on a Hilbert space $\mathcal{H}_\pi$, and conversely.

Let $\mathcal{U}$ be the set of all possible observables. Then the following are true,

1. For any $\{A_i\} \subseteq \mathcal{U}$ and $\lambda_i \in \mathbb{R}$: $\sum_i \lambda_i \pi(A_i) \in \pi(\mathcal{U})$
2. For any real function f , $f(\pi(A)) \in \pi(\mathcal{U})$
3. For $A, B \in \mathcal{U}$: $\pi(A)\pi(B) \notin \pi(\mathcal{U})$
4. $\{\pi(A), \pi(B)\} := \frac{1}{2}(\pi(A)\pi(B) + \pi(B)\pi(A)) \in \pi(\mathcal{U})$

The bilinear and commutative structure $\{\cdot, \cdot\}$ can be defined even without defining the product (i.e., only using 1 and 2 above) as,

$$\{\pi(A), \pi(B)\} = \frac{1}{4}[(\pi(A) + \pi(B))^2 - (\pi(A) - \pi(B))^2]$$

This symmetrised product is not associative in general.

$$\begin{aligned} \{\pi(A), \pi(B), \pi(C)\} &:= \{\{\pi(A), \pi(B)\}, \pi(C)\} - \{\pi(A), \{\pi(B), \pi(C)\}\} \\ &= \frac{1}{4}[\pi(B), [\pi(A), \pi(C)]] \end{aligned}$$

Theorem 6.3.1. *For bounded operators $A, C \in \mathcal{U}$, the following are equivalent.*

1. $\{\pi(A), \pi(B), \pi(C)\} = 0 \quad \forall B \in \mathcal{U}$

$$2. \{ \pi(A), \pi(A), \pi(C) \} = 0$$

$$3. \{ \pi(A), \pi(C), \pi(C) \} = 0$$

$$4. [\pi(A), \pi(C)] = 0$$

Proof. $1 \Rightarrow 2, 3$ and $4 \Rightarrow 1$ are trivial. ■

We use the above theorem to define the compatibility of two observables using the symmetrised product.

Definition 6.3.1. $A, C \in \mathcal{U}$ are compatible if ,

$$\{ \pi(A), \pi(B), \pi(C) \} = 0 \quad \forall B \in \mathcal{U}$$

The symmetric product satisfy,

$$\{ \pi(A)^2, \pi(B), \pi(A) \} = 0 \quad \forall B \in \mathcal{U}$$

which is called the Jordan Identity. Hence $(\pi(\mathcal{U}), \{ \cdot, \cdot \})$ is a real commutative Jordan algebra.

Now, the next task is defining the states. We do that via another postulate.

Postulate 2. To each state ϕ of the system, there is a corresponding density matrix $\rho = \pi(\phi)$ acting on the Hilbert space $\mathcal{H}_\pi$, and the expectation value is given by, $\langle \phi : A \rangle := \text{Tr}\{\pi(\phi)\pi(A)\}$

Here are the properties of states,

$$1. \text{ For } \lambda_i \in \mathbb{R}, \text{Tr}\{\pi(\phi) \sum_i \lambda_i A_i\} = \sum_i \lambda_i \langle \phi : A_i \rangle$$

$$2. \text{Tr}\{\pi(\phi)\pi(A)^2\} \geq 0$$

$$3. \text{Tr}\{\pi(\phi)\pi(\mathbb{1})\} = 1$$

If we have a sequence of states ϕ_i and $\lambda_i \in \mathbb{R}_0^+ : \sum_i \lambda_i = 1$, then we can create a new state ϕ via,

$$\langle \phi : A \rangle := \sum_i \lambda_i \langle \phi_i : A \rangle$$

The above equation is called the decomposition of state. If it cannot be done, then we call the state pure.

Example 6.3.1.1. any $\phi_\pi \in \mathcal{H}_\pi$ is a pure state and the expectation values is given by $(\phi_\pi, \pi(A)\phi_\pi)$

Denote the set of states which by definition satisfy the above three properties by $\mathfrak{G}$. Now we seek for a suitable algebraic structure on $(\mathcal{U}, \mathfrak{G})$ such that it can reproduce quantum mechanics. This has no reference to a Hilbert space in the first place! We achieve it axiomatically as we will see in the next section.

6.4 Algebraic Formulation of Quantum Field Theory [4]

In this section, we will build the structure necessary to have an algebraic quantum field theory.

6.4.1 Algebraic Structure

The first six axioms lay out the algebraic structure.

Axiom 1. To each physical system, we associate a triple $(\mathcal{U}, \mathfrak{G}, \langle \cdot : \cdot \rangle)$ where $\mathcal{U}$ are the observables, $\mathfrak{G}$ are the states and $\langle \cdot : \cdot \rangle : (\mathcal{U}, \mathfrak{G}) \rightarrow \mathbb{R}$ the expectation value.

A few definitions are in order.

Definition 6.4.1. Let $\mathfrak{J} \subseteq \mathfrak{G}$. Then

$$A \Big|_{\mathfrak{J}} \leq B \Big|_{\mathfrak{J}} \iff \forall \phi \in \mathfrak{J} : \langle \phi : A \rangle \leq \langle \phi : B \rangle$$

We will just write $A \leq B$ whenever $A \Big|_{\mathfrak{G}} \leq B \Big|_{\mathfrak{G}}$.

Definition 6.4.2. $\mathfrak{J} \subseteq \mathfrak{G}$ is said to be **full** w.r.t. $\mathcal{B} \subseteq \mathcal{U}$ if $A \Big|_{\mathfrak{J}} \leq B \Big|_{\mathfrak{J}}$ for any $A, B \in \mathcal{B} \Rightarrow A \leq B$

Axiom 2. “ $\leq$ ” is a partial ordering relation on $\mathcal{U}$. Which means $A \leq B$ and $B \leq A \Rightarrow A = B$

This axiom tells us that $\forall \phi \in \mathfrak{G} : \langle \phi : A \rangle = \langle \phi : B \rangle \Rightarrow A = B$. i.e, the observables are completely determined by the expectation values.

Axiom 3. .

1. $\exists 0, 1 \in \mathcal{U} : \forall \phi \in \mathfrak{G} : \langle \phi : 0 \rangle = 0, \langle \phi : 1 \rangle = 1$
2. For $A \in \mathcal{U}$ and $\lambda \in \mathbb{R}$, $\exists \lambda A \in \mathcal{U} : \forall \phi \in \mathfrak{G} : \langle \phi : \lambda A \rangle = \lambda \langle \phi : A \rangle$
3. For $A, B \in \mathcal{U} : \exists A + B \in \mathcal{U} : \forall \phi \in \mathfrak{G} : \langle \phi : A + B \rangle = \langle \phi : A \rangle + \langle \phi : B \rangle$

The uniqueness of these observables are immediate once we use axiom 2. Axiom 3 equips $\mathcal{U}$ with the structure of a real vector space.

We will now assume that a well defined procedure for measuring the expectation values exist. Whenever we make a measurement of an observable on a particular state, we get a real number. If we take a very large number of measurements, we get a distribution, whose mean we call the expectation value. Now, we call a state **dispersion free** if the distribution is just a peak at one real number.

Definition 6.4.3. $\phi \in \mathfrak{G}$ is called dispersion free w.r.t. $A \in \mathcal{U}$ if the distribution of measurement is just a peak around a real number we call $\langle \phi : A \rangle$.

$\mathfrak{G}_A :=$ set of all dispersion free states of A

$$\sigma_A := \left\{ \langle \phi : A \rangle \in \mathbb{R} \mid \phi \in \mathfrak{G}_A \right\}$$

We call σ_A the spectrum of A

To tackle the notion of compatibility, we take the following route.

Definition 6.4.4. $\mathfrak{I} \subseteq \mathfrak{G}$ is called **complete** if it is full w.r.t. $\mathcal{U}_{\mathfrak{I}} := \left\{ A \in \mathcal{U} \mid \mathfrak{G}_A \supseteq \mathfrak{I} \right\} \subseteq \mathcal{U}$

Definition 6.4.5. $\mathfrak{I} \subseteq \mathfrak{G}$ is called a **deterministic set** for $\mathcal{B} \subseteq \mathcal{U}$ if $\mathfrak{I}$ is complete and $\mathcal{B} \subseteq \mathcal{U}_{\mathfrak{I}}$

Definition 6.4.6. $\mathcal{B} \subseteq \mathcal{U}$ is called **compatible** if $\mathfrak{G}_{\mathcal{B}} := \bigcap_{B \in \mathcal{B}} \mathfrak{G}_B$ is complete

Notice that if we say $\mathfrak{I}$ is complete, then $\mathcal{U}_{\mathfrak{I}}$ contains roughly all observables whose “eigenstates” are in $\mathfrak{I}$. And as $\mathcal{U}_{\mathfrak{I}}$ is full, $\mathfrak{I}$ is enough to distinguish between observables.

Lemma 6.4.1. $\mathcal{B} \subseteq \mathcal{U}$ is compatible iff it admits a deterministic set of states.

Proof. If $\mathcal{B}$ is compatible, then $\mathfrak{G}_{\mathcal{B}}$ is complete. Now,

$$\mathcal{U}(\mathcal{B}) := \mathcal{U}_{\mathfrak{G}_{\mathcal{B}}} = \left\{ A \in \mathcal{U} \mid \mathfrak{G}_A \supseteq \mathfrak{G}_{\mathcal{B}} \right\}$$

$\mathcal{B} \subseteq \mathcal{U}(\mathcal{B}) \Rightarrow \mathfrak{G}_{\mathcal{B}} =: \mathfrak{I}$ is the deterministic set of states.

Now conversely, if $\mathfrak{I}$ is the deterministic set of states of $\mathcal{B}$, then $\mathfrak{I}$ is complete and $\mathcal{B} \subseteq \mathcal{U}_{\mathfrak{I}}$. Hence,

$$\forall B \in \mathcal{B} : \mathfrak{G}_B \supseteq \mathfrak{I}$$

Hence $\mathfrak{G}_{\mathcal{B}} \supseteq \mathfrak{I}$. Hence,

$$\mathcal{U}(\mathcal{B}) = \left\{ A \in \mathcal{U} \mid \mathfrak{G}_A \supseteq \mathfrak{G}_{\mathcal{B}} \supseteq \mathfrak{I} \right\} \subseteq \mathcal{U}_{\mathfrak{I}}$$

Since $\mathfrak{I}$ is full w.r.t. $\mathcal{U}_{\mathfrak{I}}$, $\mathfrak{G}_{\mathcal{B}} \supseteq \mathfrak{I}$ is also full w.r.t. $\mathcal{U}_{\mathfrak{I}}$. Hence $\mathfrak{G}_{\mathcal{B}}$ is full w.r.t. $\mathcal{U}(\mathcal{B}) \subseteq \mathcal{U}_{\mathfrak{I}}$. Hence $\mathfrak{G}_{\mathcal{B}}$ is complete, which means $\mathcal{B}$ is compatible. $\blacksquare$

To understand these definitions better, we take a simple example.

Example 6.4.6.1. Consider the spin $\frac{1}{2}$ system. Choose,

$$\begin{aligned} \mathfrak{I} &= \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \\ \mathcal{B}_1 &= \{id, \sigma_1\} \\ \mathcal{B}_2 &= \{\sigma_1, \sigma_3\} \end{aligned}$$

where σ_i are the Pauli spin matrices. Now,

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} : id|_{\mathfrak{I}} \right\rangle = 1 \quad \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} : id|_{\mathfrak{I}} \right\rangle = 1$$

and

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \sigma_1|_{\mathfrak{I}} \right\rangle = 0 \quad \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \sigma_1|_{\mathfrak{I}} \right\rangle = 0$$

Which tells us that,

$$\sigma_1|_{\mathfrak{I}} \leq id|_{\mathfrak{I}}$$

To check if $\mathfrak{I}$ is full w.r.t. $\mathcal{B}_1$,

$$\text{for any } v \in \mathbb{C}^2 : v = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\langle v : id \rangle = |a|^2 + |b|^2 \quad \langle v : \sigma_1 \rangle = \frac{1}{2}(a^*b + ab^*) = \Re(a^*b) \leq |ab|$$

which tells us that $\sigma_1 \leq id$. Hence $\mathfrak{I}$ is full w.r.t. $\mathcal{B}_1$. Is $\mathfrak{I}$ complete? For that,

$$\mathcal{U}_{\mathfrak{I}} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \in \mathcal{U} \mid a, b \in \mathbb{R} \right\}$$

we should check if $\mathfrak{I}$ is full w.r.t. $\mathcal{U}_{\mathfrak{I}}$. Take any $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ and $\begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}$ such that,

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \Big|_{\mathfrak{I}} \leq \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \Big|_{\mathfrak{I}}$$

using

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right\rangle = a \quad \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right\rangle = b$$

this condition yields,

$$a \leq c \quad \text{and} \quad b \leq d$$

take any,

$$v = \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$$

Then,

$$\left\langle \begin{pmatrix} \lambda \\ \mu \end{pmatrix} : \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right\rangle = |\lambda|^2 a + |\mu|^2 b \quad \text{and} \quad \left\langle \begin{pmatrix} \lambda \\ \mu \end{pmatrix} : \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \right\rangle = |\lambda|^2 c + |\mu|^2 d$$

which gives,

$$|\lambda|^2 a + |\mu|^2 b \leq |\lambda|^2 c + |\mu|^2 d$$

Hence,

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \leq \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix}$$

which gives $\mathfrak{I}$ is complete. Now, is $\mathcal{B}_1$ compatible? For that,

$$\mathfrak{G}_{\mathcal{B}_1} = \mathfrak{G}_{id} \cap \mathfrak{G}_{\sigma_1} = \mathbb{C}^2 \cap \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} = \left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

It is complete (it is equivalent to what we just did above), hence $\mathcal{B}_1$ is compatible.

What about $\mathcal{B}_2$?

$$\mathfrak{G}_{\mathcal{B}_2} = \mathfrak{G}_{\sigma_3} \cap \mathfrak{G}_{\sigma_1} = \emptyset$$

Hence, no deterministic set of states, which means it is not compatible!

Axiom 4. $\forall A \in \mathcal{U}$ the set $\mathfrak{G}_A$ is deterministic for the one dimensional subspace generated by A in $\mathcal{U}$, and for $A, B \in \mathcal{U} : \mathfrak{G}_{A+B} \supseteq \mathfrak{G}_A \cap \mathfrak{G}_B$ with $\mathfrak{G}_1 = \mathfrak{G}$

This means that $\mathfrak{G}_A$ is complete and

$$\left\{ \alpha A \in \mathcal{U} \mid \alpha \in \mathbb{R} \right\} \subseteq \mathcal{U}_{\mathfrak{G}_A}$$

Also, since $\mathfrak{G}_A$ is complete, there won't be any $B \neq A : \mathfrak{G}_B \supseteq \mathfrak{G}_A$ and $A|_{\mathfrak{G}_A} = B|_{\mathfrak{G}_A}$. This tells us that an observable is completely determined by its deterministic set!

Axiom 5. For any $A \in \mathcal{U}$, and $n \in \mathbb{N}_0$, there is an element in $\mathcal{U}$ denoted by A^n such that,

1. $\mathfrak{G}_{A^n} \supseteq \mathfrak{G}_A$
2. $\langle \phi : A^n \rangle = \langle \phi : A \rangle^n \quad \forall \phi \in \mathfrak{G}_A$

It has the following properties,

1. A^n is unique.

Proof. $\forall \phi \in \mathfrak{G}_A : \langle \phi : \tilde{A}^n \rangle = \langle \phi : A^n \rangle \Rightarrow \langle \phi : \tilde{A}^n - A^n \rangle = 0 \Rightarrow \tilde{A}^n|_{\mathfrak{G}_A} = A^n|_{\mathfrak{G}_A}$ which combined with axiom 4 gives us the uniqueness. ■

2. $(A^n)^m = A^{nm}$ and $(\lambda A)^n = \lambda^n A^n$

Proof. Trivial using the definition itself. ■

3. $\forall \phi \in \mathfrak{G}_A : \langle \phi : A^2 \rangle = \langle \phi : A \rangle^2 \geq 0$. Now since $\mathfrak{G}_A$ is deterministic for a set formed by the linear combination of A and A^2 and $\mathfrak{G}_0 \supseteq \mathfrak{G}_A$, we have $\langle \phi : A^2 \rangle \geq 0 \Rightarrow A^2 \geq 0$.

4. Define,

$$\mathfrak{P}(A) := \{ \text{Linear combination of powers of } A \} \subseteq \mathcal{U}$$

For $B \in \mathfrak{P}(A) : \mathfrak{G}_B \supseteq \mathfrak{G}_A$. Since,

$$\mathcal{U}_{\mathfrak{G}_A} = \{ B \in \mathcal{U} \mid \mathfrak{G}_B \supseteq \mathfrak{G}_A \}$$

we have, $\mathfrak{P}(A) \subseteq \mathcal{U}_{\mathfrak{G}_A}$ which means $\mathfrak{G}_A$ is deterministic on $\mathfrak{P}(A)$. Hence the lemma we proved tells us that, $\mathfrak{P}(A)$ is compatible.

The last property is easily generalised to a theorem.

Theorem 6.4.2. *If $\mathcal{B}$ is compatible, then $\mathfrak{P}(\mathcal{B})$ is compatible.*

Definition 6.4.7. For $A, B \in \mathcal{U}$, define the “symmetrized product”;

$$A \circ B := \frac{1}{2}[(A + B)^2 - A^2 - B^2]$$

It has the following properties,

1. $A \circ B = B \circ A$
2. For $\phi \in \mathfrak{G}_A \cap \mathfrak{G}_B : \langle \phi : A \circ B \rangle = \langle \phi : A \rangle \langle \phi : B \rangle$

The proof is straight forward, applying the definition.

3. $A^n \circ A^m = A^{mn}$

Proof is obtained applying property 2.

Definition 6.4.8. $\{A, B, C\} := (A \circ B) \circ C - A \circ (B \circ C)$

Take any $\mathcal{B} \subseteq \mathcal{U}$. Then for any $A, B, C \in \mathcal{U}(\mathcal{B})$ and $\phi \in \mathfrak{G}_{\mathcal{B}}$ we have,

$$\langle \phi : \{A, B, C\} \rangle = 0$$

If $\mathcal{B}$ is compatible, meaning $\mathcal{U}(\mathcal{B}) \equiv \mathfrak{G}_{\mathcal{B}}$ is complete, then

$$\forall \phi \in \mathfrak{G}_{\mathcal{B}} : \langle \phi : \{A, B, C\} \rangle = 0 \Rightarrow \{A, B, C\} \Big|_{\mathfrak{G}_{\mathcal{B}}} = 0 \Rightarrow \{A, B, C\} = 0$$

In the same way, we can prove that,

$$A \circ (\lambda B + \mu C) = \lambda(A \circ B) + \mu(A \circ C)$$

We summarise these results as a theorem,

Theorem 6.4.3. $\circ : \mathcal{U} \times \mathcal{U} \rightarrow \mathcal{U}$ is associative and distributive over $\mathfrak{G}_{\mathcal{B}}$ where $\mathcal{B}$ is compatible.

Corollary 6.4.3.1. *If $\mathcal{B}$ is compatible, then $\mathfrak{P}(\mathcal{B})$ is associative*

Proof. $\mathfrak{P}(\mathcal{B}) \subseteq \mathcal{U}(\mathcal{B})$ ■

Remark. If $\mathcal{B}$ is compatible, then $\mathcal{U}(\mathcal{B})$ (hence $\mathfrak{P}(\mathcal{B})$) is equipped with an abelian distributive, associative algebra.

Axiom 6. For any $A, B, C \in \mathcal{U}$, A and C are said to be compatible if $\{A, B, C\} = 0$

Remark. .

1. For any A, B

$$\{A, B, \lambda \mathbb{1}\} = 0 \iff (A \circ B) \circ (\lambda \mathbb{1}) = A \circ (B \circ (\lambda \mathbb{1})) \iff \lambda(A \circ B) = A \circ (\lambda B) = (\lambda A) \circ B$$

2. $(A + B)^2 = A^2 + B^2 + 2(A \circ B)$

Proof.

$$\begin{aligned} \langle \phi : (A + B)^2 \rangle &= \langle \phi : A + B \rangle^2 \\ &= \langle \phi : A \rangle^2 + \langle \phi : B \rangle^2 + 2 \langle \phi : A \rangle \langle \phi : B \rangle \\ &= \langle \phi : A^2 \rangle + \langle \phi : B^2 \rangle + \langle \phi : 2(A \circ B) \rangle \\ &= \langle \phi : A^2 + B^2 + 2(A \circ B) \rangle \end{aligned}$$

■

Hence we can write,

$$A \circ B = \frac{1}{4}[(A + B)^2 - (A - B)^2]$$

3. Use the last remark to get,

$$((A \pm C) + B)^2 + ((A \pm C) - B)^2 = 2(A \pm C)^2 + 2B^2$$

Subtract the ‘ $-$ ’ equation from the ‘ $+$ ’ equation, and again use the last remark to get,

$$(A + B) \circ C + (A - B) \circ C = 2(A \circ C)$$

Change $A + B \rightarrow A$ and $A - B \rightarrow B$, hence $A \rightarrow \frac{A+B}{2}$ to get,

$$A \circ C + B \circ C = (A + B) \circ C$$

which is distributivity!

4. We have $\{A, B, A^2\} = 0$ for any $B \in \mathcal{U}$ which is equivalent to the Jordan Identity,

$$(A \circ B) \circ A^2 = A \circ (B \circ A^2)$$

The above remarks prove the following theorem;

Theorem 6.4.4. *$\mathcal{U}$ is an abelian, distributive (not necessarily associative) real Jordan algebra.*

Remark. A real commutative Jordan algebra is also called an r -number algebra.

6.4.2 Topological Structure

Let $A \in \mathcal{U}$ and $\mathfrak{J} \subseteq \mathfrak{G}_A$ be complete.

Definition 6.4.9.

$$\tau(A, \mathfrak{J}) := \left\{ \lambda \in \mathbb{R}^+ \mid \forall \phi \in \mathfrak{J} : |\langle \phi : A \rangle| < \lambda \right\}$$

and ,

$$\|A\|_{\mathfrak{J}} := \text{glb}\{\tau(A, \mathfrak{J})\}$$

Lemma 6.4.5. *$\|A\|_{\mathfrak{J}}$ is independent of $\mathfrak{J}$ and is equal to $\|A\| := \sup_{\phi \in \mathfrak{G}} |\langle \phi : A \rangle|$*

Proof. $\forall \lambda \in \tau(A, \mathfrak{J})$, we have $-\lambda \mathbb{1} \Big|_{\mathfrak{J}} < A \Big|_{\mathfrak{J}} < \lambda \mathbb{1} \Big|_{\mathfrak{J}}$. Since $\mathfrak{J}$ is complete, it is full w.r.t. $\mathcal{U}_{\mathfrak{J}}$ and $A, \mathbb{1} \in \mathcal{U}_{\mathfrak{J}}$. Hence,

$$-\lambda \mathbb{1} < A < \lambda \mathbb{1}$$

Hence, $|\langle \phi : A \rangle| < \lambda$ for all $\phi \in \mathfrak{G}$. Which gives,

$$\text{glb}\{|\langle \phi : A \rangle|\} = \sup_{\phi \in \mathfrak{G}} |\langle \phi : A \rangle| = \|A\| = \|A\|_{\mathfrak{J}}$$

■

Notice that $\|A\| = \sup_{\phi \in \mathfrak{G}} |\langle \phi : A \rangle|$ obeys all the properties of a norm.

Lemma 6.4.6. *$(\mathcal{U}, \|\cdot\|)$ is Banach*

Proof. Take a Cauchy sequence,

$$\|A_n - A_m\| < \epsilon \Rightarrow |\langle \phi : A_n \rangle - \langle \phi : A_m \rangle| < \epsilon \quad \forall \phi \in \mathfrak{G}$$

Since $\mathbb{R}$ is Banach, $\lim \langle \phi : A_n \rangle$ exists. Define A as,

$$\langle \phi : A \rangle := \lim_{n \rightarrow \infty} \langle \phi : A_n \rangle$$

■

Axiom 7. The norm of any $A \in \mathcal{U}$ is finite. $\mathcal{U}$ is topologically complete as a metric space where $d(A, B) := \|A - B\|$. $\mathfrak{G}$ is the set of all continuous positive linear functionals ϕ on $\mathcal{U}$ with $\langle \phi : \mathbb{1} \rangle = 1$.

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